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Part 1: World Geography

01 Universe and Solar System (Astronomy)

Origin of the Universe and Stars

- **Big Bang Theory:** Propounded by Belgian astronomer and priest **Georges Lemaître** • Detailed explanation — Robert Wagoner PYQ 1
- **Galaxy:** Earth's galaxy — 'Milky Way' • Shape **spiral** • Nearest galaxy — **Andromeda**
- **Proxima Centauri:** Nearest star to Earth after the Sun • about 4.25 light years from the Sun
- **Chandrasekhar Limit:** Subrahmanyan Chandrasekhar • **1.44 solar masses** • Lower mass → 'White Dwarf' = fossil star PYQ 2
- **Mass above 1.44:** 'Supernova' explosion → Neutron star (Pulsar) or 'Black Hole'
- **Black Hole:** Extremely high density and gravity • even light cannot escape • First described by John Wheeler and S. Chandrasekhar PYQ 2
- **Colour of a Star:** Indicator of 'surface temperature' • **Blue** = highest temperature • **Red** = lowest temperature PYQ 2

Solar System and Features of Planets

- **Sun's Energy:** 'Nuclear fusion' in the core (Hydrogen → Helium)
- **Photosphere:** Visible part of the Sun • temperature about 6000°C
- **Corona:** Outer part visible only during a solar eclipse • emits **X-rays**
- **Goldilocks Zone:** Habitable zone around a star • reasonable chance of liquid water on a planet's surface • neither too hot nor too cold PYQ 1
- **Astronomical Unit (AU):** Mean Earth-Sun distance • about **149.6 million km** (14.96 crore km) PYQ 1
- **Venus:** **Hottest and brightest** planet • 96% carbon dioxide in atmosphere → intense greenhouse effect PYQ 2
- **Nicknames of Venus:** 'Morning Star' • 'Evening Star' • 'Earth's Twin Sister'
- **Retrograde Rotation:** Venus and Uranus — **east to west** (clockwise) • Sun rises in the west, sets in the east PYQ 1
- **Other Planets:** Rotate west to east (anticlockwise)
- **Uranus:** Extreme axial tilt → **'Lying-down Planet'** (sideways planet)

Major Planets of the Solar System — Comparison

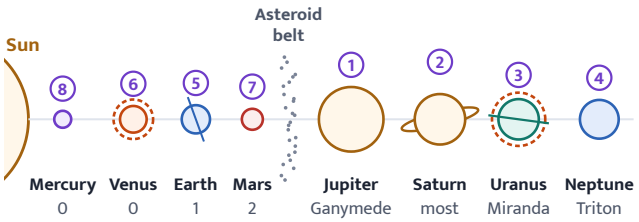
- **Mercury:** 0 satellites • Closest to the Sun • Smallest planet • Shortest revolution period (**88 days**) • Highest orbital speed • Greatest temperature range PYQ 2
- **Venus:** 0 satellites • Hottest • Closest to Earth • Slowest rotation (**243 days**) • East-to-west rotation PYQ 2
- **Earth:** 1 satellite (Moon) • Blue Planet • Highest density (**5.52 g/cm^3**) • Axial tilt 23.5° PYQ 2
- **Mars:** 2 satellites (Phobos, Deimos) • Red Planet (iron oxide) • **Olympus Mons (Nix Olympica)** — highest mountain and largest volcano of the Solar System PYQ 2
- **Jupiter:** Main satellite Ganymede • Largest and heaviest planet • Fastest rotation (about **9 hours 55 minutes**) • Ganymede — largest satellite of the Solar System • Rings present PYQ 2
- **Saturn:** Most satellites (Titan main) • Lowest density (**0.69 g/cm^3** — less than water, can float in water) • 7 distinct rings • Main satellites 'Titan' and 'Phoebe' PYQ 2
- **Uranus:** Satellites Miranda, Ariel • Discovered by **William Herschel** • Excess methane → appears green • East-to-west rotation • Alpha, Beta, Gamma rings PYQ 1
- **Neptune:** Main satellite Triton • Farthest from the Sun • Coldest • Longest revolution period (about **165 years**) PYQ 1

Motions of the Earth, Latitudes and Longitudes

- **Perihelion:** **3 January** • Earth closest to the Sun • about 147.0 million km (14.70 crore km)

Planets: Order, Size and Records

in order from the Sun



① = size rank (big → small) ⇢ = reverse rotation (east → west)
Below names: satellites (number / main) · sizes not to scale

Mercury: smallest · revolution 88 days	**Venus:** hottest · rotation 243 days
Earth: density 5.52 (highest)	**Mars:** Olympus Mons (highest)
Jupiter: largest · rotation 9 h 55 m	**Saturn:** density 0.69 (below water)
Uranus: 'lying planet' · Herschel	**Neptune:** coldest · revolution 165 yrs

- **Aphelion:** **4 July** • Earth farthest from the Sun • about 152.1 million km (15.21 crore km)
- **Equinox:** 21 March (Vernal/Spring Equinox) • 23 September (Autumnal Equinox) • rays vertical over the Equator • equal day and night all over the Earth PYQ 3
- **Summer Solstice:** **21 June** • Sun vertical over the Tropic of Cancer • longest day in the Northern Hemisphere
- **Winter Solstice:** **22 December** • Sun vertical over the Tropic of Capricorn • longest day in the Southern Hemisphere
- **International Date Line (180° longitude):** Through the Pacific Ocean (parallel to the Bering Strait) • **bent 8 times** — so that one country does not have two dates PYQ 2
- **Crossing the Date Line:** East → West = **one day added** • West → East = one day lost
- **Prime Meridian (0° longitude / Greenwich):** 8 countries — United Kingdom • France • Spain • Algeria • Mali • Burkina Faso • Togo • Ghana • also passes through Antarctica
- **Meeting point of 0° latitude and 0° longitude:** **'Gulf of Guinea'** in the Atlantic Ocean (near West Africa) PYQ 2

Points to Note

1. Mercury (closest to the Sun) is not the hottest — **Venus** is the hottest; Mercury's atmosphere is lost, while Venus's 96% carbon dioxide traps heat [1]
2. Earth's nearest planet is not Mars — **Venus** (by average distance Mercury is sometimes considered, but in Commission options 'Venus' is always correct)
3. The asteroid belt is not between Earth and Mars — it lies between the orbits of **Mars and Jupiter** [2]
4. Cancer, Capricorn and the Equator do not all pass through Asia — only **Africa** has all three [2]

Countries on Major Latitudes and Mnemonics

- **Equator (0°):** Total **13 countries**
- **Equator — South America (3):** Ecuador • Colombia • Brazil
- **Equator — Africa (7):** São Tomé and Príncipe • Gabon • Republic of the Congo • Democratic Republic of the Congo (Zaire) • Uganda • Kenya • Somalia
- **Equator — Asia/Oceania (3):** Maldives • Indonesia • Kiribati

Mnemonic (Equator, South America): "Eco Bra" — E = Ecuador, Co = Colombia, Bra = Brazil

Mnemonic (Equator, Asia): "Mai Ki" — Ma = Maldives, I = Indonesia, Ki = Kiribati

- **Tropic of Capricorn (23.5° S):** Total **10 countries**
- **Capricorn — South America (4):** Chile • Argentina • Paraguay • Brazil
- **Capricorn — Africa (5):** Namibia • Botswana • South Africa • Mozambique • Madagascar
- **Capricorn — Oceania (1):** Australia

Mnemonic (Capricorn, South America): "Cheel aur Baaz Parag laye" (Kite and Hawk brought pollen) — Cheel = Chile, Aur = Argentina, Baaz = Brazil, Parag = Paraguay

- **Tropic of Cancer (23.5° N):** **17 countries/territories** (including Western Sahara) PYQ 2
- **Cancer — Countries:** Mexico • Bahamas • Western Sahara • Mauritania • Mali • Algeria • Niger • Libya • Egypt • Saudi Arabia • UAE • Oman • India • Bangladesh • Myanmar • China • Taiwan
- **Cancer — Exceptions:** Only touches Chad's northern border, not its mainland • does not pass through **Iran and Pakistan**
- **Order by Size (largest to smallest):** Jupiter > Saturn > Uranus > Neptune > Earth > Venus > Mars > Mercury • Earth is the **fifth** largest

Mnemonic (Size Order): "Brihaspati Shani ki Aruna Varuna ne Prithvi par Shukra Mangal Budh kiya" — Jupiter, Saturn, Uranus, Neptune, Earth, Venus, Mars, Mercury

One-liner Facts

- **Sunlight to Earth:** On average **8 minutes 19 seconds** (about 500 seconds) PYQ 1
- **Titan (Saturn's moon):** Only body other than Earth with a dense atmosphere • rivers and lakes of liquid (hydrocarbon/methane) on the surface
- **Longitude and Time:** 1° longitude = **4 minutes** • 15° = 1 hour PYQ 1
- **'Sea of Tranquillity':** Vast dusty plain on the Moon • **Apollo-11** landed here PYQ 1
- **Comet:** Tail always points **away from the Sun** due to 'solar wind' pressure • **Halley's Comet** appears every 76 years PYQ 1

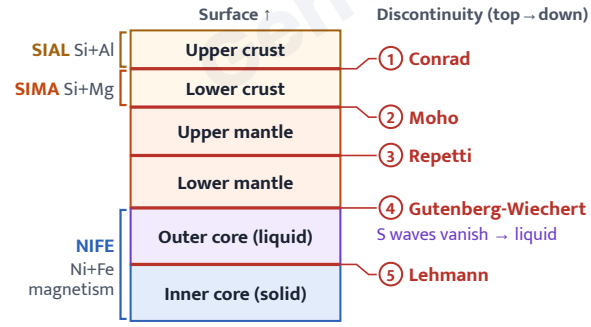
02 Geomorphology (Earth's Interior, Rocks, Volcanoes and Earthquakes)

Interior Structure of the Earth and Discontinuities

- **Three Main Layers:** **Crust** • **Mantle** • **Core**
- **Eduard Suess — SIAL (Silica + Aluminium):** Continental crust • acidic nature • low density
- **Eduard Suess — SIMA (Silica + Magnesium):** Oceanic crust and lower layer • basic nature
- **Eduard Suess — NIFE (Nickel + Ferrum/Iron):** Central part (core) • cause of **Earth's magnetism** PYQ 1
- **Elements in the Crust (by weight %):** Oxygen 46.6% • Silicon 27.7% • Aluminium 8.1% • Iron 5.0% • Calcium 3.6% PYQ 2
- **Elements in the Whole Earth (%):** Iron 35.5% • Oxygen 30.0% • Silicon 15.0% • Magnesium 13.0%

Discontinuities between Interior Layers

Earth's Layers and Discontinuities (thickness not to scale)



Kamre mein Gulel = Conrad · Moho · Repetti · Gutenberg · Lehmann
Most abundant: crust — oxygen 46.6% · whole Earth — iron 35.5%

- **Conrad Discontinuity:** Upper crust ↔ Lower crust PYQ 1
- **Mohorovičić (Moho) Discontinuity:** Lower crust ↔ Upper mantle PYQ 2
- **Repetti Discontinuity:** Upper mantle ↔ Lower mantle PYQ 1
- **Gutenberg-Wiechert Discontinuity:** Lower mantle ↔ Outer core PYQ 2
- **Lehmann Discontinuity:** Outer core (liquid) ↔ Inner core (solid) PYQ 1

Mnemonic (top to bottom): "Kamre mein Gulel" (Catapult in the room) — Ka = Conrad, Ma = Moho, Re = Repetti, Gu = Gutenberg, Lel = Lehmann

Rocks and their Metamorphism

- **Igneous Rocks:** Formed by cooling of magma/lava • crystalline • no layers • **no fossils at all** PYQ 2
- **Igneous — Examples:** Granite • Basalt • Gabbro • Pegmatite • Diorite
- **Sedimentary Rocks:** Formed by deposition by agents of weathering and erosion • layered • **fossils and mineral oil (petroleum, coal)** found in these PYQ 3
- **Sedimentary — Extent:** Only 5% of the crust's volume • but cover **75%** of the lithosphere
- **Sedimentary — Examples:** Sandstone • Limestone • Shale • Coal • Dolomite • Conglomerate • Chalk
- **Metamorphic Rocks:** Transformation of igneous and sedimentary rocks by heat and pressure • **hardest**

Metamorphism of Rocks

- **Granite (igneous):** → Gneiss PYQ 2
- **Basalt (igneous):** → Amphibolite / Schist PYQ 1
- **Gabbro (igneous):** → Serpentine PYQ 1
- **Sandstone (sedimentary):** → **Quartzite** PYQ 2
- **Limestone / Dolomite (sedimentary):** → **Marble** PYQ 2
- **Shale / Mudstone (sedimentary):** → Slate PYQ 1
- **Coal (sedimentary):** → Graphite / Diamond PYQ 1

- **Slate (metamorphic):** → Phyllite (further metamorphism → Schist) PYQ 1

Earthquakes and Seismic Waves

- **Focus (Hypocentre):** Place inside the Earth where seismic energy is released
- **Epicentre:** Point on the surface directly above the focus (at 90°) • waves reach here first • **maximum destruction** PYQ 2
- **P-waves (Primary/Longitudinal):** Like sound waves • **fastest** (arrive first) • least intensity • pass through **solid, liquid and gas**
- **S-waves (Secondary/Transverse):** Like light/water waves • **only through solids** • vanish in liquids → proof: outer core is **liquid**
- **L-waves (Surface/Love waves):** Slowest • largest amplitude and longest travel path → **most destructive** PYQ 2
- **Richter Scale:** Measures **magnitude** (energy released) • logarithmic • each next unit: amplitude 10 times, energy about 32 times PYQ 2
- **Mercalli Scale:** Measures **intensity** (effects of destruction) • 1 to 12

Major Volcanoes of the World

Major Volcanoes of the World

Pacific-centred map



▲ active ▲ dormant / active-dormant ▲ extinct

1. Kilauea (Hawaii) — most active
2. Etna, Stromboli, Vesuvius (Italy)
3. Cotopaxi — highest active
4. Ojos del Salado (Andes)
5. Aconcagua — highest extinct
6. Fujiyama (Japan)
7. Krakatoa (Indonesia)
8. Kilimanjaro (Tanzania)
9. Popa (Myanmar)
10. Damavand (Iran)
11. Rainier / St. Helens (USA)
12. Taal / Pinatubo (Philippines)

- **Mount Kilauea:** Hawaii Island (USA) • world's **most active** volcano PYQ 2
- **Mount Etna:** Sicily Island (Italy) • highly active volcano of Europe PYQ 2
- **Stromboli:** Lipari Islands (Italy — Mediterranean Sea) • **'Lighthouse of the Mediterranean'** PYQ 2
- **Cotopaxi:** Ecuador • world's **highest active** volcanic mountain PYQ 2
- **Ojos del Salado:** Argentina-Chile border (Andes) • highest-elevation active volcano on Earth PYQ 1
- **Mount Vesuvius:** Italy (Bay of Naples) • dormant PYQ 1
- **Fujiyama:** Japan • dormant PYQ 1
- **Krakatoa:** Indonesia (Sunda Strait) • dormant/active • Mount Merapi and Sinabung also in Indonesia PYQ 2
- **Mount Kilimanjaro:** Tanzania (Africa) • extinct • **highest peak of Africa** PYQ 2
- **Mount Popa:** Myanmar • extinct PYQ 2
- **Damavand and Koh-e-Sultan:** Iran (Koh-e-Sultan also in Balochistan/Pakistan) • extinct PYQ 2
- **Mount Rainier / St. Helens:** USA • active/dormant PYQ 2
- **Mount Taal / Pinatubo:** Philippines • active PYQ 2

Points to Note

1. Oxygen is not the most abundant in both crust and whole Earth — crust: **Oxygen** (46.6%), whole Earth: **Iron** (35.5%) [2]
2. If the most abundant 'metal' in the crust is asked — **Aluminium**
3. Acidic lava is not thin — high silica (65-85%) → **highly viscous and sticky**, violent explosion, forms cones
4. Basic lava — low silica → thin, spreads far to form plateaus (e.g. **Deccan Plateau**)
5. Mount Aconcagua is not active — highest peak of Argentina/Andes, world's highest **extinct volcano**; Cotopaxi and Ojos del Salado are active

One-liner Facts

- **'Ring of Fire':** Coastal belt of the Pacific Ocean • about **two-thirds** of the world's volcanoes and earthquakes • convergent plate margins PYQ 2
- **Isoseismal Line:** Joins places of equal intensity • **Homoseismal Line:** joins places experiencing the quake at the same time PYQ 1
- **Intrusive Forms:** Dome-shaped magma bodies — **'Batholith'** (largest) and **'Laccolith'** • saucer-shaped body — **'Lopolith'**
- **'Old Faithful':** Yellowstone National Park (USA) • world's most famous geyser • Alaska's **'Katmai Valley'** = 'Valley of Ten Thousand Smokes' PYQ 1
- **Continental Drift Theory:** 1912 • **Alfred Wegener** • 'Plate Tectonics' — Harry Hess, McKenzie and Morgan PYQ 1

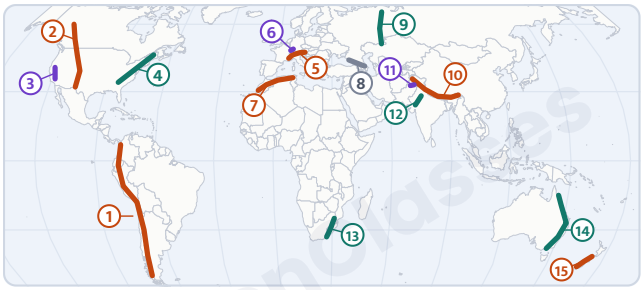
03 Major Mountains, Plateaus, Plains, Deserts and Valleys of the World

Major Mountain Ranges and Highest Peaks of the World

- **Andes:** South America (7 countries) • Young fold • highest **Aconcagua (6960 m)** PYQ 2
- **Rockies:** North America • Young fold • Mount Elbert PYQ 2
- **Himalayas:** Asia • Young fold • **Mount Everest (8848.86 m)** PYQ 1
- **Great Dividing Range:** Australia (east coast) • Old fold/residual • Mount Kosciuszko PYQ 2
- **Alps:** Europe (France, Switzerland, Italy, Austria) • Young fold • Mont Blanc (in France) PYQ 2
- **Atlas:** Africa (Morocco, Algeria, Tunisia) • Young fold • Mount Toubkal PYQ 2
- **Caucasus:** Europe-Asia border • Fold • **Mount Elbrus** (highest peak of Europe) PYQ 1
- **Urals:** Russia (separates Europe and Asia) • Old fold • Gora Narodnaya PYQ 1
- **Appalachians:** North America (eastern part) • Old fold • Mount Mitchell PYQ 1

Major Mountain Ranges of the World

colour = type



young fold — old fold / residual — block — fold

1 Andes	6 Black Forest / Vosges	11 Salt Range
2 Rockies	7 Atlas	12 Aravalli
3 Sierra Nevada	8 Caucasus	13 Drakensberg
4 Appalachians	9 Urals	14 Great Dividing
5 Alps	10 Himalayas	15 Southern Alps

Length (km): Andes 7000 › Rockies 4800 › Gt. Dividing 3500 › **Himalayas 2400**

- **Aravalli:** India • **world's oldest** fold/residual mountain • Guru Shikhar (1722 m) PYQ 2
- **Black Forest / Vosges:** Germany (Black Forest) and France (Vosges) • **Block mountain (horst)** • Feldberg PYQ 3
- **Sierra Nevada:** California (USA) • **world's most extensive block mountain** • Mount Whitney PYQ 1
- **Salt Range:** Pakistan • Block mountain • Sakesar PYQ 1
- **Southern Alps:** **New Zealand** (South Island) • Young fold • Mount Cook (Aoraki) PYQ 2
- **Drakensberg:** South Africa • Old fold/escarpment • Thabana Ntlenyana (Lesotho) PYQ 2
- **Decreasing order of length (standard):** **Andes (7000 km) > Rockies (4800 km) > Great Dividing Range (3500 km) > Himalayas (2400 km)** PYQ 2

Points to Note

1. The Himalayas are sometimes placed third, sometimes fourth — in the standard order the **Great Dividing Range is third, the Himalayas fourth**

Major Plateaus and Valleys of the World

- **Tibetan Plateau (Changtang):** **Intermontane plateau** between the Himalayas and Kunlun • world's **highest** (average 4500 m) and largest plateau PYQ 1
- **Pamir Plateau:** **'Roof of the World'** PYQ 1
- **Anatolian Plateau (Turkey):** Between the Pontic and Taurus ranges (intermontane) • also called **'Asia Minor'** PYQ 2
- **Chiapas Plateau:** Southern Mexico • Pacific Ocean coast PYQ 1
- **Meseta Plateau:** Spain and Portugal (Iberian Peninsula) • capital **Madrid** lies on it PYQ 1
- **Shan Plateau:** Myanmar • silver and tin production PYQ 1
- **Katanga Plateau:** Democratic Republic of the Congo (Zaire) • **copper and diamonds** • source of the Congo and Zambezi rivers PYQ 2
- **Death Valley:** California (Mojave Desert), USA • **Rift valley** • extreme heat • **lowest point of North America (−86 m)** PYQ 3
- **Panjshir Valley:** Afghanistan • near the Hindu Kush • **emerald** mining PYQ 2
- **Rhine Valley:** Famous **rift valley** between Germany's Black Forest and France's Vosges (block mountains) PYQ 1
- **Valley of the Kings:** Egypt • **west bank** of the Nile PYQ 1
- **Silicon Valley (Santa Clara Valley):** Northern California (USA) • southern part of San Francisco Bay • computer chip and software industry PYQ 1
- **Taylor Valley:** Antarctica • discovered by **Robert Falcon Scott** • named by him the 'Valley of the Dead' PYQ 1

Major Deserts of the World and Their Features

- **Sahara:** North Africa (11 countries) • world's **largest non-polar (hot) desert** PYQ 1
- **Atacama:** Northern Chile and Peru (South America) • **world's driest** • Humboldt cold current aids its formation • nitrate and copper reserves PYQ 3
- **Kalahari:** Botswana (mainly), Namibia, South Africa • Hot • **Bushman** tribe PYQ 3
- **Taklamakan:** China (Tarim Basin, Xinjiang province) • Temperate/cold PYQ 2
- **Gobi:** Mongolia and China • Temperate/cold • two-humped **Bactrian camel** PYQ 2
- **Dasht-e-Lut / Dasht-e-Kavir:** Iran • Hot and saline PYQ 2
- **Karakum / Kyzylkum:** Turkmenistan (Karakum) • Uzbekistan-Kazakhstan (Kyzylkum) • Central Asia PYQ 2
- **Sonoran / Mojave:** USA and Mexico • Hot PYQ 2
- **Great Victoria / Gibson / Simpson:** Australia ('land of the thirsty') • Sturt Stony and Tanami also here PYQ 2
- **Patagonia:** Argentina (partly Chile) • **cold desert** in the **rain-shadow** of the Andes PYQ 2
- **Rub' al Khali / An-Nafud:** Saudi Arabia • world's largest **continuous sand (erg)** desert PYQ 1

Major Grasslands of the World

- **Savanna:** East and Central Africa (Sudan, Kenya, Tanzania) • **Tropical** PYQ 2
- **Llanos:** Venezuela and Colombia (Orinoco Basin) • **Tropical** PYQ 3
- **Campos:** Brazil (Brazilian Highlands) • **Tropical** PYQ 2
- **Selvas:** Amazon Basin (Brazil) • Equatorial evergreen rainforest (not a grassland, but appears in matching questions) PYQ 2
- **Prairies:** USA and Canada • Temperate PYQ 2
- **Pampas:** Argentina, Uruguay, southern Brazil • Temperate • **alfalfa grass** PYQ 3
- **Steppes:** Eurasia (Ukraine, Russia to Central Asia) • Temperate PYQ 1
- **Veld:** South Africa • Temperate PYQ 2
- **Downs:** Australia (Murray-Darling Basin) • Temperate PYQ 2
- **Canterbury:** New Zealand • Temperate PYQ 2
- **Puszta:** Hungary • Temperate PYQ 2
- **Manchuria:** China • Temperate PYQ 1

Mnemonic (tropical grasslands): "Saloni ka Kaam Pass hua" — Sa = Savanna, Loni = Llanos, Kaam Pass = Campos

- **Other major grasslands:** Prairies, Pampas, Veld, Downs, Steppes, Puszta, Canterbury — all **temperate**

Points to Note

1. Sahara is not the largest desert — overall largest is **Antarctica** (polar cold), second the Arctic; Sahara is third, but the **largest hot (non-polar)**

2. Llanos and Pampas are both in South America, but **Llanos (Venezuela) is tropical, Pampas (Argentina) temperate** [2]
3. The 'Alps' are in Europe, but the **'Southern Alps' are in New Zealand** (western part of South Island) — not in Europe or Australia [2]

One-liner Facts

- **Tropical deserts:** **western margins** of continents • 15°-30° latitudes • caused by the **subtropical high-pressure belt (anticyclonic conditions)** and **cold ocean currents** PYQ 2
- **Pyrenees:** natural boundary between Spain and France • **Jura Mountains:** between France and Switzerland PYQ 2
- **Grand Canyon (USA):** deep gorge carved by the **Colorado River** in the Colorado Plateau
- **Bolivian Plateau (Altiplano):** within the Andes • world-famous for **tin** production PYQ 1
- **Australian deserts (north → south):** **Tanami → Simpson → Gibson → Great Victoria**

04 Climatology (Atmosphere, Winds, Cyclones and Climatic Regions)

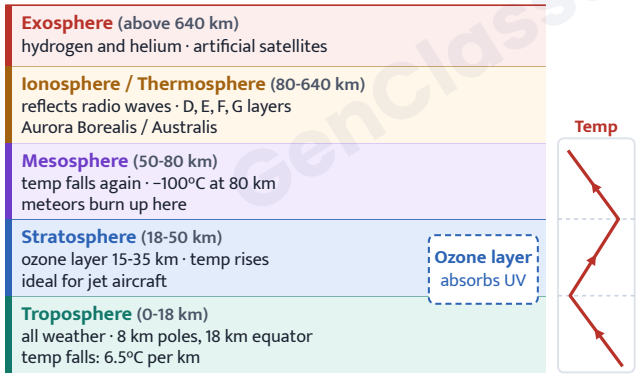
Composition and Layers of the Atmosphere

- **Major gases (by volume):** Nitrogen 78.08% • Oxygen 20.95% • **Argon 0.93%** (most abundant inert gas) • Carbon dioxide 0.036% or 0.04% • Neon 0.002% • Helium 0.0005% PYQ 2
- **5 layers (bottom → top):** Troposphere → Stratosphere → Mesosphere → Ionosphere/Thermosphere → Exosphere
- **Troposphere:** 8 km at poles • **18 km at equator** (highest due to convection currents) • all weather phenomena (clouds, rain, cyclones, fog) PYQ 2
- **Troposphere lapse:** 1°C per 165 m • normal lapse rate **6.5°C per km**
- **Stratosphere:** 18-50 km • no weather in lower part → **ideal for jet aircraft** PYQ 3
- **Ozone layer:** 15-35 km • absorbs ultraviolet rays → temperature **rises with height** in the stratosphere
- **Mesosphere:** 50-80 km • temperature falls again with height • **−100°C at 80 km (lowest in the atmosphere)** • **meteors burn up here**
- **Ionosphere / Thermosphere:** 80-640 km • electrically charged particles (ions) • **reflects radio waves** → communication • D, E, F, G layers PYQ 2
- **Polar lights:** in the ionosphere • Northern Hemisphere **Aurora Borealis** • Southern **Aurora Australis**
- **Exosphere:** above 640 km • hydrogen and helium dominate • artificial satellites placed here (geostationary satellites about 36,000 km)
- **Albedo:** ratio of solar radiation received by a surface to that reflected straight back to space without absorption PYQ 2
- **Albedo values:** Earth's average 30-35% • **fresh snow highest 75-90%**

Planetary Winds and Ferrel's Law

- **Coriolis force / Ferrel's Law:** deflecting force due to Earth's rotation • winds turn **right in the Northern Hemisphere, left in the Southern** • zero at equator, maximum at poles PYQ 1
- **Doldrums (calm belt):** 5° (sometimes 10°) on either side of the equator • low-pressure belt
- **Horse Latitudes:** 30°-35° in both hemispheres • subtropical high-pressure belt • air subsides, winds calm PYQ 1
- **Ferocity of Westerlies:** Southern Hemisphere has less land, more ocean → very strong

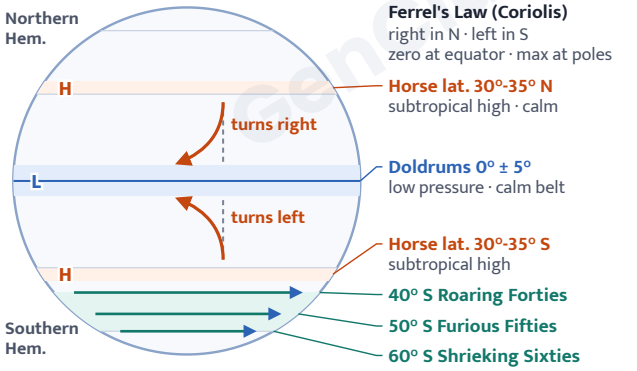
5 Layers of the Atmosphere (bottom → top · not to scale)



- **Names of Westerlies (Southern Hemisphere):** 40° S **Roaring Forties** • 50° S **Furious Fifties** • 60° S **Shrieking Sixties** PYQ 2

Major Local Winds of the World (Hot and Cold)

Pressure Belts and Winds (H = high · L = low pressure)



Westerlies fiercer in S: less land, more ocean

- **Chinook:** Hot-dry • eastern slopes of the Rockies (USA and Canada) • **'Snow-eater'** • beneficial for ranchers PYQ 3
- **Foehn (Föhn):** Hot-dry • northern slopes of the Alps (greatest effect in **Switzerland**) • helps ripen grapes PYQ 3
- **Sirocco:** Hot, dry, dusty • from the Sahara across the Mediterranean to Italy and Spain • red rain in Italy = **'Blood Rain'** PYQ 2
- **Other names of Sirocco:** Egypt **Khamsin** • Libya **Ghibli** • Tunisia **Chili** • Arabia **Simoom** — all hot winds of the Sahara/Arabia PYQ 3
- **Harmattan:** Hot-dry • from the Sahara to the Guinea Coast (West Africa) • relief from humid, sultry climate → **'Doctor wind'** PYQ 3
- **Brickfielder:** Hot-dry • Australia (Victoria) • hot desert wind PYQ 2
- **Black Roller / Santa Ana:** Hot-dry • North American plains (Black Roller) • southern California (Santa Ana) • Santa Ana notorious for spreading **wildfires** PYQ 2
- **Nor'wester:** Hot-dry • New Zealand (South Island) • descends from mountains PYQ 1

- **Zonda:** Hot-dry • Argentina and Uruguay (eastern slopes of the Andes) • warm wind in winter PYQ 2
- **Mistral:** **Cold**-dry • Spain and **France** (Rhone Valley to the Mediterranean) • strong cold wind from the Alps PYQ 3
- **Bora:** **Cold**-dry • eastern coast of the Adriatic Sea (**Italy**, Yugoslavia/Croatia) • icy wind from Central Europe PYQ 2
- **Blizzard:** **Cold** (snowstorm) • USA, Canada, Siberia (tundra) • polar wind laden with ice crystals PYQ 1
- **Pampero:** **Cold** polar • Argentina, Uruguay (Pampas region) • South America PYQ 1
- **Purga / Buran:** **Cold** icy • Russia and Siberia • tundra region PYQ 1
- **Levanter:** **Cold** • southern Spain and Strait of Gibraltar • cold humid easterly wind PYQ 2

Cyclones and Anticyclones

Basis	Cyclone	Anticyclone
Centre	**Low pressure** (high outside)	**High pressure**
Wind direction	Periphery to centre	Centre outward
Northern Hemisphere	**Anti-clockwise**	Clockwise
Southern Hemisphere	**Clockwise**	Anti-clockwise
Weather	Heavy rain and storms	**Clear sky, calm weather**
Previous questions	[3]	[1]

- **Eye of the cyclone:** centre of a tropical cyclone • **clear sky, calm wind, no rain**
- **Hurricane:** Caribbean Sea, Gulf of Mexico, USA (Atlantic Ocean)
- **Typhoon:** China Sea and Philippines (western Pacific Ocean)
- **Tornado:** USA (Mississippi Valley) and Mexico • smallest but **most destructive and fastest** land/coastal cyclone
- **Intensity scales:** Tornado **Fujita scale** • Hurricane **Saffir-Simpson scale**
- **Willy-willy:** north-western **Australia** • **Baguio:** Philippines • **Taifu:** Japan
- **Cyclone:** Indian Ocean (India, Bangladesh) PYQ 5

Major Climatic Regions of the World

- **Mediterranean climate:** 30°-45° latitudes • **western parts** of continents • countries around the Mediterranean, central California, central Chile, Cape Province of South Africa, southern and south-western Australia
- **Mediterranean feature:** **dry summers, rain only in winter** (Westerlies) • **citrus and juicy fruits** (grapes, lemon, orange, olive) • wine industry PYQ 4
- **Equatorial climate (10° N to 10° S):** high temperature year-round • **daily afternoon convectional rain (4 o'clock rain)** • minimum annual range • evergreen rainforests (Selvas) PYQ 2
- **Savanna / Sudan type climate:** summer rain, dry winter • **land of big game**

Points to Note

1. Ozone layer is not in the troposphere — 90% lies in the **stratosphere**; concentration/thickness measured in **Dobson Units** (decibel measures sound) [2]
2. Willy-willy is not a cold wind of the USA — it is a **tropical cyclone of north-western Australia** [2]
3. Relative humidity does not rise with temperature — **inversely related**; temperature rises → moisture-holding capacity rises → relative humidity **falls** [2]
4. Relative humidity is measured by a **hygrometer** — a **hydrometer** measures relative density of liquids

One-liner Facts

- **Most abundant greenhouse gas:** **water vapour** • then carbon dioxide and methane PYQ 1
- **Highest annual temperature range:** **Verkhoyansk**, Siberia (Russia) • Earth's 'Cold Pole'
- **Cirrus clouds:** highest (8-12 km) • made of ice crystals • **indicate approaching cyclone** PYQ 2
- **Cumulonimbus clouds:** torrential rain, thunderstorms, hailstorms
- **Front:** sloping boundary zone where air masses of opposite nature (warm and cold) meet → origin of **temperate cyclones**
- **Isobar:** joins places of equal pressure • **Isoneph:** joins places of equal cloud cover PYQ 1

05 Oceanography (Ocean Relief, Salinity, Currents, Coral Reefs and Tides)

Ocean Relief and Major Trenches

- **Continental Shelf:** shallow submerged part of continents • depth up to **100 fathoms / 200 m** • slope less than 1°
- **Extent of Shelf:** about **8.6%** of total ocean area • **20% of world petroleum and natural gas** • major fishing grounds lie here
- **Abyssal Plain:** most extensive part of the ocean floor • about **75.9%** of area • depth 3000-6000 m
- **Pacific Ocean Trenches:** **Mariana Trench** (near Philippines) • **Tonga** • **Kuril** • **Japan** • **Mindanao** • **Atacama (Peru-Chile)** • **Aleutian** PYQ 4
- **Mariana Trench:** **deepest trench** in the world • depth **11,022 m** • deepest point **Challenger Deep**
- **Atlantic Ocean Trenches:** **Puerto Rico Trench** (deepest in Atlantic) • **Romanche Trench** (South Atlantic) • **South Sandwich Trench** • **Cayman Trench** PYQ 4
- **Indian Ocean Trenches:** **Sunda (Java) Trench** (deepest in Indian Ocean) • **Diamantina Trench** (exam-relevant special trench) PYQ 4
- **Indian Ocean Ridges:** **Ninety East Ridge (90° East Ridge)** • **Carlsberg Ridge** PYQ 3
- **Mid-Atlantic Ridge:** shaped like the English letter 'S' • parts: **Dolphin Rise** (northern) • **Challenger Rise** (southern) PYQ 3
- **Telegraphic Plateau:** North Atlantic Ocean PYQ 3

Ocean Salinity

- **Average Salinity:** **35 parts per thousand (35‰)**
- **Isohaline:** line joining places of equal salinity
- **Salts in Sea Water (decreasing order):** **Sodium Chloride (77.8%)** > **Magnesium Chloride (10.9%)** > Magnesium Sulphate (4.7%) > Calcium Sulphate (3.6%) > Potassium Sulphate (2.5%) PYQ 3
- **Most Saline Water Bodies (decreasing order):** Lake Assal → Lake Van → Dead Sea → Great Salt Lake PYQ 4
- **Lake Assal (Djibouti):** 34.8% (**348‰**) PYQ 4

- **Lake Van (Turkey): 330‰** • usually placed first in Commission options PYQ 4

Salinity: average 35‰ — isohaline = line of equal salinity

Most saline water bodies (parts per thousand, ‰)

Lake Assal (Djibouti)	highest	348‰
Lake Van (Turkey)	often first in Commission options	330‰
Dead Sea	Jordan-Israel	238‰
Great Salt Lake (USA)		220‰
Ocean average		35‰

Salts in sea water (%, decreasing)

Sodium chloride 77.8 | 10.9

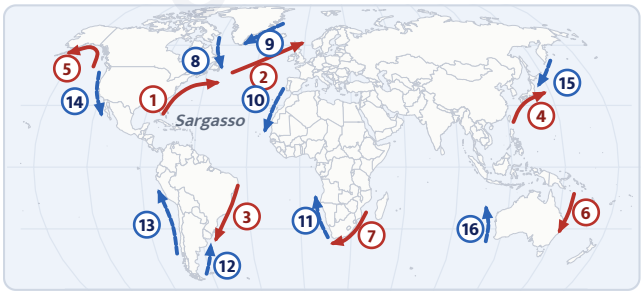
Magnesium chloride 10.9 > magnesium sulphate 4.7 > calcium sulphate 3.6 > potassium sulphate 2.5

Trap: highest salinity not at Equator — 20°-40° N and 10°-30° S (subtropical high: high evaporation, low rain)

- **Dead Sea (Jordan-Israel border): 238‰** PYQ 4
- **Great Salt Lake (USA): 220‰** PYQ 4
- **Latitudinal Distribution:** highest salinity not at Equator — **20°-40° N and 10°-30° S (Subtropical High Pressure Belt)** • reason: high evaporation, very low rainfall PYQ 2
- **Lower Salinity at Equator:** heavy daily rainfall • inflow of fresh water from large rivers PYQ 2

Ocean Currents (Warm and Cold)

Major Ocean Currents — red solid = warm · blue dashed = cold



Warm: Equator → Poles
1 Gulf Stream · **2** N. Atlantic Drift · **3** Brazil · **4** Kuroshio · **5** Alaska · **6** E. Australian · **7** Agulhas

Warm-cold meeting = fishing:
1 + 8 Grand Banks · 4 + 15 Japan coast

Cold: Poles → Equator
8 Labrador · **9** E. Greenland · **10** Canary · **11** Benguela · **12** Falkland · **13** Humboldt (Peru) · **14** California · **15** Oyashio · **16** W. Australian

- **Warm Current:** from Equator towards Poles • **Cold Current:** from Poles towards Equator • exam mainly asks to identify **cold currents**
- **Atlantic - Cold Currents: Labrador** (North) • **Canary** (North, NW of Africa) • **Benguela** (South, SW of Africa) • **Falkland** (South) • **East Greenland** PYQ 5
- **Atlantic - Warm Currents: Gulf Stream** (warm blanket of Europe) • North Atlantic Drift • Brazil (South) • Florida and Antilles • Irminger and Rennell • Guinea PYQ 5
- **Pacific - Cold Currents: Humboldt / Peru** (South) • **California** (North) • **Oyashio / Kuril** (North) PYQ 4
- **Pacific - Warm Currents: Kuroshio** (Black Current of Japan) • Alaska (North) • **El Nino** (counter warm current) • East Australian • Tsushima PYQ 4
- **Indian - Cold Current: West Australian Current** — the only major permanent cold current of the Indian Ocean PYQ 4
- **Indian - Warm Currents: Agulhas** (SW Indian Ocean) • Mozambique and Madagascar • **Somali** (cold/upwelling in summer) PYQ 4

Mnemonic (major cold currents): "Hum bole Green bagula kela kyon phek raha hai" — Hum bole = Humboldt (Peru), Green = East Greenland, Bagula = Benguela, Kela = California, Kyon = Kuril (Oyashio) and Canary, Phek = Falkland

- **Other Cold Currents:** Labrador • West Australian
- **Sargasso Sea:** North Atlantic (20°-40° N) • **sea without coast** • rootless seaweed **Sargassum** floats PYQ 3
- **Currents Enclosing Sargasso: Gulf Stream** (west) • **North Atlantic Drift** (north) • **Canary cold current** (east) • **North Equatorial Current** (south) PYQ 3
- **Warm-Cold Current Meeting:** dense fog • abundant **plankton** (fish food) • major fishing grounds PYQ 2
- **Grand Banks:** near Newfoundland (Canada) • **Gulf Stream (warm) + Labrador (cold)** PYQ 2
- **Japan Coast: Kuroshio (warm) + Oyashio (cold)** meet PYQ 2
- **Dogger Bank:** North Sea (Europe) • famous fishing ground PYQ 2

Coral Reefs and Tides

- **Coral:** marine organism (polyp) of phylum **Coelenterata** • secretes lime (**calcium carbonate**) • symbiotic with alga **Zooxanthellae**
- **Coral - Ideal Conditions:** tropical oceans (30° N-30° S) • water temperature **20°-21° C** • depth **less than 200 ft (60-70 m)** (no sunlight below) PYQ 2
- **Coral - Water Conditions:** salinity **27‰-30‰** • **sediment-free clear water** • cannot survive in fully fresh water or river mouths PYQ 2
- **Great Barrier Reef:** NE coast of Australia (parallel to Queensland, Pacific/Coral Sea) • world's **largest barrier reef** • length about **1900 km** PYQ 2
- **Coral Bleaching:** rise in sea temperature (global warming, El Nino) and pollution → loss of Zooxanthellae → corals turn **white** PYQ 3
- **Tides:** attraction of Sun and Moon + Earth's **centrifugal force** • rise of water = high tide • fall = low tide (ebb)
- **Moon's Tide-generating Force:** being closer to Earth, **2.17 times** that of Sun (more than double)
- **Spring Tide: New Moon and Full Moon** • Sun-Moon-Earth in a straight line (**Syzygy**) • **20% higher** than normal PYQ 2
- **Neap Tide: 7th / 8th day** of Shukla and Krishna Paksha • Sun and Moon at **right angle (90°)** at Earth's centre • **20% lower** than normal PYQ 1
- **Interval Between Two Successive Tides: 12 hours 26 minutes** (Moon moves ahead) PYQ 2
- **Highest Tide: Bay of Fundy** (Nova Scotia, Canada) PYQ 2
- **Southampton (England):** tides **4 times** a day PYQ 2

Points to Note

1. **Alaska Current and North Atlantic Drift:** both **warm currents** — flow from low to high latitudes (myth: cold because they flow in high latitudes) • Brazil and Agulhas also warm [2]

2. **Agulhas:** warm current • **SW Indian Ocean**, east coast of Africa (formed by Mozambique + Madagascar currents) — not South Atlantic [2]
3. **West Coast of Africa (South Atlantic): Benguela** cold current [2]
4. **Coral and Water:** sediment-free water needed (sediment clogs coral mouths) — but **fully fresh water is fatal** • salinity 27‰-30‰ essential

One-liner Facts

- **Antarctic Circumpolar Current (West Wind Drift):** south of 60° S, around all three oceans, west to east • world's **largest (highest water volume) cold current**
- **Atoll (ring-shaped coral reef):** horseshoe or ring shape around an island/submerged plateau • e.g. **Lakshadweep** (India) • **Funafuti Atoll** (Pacific)
- **Tidal Bore:** tidal waves enter river mouth like a wall against river flow • in India **Hooghly and Mahi**
- **El Nino:** warm current off Peru coast • **weakens** Indian monsoon (drought) PYQ 2
- **La Nina:** cold current • **strengthens** Indian monsoon (good rainfall) PYQ 2
- **Mean Annual Surface Temperature of Oceans:** about **17.2° C** (about 63° F) • Northern Hemisphere oceans warmer than Southern due to more land

06 Major Rivers, Lakes, Waterfalls, Canals and Straits of the World

Major Rivers of the World

- **Nile:** source Lake Victoria (White Nile) and Lake Tana, Ethiopia (Blue Nile) • mouth Mediterranean Sea • **longest river in the world (6650 km)** PYQ 3
- **Nile - Special:** both streams meet at **Khartoum** (capital of Sudan) • in Egypt **Aswan Dam** and **Lake Nasser** PYQ 3
- **Amazon:** Andes (Peru) → Atlantic • **largest river in the world** by drainage basin and water volume • largest tributary **Madeira** PYQ 2
- **Congo (Zaire):** confluence of Lualaba and Luapula → Atlantic • **crosses Equator twice** • **Stanley** and **Livingstone** Falls PYQ 3
- **Limpopo:** Highveld (South Africa) → Indian Ocean • **crosses Tropic of Capricorn twice** PYQ 3
- **Mahi:** Vindhya Range (Dhar, Madhya Pradesh) → Gulf of Khambhat (Arabian Sea) • **only river in the world crossing Tropic of Cancer twice** PYQ 2
- **Niger:** Guinea Highlands → Gulf of Guinea • **Oil River (Palm Oil River)** • **Kainji Dam** PYQ 2
- **Zambezi:** Katanga Plateau → Indian Ocean (Mozambique Channel) • **Victoria Falls** • man-made **Kariba Dam** (among the dams with highest hydropower capacity) PYQ 2
- **Mississippi-Missouri:** Lake Itasca (USA) → Gulf of Mexico • largest river of North America • **Bird-foot Delta** PYQ 3
- **St. Lawrence:** Lake Ontario → Gulf of St. Lawrence • world's **busiest inland waterway** • Niagara Falls in this system PYQ 1
- **Rhine:** Alps (Switzerland) → North Sea (near Rotterdam) • flows in rift valley • **Europe's busiest inland waterway** • **Coal River** PYQ 3
- **Danube:** Black Forest (Germany) → Black Sea • second longest in Europe • flows through **most countries (10)** in the world PYQ 2
- **4 Capitals on Danube: Vienna** • **Bratislava** • **Budapest** • **Belgrade** PYQ 2
- **Volga:** Valdai Hills (Russia) → Caspian Sea • **longest river of Europe** PYQ 2
- **Yangtze (Chang Jiang):** Tibetan Plateau → East China Sea • **longest in Asia** • world's largest hydropower dam **Three Gorges** PYQ 2
- **Hwang Ho (Yellow River):** Kunlun Mountains (China) → Bohai Gulf (Yellow Sea) • yellow soil from Loess Plateau • **Sorrow of China** PYQ 1
- **Mekong:** Tibetan Plateau → South China Sea • longest in Southeast Asia • 6 countries: China, Myanmar, Laos, Thailand, Cambodia, Vietnam PYQ 3
- **Amur:** Manchurian mountains → Sea of Okhotsk / Strait of Tartary • **natural international boundary between Russia and China** PYQ 2
- **Rio Grande: San Juan Mountains (Colorado, USA)** → Gulf of Mexico • **USA-Mexico international boundary** PYQ 2

Mnemonic (rivers crossing lines twice): "Kali makdi ne Kark ko Mahi nadi me kata" — Ka = Congo → Bhoomadhya (Equator), Li = Limpopo → Makdi (Tropic of Capricorn), Mahi = Mahi → Kark (Tropic of Cancer)

Mnemonic (short): "Mali kam bhuka" — Ma = Mahi-Cancer, Li = Limpopo-Capricorn, Ka = Congo-Equator

Mnemonic (rivers draining into Black Sea): "Don ne Denim jeans pahankar Dnieper aur Dniester ko kala kiya" — Don, Danube, Dnieper, Dniester → Black Sea PYQ 3

- **Draining into Caspian Sea: Volga** and **Ural** PYQ 3

Major Lakes of the World and Littoral Countries

- **Caspian Sea (saltwater lake): largest lake in the world** by both area and volume PYQ 3
- **5 Littoral Countries of Caspian: Kazakhstan** • **Russia** • **Azerbaijan** • **Iran** • **Turkmenistan** PYQ 3

Mnemonic (Caspian countries): "TARIK" — T = Turkmenistan, A = Azerbaijan, R = Russia, I = Iran, K = Kazakhstan (or "KRATI" — K-R-A-T-I)

- **Lake Superior (USA and Canada): largest freshwater lake** in the world by area • city of **Duluth** on its shore PYQ 2
- **Lake Baikal (Siberia, Russia): deepest (1642 m)** and oldest lake in the world PYQ 3
- **Baikal - Volume: largest freshwater lake** by volume • **20%** of world's liquid surface fresh water • **Pearl of Siberia** PYQ 3
- **Lake Victoria:** largest in Africa • borders: **Uganda, Kenya, Tanzania** (not Rwanda/Burundi) • **Equator passes through it** • not in a rift valley PYQ 2
- **Lake Titicaca: Peru-Bolivia** border • world's **highest navigable lake** PYQ 2
- **Aral Sea: Kazakhstan-Uzbekistan** border • fed by **Amu Darya** and **Syr Darya** • up to **90%** dried due to withdrawal for irrigation PYQ 3
- **Great Lakes (west → east): Superior → Michigan → Huron → Erie → Ontario** PYQ 4
- **Lake Michigan:** only one of the 5 **entirely in USA** • other 4 on USA-Canada border PYQ 4
- **Soo Canal:** joins Superior-Huron • **Welland Canal:** joins Erie-Ontario PYQ 4

- **Finger Lakes:** USA (New York region) PYQ 1
- **Lakes of Canada:** Great Bear • Great Slave • Athabasca • Reindeer • Winnipeg PYQ 3
- **Great Bear Lake:** Arctic Circle passes through • **Athabasca: Uranium City** nearby PYQ 3
- **Ladoga and Onega:** Russia • **Ladoga = largest freshwater lake of Europe** PYQ 1
- **Lake Balkhash:** Kazakhstan PYQ 1
- **Lake Eyre:** Australia • its lowest point
- **Lake Biwa:** largest lake of Japan
- **Lake Tanganyika:** Africa (Tanzania, Zaire, Burundi, Zambia) • world's **longest freshwater lake** • **second deepest** after Baikal PYQ 1
- **Salton Sea and Boiling Lake:** Boiling Lake in **Dominica** • Five Flower Lake in **China** PYQ 1

Major Waterfalls and Canals of the World

- **Angel Falls:** **Venezuela** • on **Caroni (Churun)** river, tributary of Orinoco • **highest in the world (979 m)** PYQ 2
- **Tugela Falls:** South Africa • world's **second highest (948 m)** PYQ 1
- **Niagara Falls:** USA-Canada border • **between Lakes Erie and Ontario** • St. Lawrence/Niagara river PYQ 2
- **Victoria Falls:** Zambia-Zimbabwe border • **Zambezi river** PYQ 2
- **Ribbon and Yosemite Falls:** California (USA)
- **Browne Falls:** New Zealand
- **Suez Canal (Egypt):** **1869** • engineer **Ferdinand de Lesseps** • joins **Mediterranean Sea-Red Sea** • present length **193.3 km** PYQ 4
- **Suez Ports:** northern end **Port Said** • southern end **Port Suez** PYQ 4
- **4 Lakes of Suez (north → south):** **Manzala → Timsah → Great Bitter → Little Bitter** PYQ 4
- **Panama Canal (Panama):** **1914** • joins **Atlantic (Caribbean Sea)-Pacific** • ports: Atlantic end **Colon**, Pacific end **Panama City** PYQ 2
- **Panama - Special:** most famous canal with **lock system** • **Gatun Lake** in the middle PYQ 2
- **Kiel Canal (Germany):** joins **North Sea-Baltic Sea** • among the busiest canals in the world PYQ 3

Major Straits of the World

- **Malacca:** Andaman Sea (Indian Ocean)-South China Sea • Malay Peninsula (Malaysia)-Sumatra (Indonesia) PYQ 2
- **Sunda:** Java Sea-Indian Ocean • Java-Sumatra (Indonesia) PYQ 2
- **Hormuz:** **Persian Gulf-Gulf of Oman** • Iran-Oman/UAE • strategic for petroleum trade PYQ 3
- **Bab-el-Mandeb:** **Red Sea-Gulf of Aden** (Arabian Sea) • Yemen (Asia)-Djibouti/Eritrea (Africa) • **Gate of Tears** PYQ 2
- **Gibraltar:** **Mediterranean Sea-Atlantic** • **Spain (Europe)-Morocco (Africa)** • **Key to the Mediterranean** PYQ 4

Major Straits

numbered west → east



1 **Bering** Alaska–Russia
2 **Yucatan** Mexico–Cuba
3 **Florida** Florida–Cuba
4 **Davis** Greenland–Baffin · widest
5 **Gibraltar** Spain–Morocco
6 **Dover** England–France
7 **Bosporus** Turkey: Europe–Asia
8 **Kerch** Crimea–Russia
9 **Bab-el-Mandeb** Yemen–Djibouti
10 **Hormuz** Iran–Oman
11 **Palk** India–Sri Lanka
12 **Malacca** Malaysia–Sumatra
13 **Sunda** Java–Sumatra
14 **Tsugaru** Hokkaido–Honshu
15 **Torres** Australia–PNG
16 **Bass** Australia–Tasmania
17 **Cook** NZ North–South Is.

- **Dover:** English Channel-North Sea • England (Britain)-France PYQ 2
- **Bering:** Bering Sea (Pacific)-Chukchi Sea (Arctic) • Alaska (USA/North America)-Russia (Asia) • **International Date Line** passes here PYQ 2
- **Palk:** Palk Bay-Gulf of Mannar (Bay of Bengal) • India (Tamil Nadu)-Sri Lanka PYQ 1
- **Tsugaru:** Sea of Japan-Pacific • Hokkaido-Honshu (Japan) PYQ 1
- **Bosphorus:** Black Sea-Sea of Marmara • separates European and Asian parts of Turkey PYQ 2
- **Kerch:** **Sea of Azov-Black Sea** • Crimea-Russia (Taman Peninsula) PYQ 2
- **Yucatan:** Gulf of Mexico-Caribbean Sea • Mexico (Yucatan Peninsula)-Cuba PYQ 1
- **Florida:** Gulf of Mexico-Atlantic • Florida (USA)-Cuba PYQ 1
- **Davis:** Baffin Bay-Atlantic • Greenland-Baffin Island (Canada) • **widest strait in the world** PYQ 2
- **Bass:** Tasman Sea-Southern Ocean • Australian mainland-Tasmania PYQ 1
- **Cook:** South Pacific • North Island-South Island of New Zealand PYQ 2
- **Torres:** Arafura Sea-Coral Sea (Gulf of Papua) • Australia (Cape York)-Papua New Guinea PYQ 1

Points to Note

1. **Great Lakes:** not all 5 on the border — only **4** • **Michigan** does not touch Canada, lies **entirely in USA** (Chicago on its shore) [3]
2. **Suez Canal:** at sea level • **no lock system** — locks are in **Panama** and **Soo Canal**
3. **Uzbekistan:** not on Caspian coast — on **Aral Sea** coast [2]
4. **Double Landlocked Countries:** only two — **Uzbekistan** and **Liechtenstein** [2]

One-liner Facts

- **Suez Lakes (north → south):** **Manzala → Timsah → Great Bitter → Little Bitter** • Commission asked this order directly PYQ 1
- **World's Largest River Island:** **Majuli** (Assam) • Brahmaputra river PYQ 1

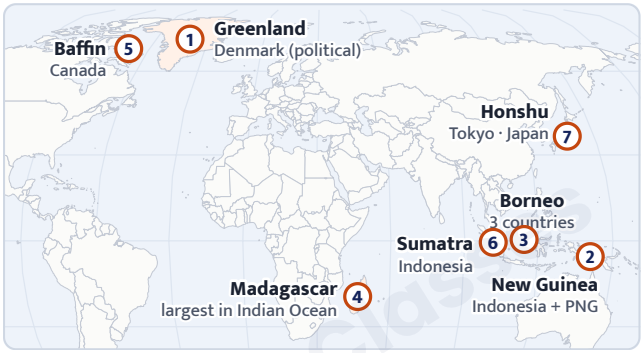
- **Dead Sea:** rift valley lake • bordered on only three sides — **Israel, Jordan, West Bank (Palestine)** • not Syria/Lebanon PYQ 2
- **Po River: Ganga of Italy** • due to agricultural utility and basin structure PYQ 1
- **Murray-Darling:** largest river system of Australia • fertile land between them **Riverina**

07 Major Islands, Peninsulas, Landlocked Countries, Tribes and Geographical Nicknames of the World

Largest Islands of the World (Descending Order of Area)

7 Largest Islands of the World

number = area rank



Borneo	Indonesia · Malaysia · Brunei (only such island)
Japan N → S	Hokkaido → Honshu → Shikoku → Kyushu
Indonesia W → E	Sumatra → Java → Bali → Lombok

- **1. Greenland:** North Atlantic/Arctic • geographically part of North America • **politically under Denmark** • **world's largest island**
- **2. New Guinea:** Pacific Ocean • divided between Indonesia and Papua New Guinea
- **3. Borneo (Kalimantan):** Pacific/Indian Ocean • the only island in the world divided among **3 countries** (Indonesia, Malaysia, Brunei)
- **4. Madagascar (Malagasy): largest island of the Indian Ocean** • 4th largest in the world
- **5. Baffin Island:** Canada
- **6. Sumatra Island:** Indonesia
- **7. Honshu Island:** largest island of Japan • capital **Tokyo** located on it PYQ 4
- **4 main islands of Japan (North → South): Hokkaido → Honshu → Shikoku → Kyushu** • order by size: Honshu > Hokkaido > Kyushu > Shikoku PYQ 2
- **Major islands of Indonesia (West → East): Sumatra → Java → Bali → Lombok** • Java: most densely populated island of Indonesia PYQ 1
- **Diego Garcia Island:** coral island in the **central Indian Ocean** (Chagos Archipelago) • under British control • US naval base PYQ 2
- **Diego Garcia (May 2025 treaty):** sovereignty of Chagos to **Mauritius** • Diego Garcia retained by Britain on a 99-year lease
- **Socotra Island:** Arabian Sea (near Gulf of Aden) • controlled by **Yemen** PYQ 1
- **Senkaku (Diaoyu) Islands dispute:** uninhabited islands in the East China Sea • between **Japan and China**
- **Kuril Islands dispute:** between **Russia and Japan**
- **Falkland Islands dispute:** South Atlantic • between Britain and Argentina

Continent-wise Major Landlocked Countries

Landlocked Countries

shaded = major landlocked



World total 44 | Africa 16 (most) | Asia 12 | S. America 2 | N. America 0 | Australia 0

Trap: Cambodia, Vietnam not landlocked · nor Uruguay, Ecuador, Guyana

- **Total landlocked countries: 44** in the world • no border with the open ocean • countries bordering closed seas like the Caspian and Aral also counted as landlocked
- **North America and Australia: no landlocked country** PYQ 2
- **South America (only 2): Bolivia** and **Paraguay** • Uruguay, Ecuador, Guyana not landlocked PYQ 3
- **Asia (12): Kazakhstan** (world's largest landlocked) • Mongolia (2nd largest) • Afghanistan • Uzbekistan • Turkmenistan • Kyrgyzstan • Tajikistan • Nepal • Bhutan • **Laos (only** landlocked country of South-East Asia) • Armenia • Azerbaijan PYQ 4
- **Europe (major):** Switzerland • Austria • Hungary • Czech Republic • Slovakia • Serbia • Belarus • Luxembourg • Vatican City • Andorra • North Macedonia • Moldova
- **Africa (highest, 16):** Chad • Niger • Mali • Burkina Faso • Ethiopia (most populous landlocked) • South Sudan • Uganda • Rwanda • Burundi • Zambia • Zimbabwe • Botswana • Malawi • Lesotho • Eswatini (Swaziland) • Central African Republic
- **Horn of Africa:** peninsular part of East Africa • **4 countries: Somalia, Ethiopia, Eritrea, Djibouti** PYQ 3

Mnemonic (Horn of Africa): "SEED" — S = Somalia, E = Ethiopia, E = Eritrea, D = Djibouti

Major Tribes of the World and Their Habitat

- **Pygmy: Congo Basin** (Zaire, Gabon, Uganda) • shortest people in the world • live in **tree huts ('Mongulus')** in dense equatorial forests PYQ 3
- **Bushmen (San): Kalahari Desert** (Botswana, Namibia) • hunters and food gatherers • **termites** = **'Bushman's rice'** (favourite food) PYQ 3
- **Masai: East Africa** (Kenya, northern Tanzania, eastern Uganda) • pastoralists • cluster of huts called **'Kraal'** PYQ 3
- **Eskimo (Inuit): Tundra** (Greenland, northern Canada, Alaska) • snow house **'Igloo'** • dog-drawn vehicle **'Sledge'** • leather boat **'Kayak'** • hunting spear **'Harpoon'** PYQ 2
- **Kirghiz: Central Asia** (Kyrgyzstan — Pamir and Tien Shan) • nomadic herders (transhumance) • tents called **'Yurt'** PYQ 2

- **Bedouin: Arabian Desert** (Saudi Arabia, Jordan, Syria) • nomadic camel and sheep herders PYQ 2
- **Maori: New Zealand** • natives of Polynesian origin PYQ 3
- **Ainu: Japan** (mainly Hokkaido) • ancient tribe of Caucasoid origin PYQ 2
- **Semang and Sakai: Malaysia** (Malay Peninsula) • equatorial forest dwellers PYQ 3
- **Zulu: South Africa** (Natal province) • warrior and pastoral tribe PYQ 2

Tribe → Homeland by continent

Africa	Asia
Pygmy → Congo Basin	**Kirghiz** → Kyrgyzstan
Bushman → Kalahari	**Bedouin** → Arabian Desert
Masai → Kenya, N. Tanzania	**Ainu** → Japan (Hokkaido)
Zulu → S. Africa (Natal)	**Semang, Sakai** → Malaysia
Hausa, Fulani → Nigeria	**Vedda** → Sri Lanka

Cold tundra	Oceania, America
Eskimo → Greenland, N. Canada, Alaska	**Maori** → New Zealand
Lapps (Sami) → Scandinavia	**Boro** → Brazil (Amazon)

Marks: Mongulus = Pygmy · Kraal = Masai · termite "rice" = Bushman · Igloo, Sledge, Kayak = Eskimo · Yurt = Kirghiz · Reindeer = Lapps

Trap: Bushman not in Sahara · Pygmy not in Amazon · Amazon = Boro

- **Boro: Brazil** (Amazon Basin) • farmers and hunters PYQ 2
- **Veddas: Sri Lanka** • ancient natives PYQ 1
- **Papuans:** Papua New Guinea • Pacific island tribe PYQ 1
- **Lapps (Sami): Scandinavian countries** (Norway, Sweden, Finland, Kola Peninsula of Russia) • reindeer-herding tundra tribe PYQ 2
- **Yukaghir, Yakut, Chukchi: Siberia (Russia)** • cold-region tribes PYQ 2
- **Fulani and Hausa: Nigeria** (West Africa) • Hausa = farmers • Fulani = pastoralists PYQ 2
- **Maya:** Central America (Guatemala, Mexico, Honduras) • linked to an ancient civilisation PYQ 1
- **Kurds:** border plateau region of Iraq, Iran, Turkey, Syria • demand an independent homeland **'Kurdistan'** PYQ 2

Major Geographical Nicknames

- **Land of Lakes / Land of a Thousand Lakes:** Finland PYQ 1
- **Land of the Midnight Sun:** Norway PYQ 1
- **Land of the Rising Sun:** Japan ('Nippon')
- **Land of White Elephants:** Thailand PYQ 1
- **Land of Thunderbolt:** Bhutan PYQ 1
- **Land of Golden Pagodas:** Myanmar PYQ 1
- **Cockpit of Europe / Battlefield of Europe:** Belgium
- **Sick Man of Europe:** Turkey PYQ 1
- **Sugar Bowl of the World:** Cuba PYQ 2
- **Bread Basket of the World:** Prairies of North America • Steppes of Ukraine
- **Playground of Europe:** Switzerland
- **Dark Continent:** Africa
- **Continent of Birds:** South America PYQ 1
- **White Continent / Continent dedicated to Science:** Antarctica
- **Land of Kangaroos / Land of the Thirsty Land / Land of the Golden Fleece:** Australia
- **Britain of the South:** New Zealand PYQ 1
- **Island of Pearls:** Bahrain PYQ 1
- **Island of Cloves:** Zanzibar and Pemba (Tanzania, East Africa) PYQ 2
- **Land of Canals:** Pakistan • 'Land of Rivers' = **Bangladesh** PYQ 1
- **City of Seven Hills / Eternal City:** Rome (Italy) PYQ 1
- **City of Skyscrapers:** New York
- **Windy City / Meat Market of the World:** Chicago (USA) PYQ 1
- **Forbidden City:** Lhasa (Tibet)
- **Manchester of the East:** Osaka (Japan) PYQ 2
- **Venice of the East:** Bangkok (Thailand)

Points to Note

1. Cambodia or Vietnam are not landlocked — in all of South-East Asia (10 ASEAN countries) **only Laos** is landlocked; Cambodia, Vietnam, Thailand, Myanmar all have a coastline [3]
2. Greenland is not controlled by Canada or the USA — geographically part of North America, but politically under the European country **Denmark** [1]
3. Bushmen are not in the Sahara nor Pygmies in the Amazon — Bushmen live in the **Kalahari** of southern Africa, Pygmies in the **Congo Basin** of Central Africa; the Amazon Basin has the **Boro** [2]

One-liner Facts

- **Indonesia:** world's largest **archipelago** country • over 17,000 islands • capital being shifted from Jakarta (Java) to **Nusantara** (Borneo/East Kalimantan)
- **Berber and Tuareg:** pastoral tribe of north-west Africa (Morocco, Algeria, Tunisia) = **Berber** • guides for tourists in the Sahara = **Tuareg**
- **Scandinavian countries:** geographically Norway, Sweden, Denmark • culturally-historically Finland and Iceland also added • collectively **'Nordic countries'**
- **World's longest international land border: USA-Canada** • largely set by the **49th Parallel** (49° N latitude) and the Great Lakes PYQ 2
- **Boundary lines: 38th Parallel** = North and South Korea • **Maginot Line** = France-Germany border (French fortification) • **Hindenburg Line** = German defensive line on the Western Front (northern France) in World War I PYQ 2

08 Economic Geography of the World (Major Minerals, Energy Resources, Agriculture and Industrial Centres)

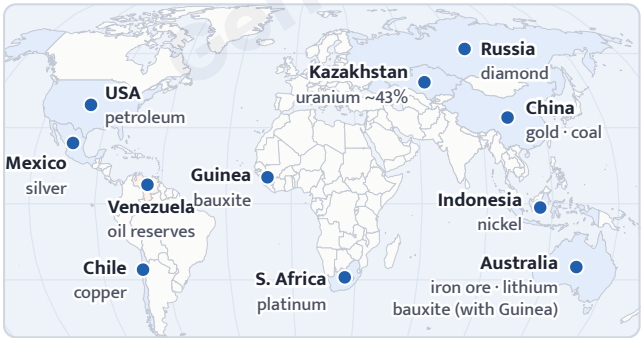
Major Mineral and Energy Producing Regions of the World

- **Iron ore — Mines:** Mesabi Range and Lake Superior region (USA) • Krivoy Rog (Ukraine) • Magnitogorsk / Kuzbass (Russia) • Kiruna and Gällivare (Sweden) • Pilbara region (Western Australia) • Itabira / Minas Gerais (Brazil) • Lorraine region (France) PYQ 5
- **Iron ore — Producers: Australia** (first) • Brazil • China • India
- **Copper — Mines: Chuquicamata** (Chile) — world's largest open-pit copper mine • Butte mine (Montana, USA) — "the richest hill on Earth" • Katanga Plateau (Congo/Zaire and Zambia) • Sudbury (Canada) PYQ 4

- **Copper — Producers:** **Chile** (highest production and reserves) • Peru • China
- **Gold — Mines:** **Johannesburg / Witwatersrand** (South Africa) — "City of Gold" • Kalgoorlie and Coolgardie (Western Australia) • Homestake mine (South Dakota, USA) • Muruntau mine (Uzbekistan) PYQ 4
- **Gold — Producers:** **China** (first) • Australia • Russia
- **Diamond — Mines:** **Kimberley** (South Africa) — world's most famous diamond mine • Argyle mine (Australia) • Mir and Udachnaya mines (Yakutia, Russia) • Orapa and Jwaneng (Botswana) PYQ 3
- **Diamond — Producers:** **Russia** (highest production and reserves of natural diamond) • Botswana • Congo • South Africa
- **Silver — Mines:** **Chihuahua** (Mexico) — world's largest silver mine • Mount Isa (Queensland, Australia) • Bawdwin mine (Myanmar — Shan Plateau) • Potosí (Bolivia) PYQ 2
- **Silver — Producers:** **Mexico** (first) • China • Peru
- **Coal — Fields (1):** **Ruhr Valley** (Germany) — Europe's foremost coal and industrial region • Saar region (Germany) • Donbas / Donetsk Basin (Ukraine) • Kuznetsk (Kuzbass) and Karaganda (Russia and Kazakhstan) PYQ 5
- **Coal — Fields (2):** Appalachian and Pennsylvania region (USA — **anthracite** coal) • Shanxi and Shaanxi (China) • Lancashire and South Wales (Britain)

Minerals: Top Producers

orange = top oil reserves



Reserves ≠ production: oil — reserves Venezuela, production USA · uranium — reserves Australia, production Kazakhstan · coal — reserves USA, production China

- **Coal — Producers:** **China** (first) • India • Indonesia • USA • Australia • Largest reserves: **USA**
- **Petroleum — Fields:** Lake Maracaibo region and Orinoco Basin (Venezuela) • Ghawar and Abqaiq (Saudi Arabia) • Kirkuk and Mosul (Iraq) • **Baku** (Azerbaijan — Caspian coast) • Abadan and Masjed-e-Soleyman (Iran) PYQ 4
- **Petroleum — Producers:** **USA** (first) • Saudi Arabia • Russia • Largest proven reserves: **Venezuela**
- **Uranium — Mines:** Olympic Dam and Ranger mine (Australia) • Athabasca Basin / Uranium City / McArthur River (Canada) • Port Radium (Canada) • Cameco / Inkai (Kazakhstan) PYQ 4
- **Uranium — Producers:** **Kazakhstan** (largest, about 43%) • Canada • Namibia • Australia • Largest reserves: **Australia**
- **Bauxite (Aluminium) — Mines:** Weipa and Gove (Northern Australia) • Boké (Guinea) PYQ 2
- **Bauxite — Producers:** **Guinea and Australia** (top producers) • China • Largest reserves: **Guinea**
- **Tin — Mines:** Kinta Valley and Ipoh (Malaysia) • Potosí (Bolivian Plateau) PYQ 3
- **Tin — Producers:** China • Indonesia • Myanmar • Peru
- **Nickel and Platinum — Mines:** **Sudbury mine** (Ontario, Canada) — world-famous for nickel and platinum • Bushveld Complex (South Africa — platinum) PYQ 2
- **Nickel and Platinum — Producers:** **Indonesia** (first in nickel) • Philippines • Russia • **South Africa** (first in platinum)
- **Lithium (White Gold) — Region:** **Lithium Triangle** — Argentina, Bolivia, Chile in South America (Salar de Uyuni and Atacama region) PYQ 2
- **Lithium — Producers:** **Australia** (first) • Chile • China • Largest reserves: Chile/Bolivia

Major Agricultural Systems and Crops of the World

Shifting (Jhum) Cultivation: Local Names

country-wise

Asia	
Ladang	Malaysia, Indonesia
Chena	Sri Lanka
Taungya	Myanmar
Caingin	Philippines
Ray	Vietnam, Laos
Tamrai	Thailand

Americas	
Roca	Brazil
Milpa	Mexico, C. America
Conuco	Venezuela

Africa	
Masole	Congo (Zaire valley)
Fang	Gabon (C. Africa)
Logan	West Africa
Tavy	Madagascar

Terms: Viticulture = grapes · Olericulture = vegetables · Sericulture = silkworm · Apiculture = honeybee · Silviculture = forests

- **Shifting (Jhum) cultivation — regional names (1):** **Ladang** — Malaysia and Indonesia • **Chena** — Sri Lanka • **Taungya** — Myanmar • **Caingin** — Philippines • **Ray** — Vietnam and Laos • **Tamrai** — Thailand • **Roca** — Brazil PYQ 5
- **Shifting (Jhum) cultivation — regional names (2):** **Milpa** — Mexico and Central America • **Conuco** — Venezuela • **Masole** — Congo (Zaire river basin) • **Fang** — Central Africa (Gabon) • **Logan** — West Africa • **Tavy** — Madagascar
- **Coffee — Producers:** **Brazil** (first) • Vietnam (second) • Colombia (third) PYQ 4
- **Fazenda:** large coffee plantations in Brazil
- **São Paulo:** world's largest **coffee market** • **Santos** port — highest coffee export, "Coffee Port" • São Paulo–Santos railway — "Coffee Railway"
- **Mocha coffee:** high-quality coffee of Yemen • **Blue Mountain Coffee** — Jamaica • Main varieties: Robusta and Arabica
- **Natural rubber:** native to the Amazon Basin (Brazil) • present largest production **Thailand** (first), Indonesia (second) • Malaysia's "Bukit Banjaran" region was famous for rubber PYQ 2
- **Crescent-shaped Wheat Belt:** Prairie region of USA and Canada • "Chernozem" (black soil) region of Ukraine • Canada's **Winnipeg** market — "Wheat market of the world" PYQ 1
- **Corn Belt:** central USA • maize mainly fodder for **fattening livestock (pigs, cattle)** • USA — world's largest producer of maize and soybean (with Brazil) PYQ 1
- **Truck farming:** intensive cultivation of vegetables, fruits, flowers near urban centres • carried overnight to market by trucks • first developed in USA PYQ 1
- **Agricultural terms (1):** **Viticulture** — grape cultivation • **Olericulture** — vegetable cultivation • **Horticulture** — gardening • **Sericulture** — silkworm rearing (on mulberry leaves) PYQ 3

- **Agricultural terms (2):** **Apiculture** — beekeeping • **Silviculture** — conservation and development of forests • **Hydroponics** — growing plants in water without soil • **Aeroponics** — growing plants in air

Major Industrial Cities of the World and their Industries

Industrial City Look-alikes

original → "... of"

Manchester cotton textile · UK	→	Osaka = of Japan/East; Ivanovo = of Russia; Shanghai = of China; Milan = of Italy (silk)
Pittsburgh iron-steel · USA	→	Yawata = of Japan; Anshan = of China
Detroit automobile · USA	→	Nagoya = of Japan; Turin = of Italy
Birmingham iron-steel · UK	→	Hamilton = of Canada

World capital: steel = Pittsburgh · rubber = Akron · diamond = Antwerp

- **Detroit:** USA • **Automobile** industry PYQ 4
- **Pittsburgh:** USA • **Iron and steel** • "Steel capital of the world" PYQ 3
- **Chicago:** USA • **Meat processing and packing** • agricultural machinery PYQ 2
- **Seattle:** USA • **Aircraft manufacturing** (main centre of Boeing) PYQ 2
- **Los Angeles:** USA • **Film industry (Hollywood)** • oil refining PYQ 1
- **Akron:** USA • **Synthetic rubber and tyres** • "Rubber capital of the world" PYQ 2
- **Osaka:** Japan • **Cotton textiles** • "Manchester of Japan/the East" PYQ 3
- **Nagoya:** Japan • **Automobile and cotton textiles** • "Detroit of Japan" PYQ 3
- **Yawata (Kitakyushu):** Japan • **Iron and steel** • "Pittsburgh of Japan" PYQ 2
- **Yokohama and Kobe:** Japan • **Shipbuilding** and engineering PYQ 1
- **Manchester:** Britain (UK) • **Cotton textiles** PYQ 1
- **Bradford and Leeds:** Britain (UK) • **Woollen textiles** PYQ 2
- **Sheffield:** Britain (UK) • **Cutlery** (knives, forks, utensils) and steel PYQ 3
- **Birmingham:** Britain (UK) • **Iron and steel** PYQ 1
- **Glasgow:** Britain (Scotland) • **Shipbuilding** and machinery PYQ 2
- **Ivanovo:** Russia • **Cotton textiles** • "Manchester of Russia" PYQ 3
- **Chelyabinsk and Magnitogorsk:** Russia • **Iron-steel and metallurgy** PYQ 2
- **Tula:** Russia • **Iron-steel and armaments** PYQ 1
- **Milan and Turin:** Italy • Milan — **silk textiles** ("Manchester of Italy") • Turin — **automobiles** ("Detroit of Italy") PYQ 3
- **Venice:** Italy • **Glass making** PYQ 2
- **Havana:** Cuba • **Cigars (tobacco)** PYQ 3
- **Anshan:** China • **Iron and steel** • "Pittsburgh of China" PYQ 3
- **Shanghai:** China • **Cotton textiles and shipbuilding** • "Manchester of China" PYQ 2
- **Hamilton:** Canada • **Iron and steel** • "Birmingham of Canada" PYQ 2
- **Ottawa:** Canada • **Paper and pulp** PYQ 2
- **Antwerp:** Belgium • **Diamond cutting** • "Diamond capital of the world" PYQ 2
- **Munich and Stuttgart:** Germany • Munich — **lens making** and beer • Stuttgart — **automobiles** PYQ 1
- **Lyon (Lyons):** France • **Silk textiles** PYQ 2
- **Multan:** Pakistan • **Pottery** PYQ 1
- **Dhaka:** Bangladesh • **Carpets and jute** PYQ 1

Major Railways and Ports of the World

- **Trans-Siberian Railway (Russia):** **world's longest railway** (about 9289 km) • **Moscow** to **Vladivostok** on the Pacific coast (linked in the west to St. Petersburg/Leningrad) PYQ 3
- **Trans-Siberian — major stations:** Moscow • Kazan • Yekaterinburg • Omsk • Novosibirsk • Irkutsk (near Lake Baikal) • Chita
- **Canadian Pacific Railway:** **Halifax** in eastern Canada (Atlantic coast) — **Vancouver** in western Canada (Pacific coast) PYQ 1
- **Trans-Australian Railway:** **Perth** in the west — **Sydney** on the east coast • Kalgoorlie and Coolgardie gold mines lie on this route PYQ 2
- **Orient Express:** famous European railway • **Paris (France)** — **Istanbul (Turkey, Constantinople)** PYQ 2
- **Rotterdam port (Netherlands):** mouth of the Rhine (North Sea) • **Europe's largest and busiest port** • "Europoort" PYQ 2
- **Alexandria:** Egypt's foremost **cotton-exporting** port • on the Nile delta PYQ 1
- **Murmansk (Russia):** Kola Peninsula (Arctic Ocean) • **ice-free all year** due to the **North Atlantic Drift** (Norwegian Current) PYQ 2
- **Gwadar:** Pakistan (Balochistan, Arabian Sea) • developed by **China** PYQ 2
- **Chabahar:** Iran (Gulf of Oman; Iran's only oceanic port) • developed by **India** • **Hambantota** — Sri Lanka

Points to Note

1. **Kimberley** = **Diamond, Johannesburg (Witwatersrand)** = **Gold** (confusion: reversed) • Kalgoorlie and Coolgardie (Australia) — **gold**, not coal [3]
2. Saudi Arabia is not both the largest reserve holder and producer — largest proven reserves **Venezuela** (Saudi Arabia second), largest producer **USA** (shale oil revolution) [2]
3. Canada/Australia are not the largest uranium producers — largest producer **Kazakhstan** (over 40%); largest reserves Australia [2]

One-liner Facts

- **Ruhr Basin (Germany):** Ruhr river valley, tributary of the Rhine • huge coal and steel centre • other rich mineral regions called "the Ruhr of that country" • Damodar Valley/Chota Nagpur Plateau — "Ruhr of India" PYQ 2
- **North Atlantic Sea Route:** world's busiest oceanic trade route • Western Europe — east coast of North America • "**Big Trunk Route**" PYQ 2
- **Chile:** **Chuquicamata** — copper • northern desert (Atacama) — **natural nitrate** reserves PYQ 1

- **Kinta Valley (Malaysia):** tin • **Chindwin and Irrawaddy valleys** (Myanmar) — mineral oil and teak wood PYQ 2
- **Collar workers:** **Red collar** — primary (agriculture, mining, hunting) • **Blue collar** — secondary (industry/manufacturing) • **Pink/White collar** — tertiary (services) • **Gold collar** — quinary (top policy makers, senior executives, scientists) PYQ 1

Part 2: Physical Geography of India

09 Physical Divisions of India (Relief and Physiographic Regions)

- **Area distribution:** High mountain ranges **10.7%** • Hills 18.6% • Plateaus 27.7% • Extensive plains **43.0%**
- **4 physiographic regions (by geological structure):** Northern Mountains • Great Northern Plains • Peninsular Plateau • Coastal Plains and Islands

Geological Rock Systems — Chronological Order

- **Archaean Rocks:** **Oldest** rocks, formed after Earth cooled • Fossil-free (Azoic) • Crystalline (gneiss, schist) PYQ 1
- **Dharwar System:** **First sedimentary** rocks, from erosion-deposition of Archaean rocks • **Richest in metallic minerals** (gold, iron ore, manganese, copper) • Aravallis belong to this period PYQ 3
- **Cuddapah System:** From erosion of Dharwar rocks • Limestone, sandstone, marble
- **Vindhyan System:** Building material (red sandstone, limestone) • **Diamond mines of Panna (MP) and Golconda (Telangana)** PYQ 1
- **Gondwana System:** **98% of India's coal (bituminous)** • Damodar, Son, Mahanadi, Godavari valleys PYQ 2
- **Deccan Trap:** Last phase of Cretaceous period • Fissure volcanic eruption (basaltic lava) • Weathering produced **Black soil (Regur)** PYQ 1

Trans-Himalaya (Tibetan Himalaya)

- **Formation:** Before the Himalaya (part of Eurasian Plate) • Sedimentary rocks • Separated from main Himalaya by **Indus-Tsangpo Suture Zone (ITSZ)** PYQ 1
- **Karakoram Range:** Northernmost range of India • 'Backbone of High Asia' • Highest peak **K2 (Godwin Austen, 8611 m)** PYQ 3
- **Karakoram glaciers:** **Siachen (76 km, longest; Nubra Valley)** • Biafo • Baltoro • Hispar
- **Ladakh Range:** **Rakaposhi**, world's steepest-slope peak • Indus flows between Ladakh and Zanskar ranges • **Leh** on the **right bank** of Indus, between Ladakh and Zanskar PYQ 2
- **Zanskar Range:** Main peak Kamet (7756 m)
- **Kailash Range:** Extension of Ladakh Range in Tibet

Three Parallel Ranges of the Main Himalaya

- **Greater Himalaya (Himadri/Inner):** Most continuous and highest (average 6000 m) • Core of Archaean **granite and gneiss** PYQ 1
- **Greater Himalaya peaks (East → West):** **Namcha Barwa** (Arunachal-Tibet border) → **Kanchenjunga** (Sikkim-Nepal) → Makalu → Everest → Manaslu → Annapurna → Dhaulagiri → **Nanda Devi** (Uttarakhand) → Kamet → **Nanga Parbat** (Gilgit-Baltistan region) PYQ 4
- **Lesser Himalaya (Middle/Himachal) — ranges (West → East):** **Pir Panjal** (J&K, longest) → Dhauladhar (HP) → Mussoorie and Nag Tibba (Uttarakhand) → Mahabharat Range (Nepal) PYQ 2
- **Temperate grasslands (Lesser Himalayan slopes):** **'Margs'** in Kashmir (Gulmarg, Sonamarg) • **'Bugyals'/'Payar'** in Uttarakhand PYQ 1
- **Valleys between Greater and Lesser Himalaya:** Kashmir • Kangra • Kullu
- **Karewa:** Glacial deposits (clay and moraine) of Kashmir Valley • World-famous for **saffron (zafran)** cultivation PYQ 2
- **Shiwalik (Outer/Sub-Himalaya):** **Youngest** and southernmost Himalayan range • Unconsolidated sediments PYQ 2
- **Duns/Duars:** Longitudinal valleys between Lesser Himalaya and Shiwalik • **'Duns'** in west (Dehradun, Kotli Dun) • **'Duars'** in east (Haridwar) PYQ 1

Regional Division of the Himalaya (Sidney Burrard — by Rivers)

- **Punjab/Kashmir Himalaya:** Between Indus and Sutlej • 560 km • Karakoram, Ladakh, Pir Panjal, Zanskar ranges • Zojila Pass • Karewa deposits PYQ 1
- **Kumaon Himalaya:** Between Sutlej and Kali • 320 km • **Shortest** • Nanda Devi (highest in Kumaon) • Kamet • Badrinath • Kedarnath • Gangotri • Dun valleys PYQ 2
- **Nepal Himalaya:** Between Kali and Teesta • 800 km • **Longest** and highest • Everest, Kanchenjunga, Makalu, Dhaulagiri, Annapurna PYQ 1
- **Assam Himalaya:** Between Teesta and Dihang (Brahmaputra) • 720 km • Namcha Barwa, Kula Kangri • Narrower width, steeper slope

Major Faults of the Himalaya — North to South

PYQ 2

- **ITSZ (Indus-Tsangpo Suture Zone):** Between Trans-Himalaya and Greater Himalaya
- **MCT (Main Central Thrust):** Between Greater and Lesser Himalaya
- **MBF (Main Boundary Fault):** Between Lesser Himalaya and Shiwalik
- **HFF (Himalayan Frontal Fault):** Between Shiwalik and Great Northern Plains

Major Mountain Passes of India — State/UT-wise

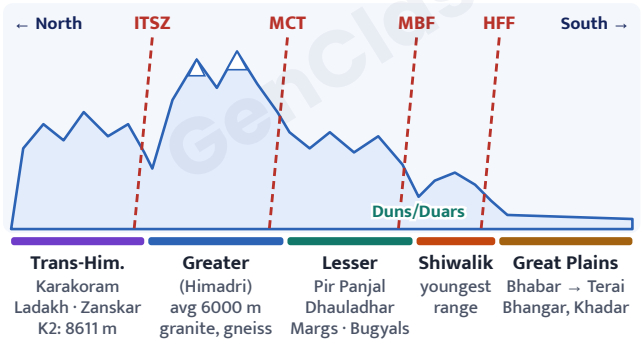
- **Ladakh:** **Karakoram** (India's highest pass) • **Zojila** (Srinagar-Leh; Zanskar Range) • Chang-La • Khardung-La • Thang-La • Aghil • Imis-La (Ladakh-Tibet) PYQ 2
- **Jammu & Kashmir:** Burzil (Srinagar-Gilgit) • Pir Panjal • **Banihal** (Jammu-Srinagar; Jawahar Tunnel) PYQ 1
- **Himachal Pradesh:** **Shipki-La** (Shimla-Tibet; Sutlej enters India) • **Rohtang** (Manali-Leh; Atal Tunnel) • **Baralacha-La** (Mandi-Leh; source of Chenab) PYQ 3
- **Uttarakhand:** Thaga-La • Mana-La • Niti-La • **Lipulekh** (India-China-Nepal tri-junction; Mansarovar Yatra route) • Muling-La • Traill's Pass PYQ 3
- **Sikkim:** **Nathu-La** (closed after 1962 war, reopened for trade in **2006**) • Jelep-La (to Tibet via Chumbi Valley) PYQ 2
- **Arunachal Pradesh:** Bomdila • **Yangyap** (Brahmaputra enters India) • **Diphu** (India-China-Myanmar border) • Pangsau PYQ 2
- **Manipur:** **Tuju Pass** (Imphal-Myanmar) PYQ 1

Hills of the Northeast (Purvanchal)

- **North → South (Arunachal to Mizoram):** Patkai Bum (Arunachal) → Naga Hills (Nagaland; highest **Mount Saramati**) → Manipur Hills (Barail Range) → Mizo/Lushai Hills (Mizoram; highest **Blue Mountain/Phawngpui**) PYQ 3

Himalaya: Cross-section

North → South · 4 faults



- **Hills of Arunachal (West → East):** Dafla → Miri → Abor → Mishmi PYQ 2
- **Meghalaya Plateau (part of Peninsular Plateau; West → East):** Garo (highest **Nokrek**) → Khasi (Shillong, Cherrapunji) → Jaintia PYQ 2

Great Plains of North India (by Relief; North to South) PYQ 3

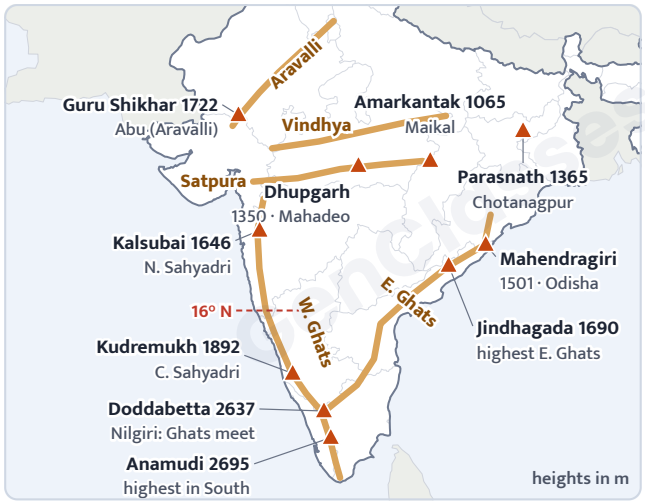
- **Bhabar:** Foothills of Shiwalik • 8-16 km wide belt from Indus to Teesta • Heavy pebbles-boulders (alluvial fans) • Rivers **disappear** • Unsuitable for agriculture PYQ 2
- **Terai:** 10-20 km wide marshy-damp belt south of Bhabar • Lost rivers **re-emerge** • Biodiversity, dense forests, sugarcane-paddy cultivation PYQ 1
- **Bhangar:** Upland of old alluvium • Floodwater does not reach annually • Calcium carbonate nodules (**'Kankar'**) PYQ 2
- **Reh/Kallar:** White salt crust in over-irrigated Bhangar areas
- **Khadar:** Lowland of new alluvium • Soil renewed by annual floods • **Most fertile** for agriculture PYQ 1

Peninsular Plateau and Major Mountain Ranges

- **Aravalli:** World's **oldest residual fold mountain** (Pre-Cambrian/Dharwar period) • 800 km from Gujarat (Palanpur) through Rajasthan to Delhi (Raisina Hills) • Highest **Guru Shikhar (1722 m)**, Mount Abu (Rajasthan) PYQ 3
- **Vindhyan Range:** Separates North India from South India • Water divide between Ganga and Peninsular drainage systems
- **Satpura:** **Block Mountain** • Narmada rift valley to north, Tapti rift valley to south • Ranges (West → East): **Rajpipla** (Gujarat) → **Mahadeo** (MP) → **Maikal** (MP-Chhattisgarh) PYQ 3
- **Dhupgarh (1350 m):** Highest peak of Satpura • Mahadeo Hills (Pachmarhi here) PYQ 2
- **Amarkantak (1065 m):** Highest peak of Maikal Range • Source of Narmada and Son
- **Western Ghats (Sahyadri):** Not true mountains — **Fault Scarp** of Peninsular Plateau • 1600 km from Tapti mouth to Kanyakumari • **16° N** latitude (through Goa) divides Northern and Southern Sahyadri PYQ 2

Peninsular Ranges and Peaks

highest in South: Anamudi



- **Northern Sahyadri:** Highest **Kalsubai (1646 m, Maharashtra)** • Krishna rises at Mahabaleshwar (1438 m)
- **Central Sahyadri (Karnataka):** Highest **Kudremukh (1892 m; iron ore)** • Baba Budan Hills (coffee)
- **Western Ghats passes (North → South):** **Thalghat** (Mumbai-Nashik) → **Bhorghat** (Mumbai-Pune) → **Palghat/Palakkad Gap** (between Nilgiri and Anaimalai; Kochi-Coimbatore) → **Shencottah Gap** (in Anaimalai/Cardamom; Thiruvananthapuram-Madurai) PYQ 3
- **Nilgiri (Blue Mountains):** **Meeting point (knot)** of Western and Eastern Ghats • Kerala-Karnataka-Tamil Nadu tri-junction • Highest **Doddabetta (2637 m)** — **second highest** in South India • Ooty (Udhagamandalam) • Toda tribe PYQ 4
- **Anaimalai Hills:** South of Nilgiri • Highest **Anamudi (2695 m)**, Idukki (Kerala) • **Highest in entire South/Peninsular India and Western Ghats** PYQ 3
- **Cardamom Hills:** India's **southernmost mountain range** • Kerala-Tamil Nadu border • Closest to Kanyakumari PYQ 2
- **Eastern Ghats:** Dissected, discontinuous, from Mahanadi to Nilgiri • Highest **Jindhagada (1690 m, Andhra Pradesh)** and **Mahendragiri (1501 m, Odisha)** PYQ 1
- **South Indian hills (North → South; UPPCS favourite):** **Nallamala** (Andhra) → Velikonda → Palkonda → Nagari → **Javadi** (Tamil Nadu) → **Shevaroy** (Tamil Nadu; Yercaud) → Panchamalai → Sirumalai → Nilgiri → Anaimalai → **Palani** (Kodaikanal) → **Cardamom** PYQ 5
- **Central Indian plateaus (West → East):** Malwa → Bundelkhand → Baghelkhand → **Chotanagpur** (Ruhr of India; highest **Parasnath**, 1365 m) → Meghalaya Plateau (separated from Chotanagpur by Malda Gap/Garo-Rajmahal Gap) PYQ 3

Coastal Plains and Islands

- **Western Coastal Plain:** Narrow and submergent coast • Estuaries formed
- **Western coast segments (North → South):** Kathiawar/Kachchh (Gujarat) → Konkan (Daman to Goa; Maharashtra and Goa) → Kannada/Kanara (Goa to Mangaluru; Karnataka) → Malabar (Mangaluru to Kanyakumari; Kerala)
- **Kayals:** Backwaters/lagoon lakes of Malabar Coast • Vembanad • Ashtamudi PYQ 2
- **Eastern Coastal Plain:** Broad and emergent coast • Deltas formed

- **Eastern coast segments (North → South):** Utkal (Odisha; Subarnarekha to Mahanadi) → Northern Circars (Mahanadi to Krishna; Odisha and Andhra Pradesh) → Coromandel (Krishna delta to Kanyakumari; Andhra Pradesh and Tamil Nadu) PYQ 2
- **Andaman-Nicobar (Bay of Bengal):** Submerged peaks of Himalaya (Arakan Yoma)
- **Andaman-Nicobar islands (North → South):** North Andaman (highest **Saddle Peak, 732 m**) → Middle Andaman (**largest**) → **South Andaman** (capital Port Blair, now Sri Vijaya Puram) → Little Andaman → Car Nicobar → Great Nicobar PYQ 4
- **Great Nicobar:** Southernmost island • Highest **Mount Thullier** • India's southernmost point **Indira Point/Pygmalion Point (6°45′ N)**
- **Volcanic islands:** **Barren** (only **active volcano** of India and South Asia) • **Narcondam** (dormant) PYQ 2
- **Lakshadweep (Arabian Sea):** 36 islands of coral reef/atoll • Largest **Androtт** (Minicoy in some sources) • Capital **Kavaratti** PYQ 2
- **8° Channel:** Between Maldives and Minicoy
- **9° Channel:** Between Minicoy and main Lakshadweep (Kavaratti)
- **10° Channel:** Between Little Andaman and Car Nicobar PYQ 3
- **Duncan Passage:** Between South Andaman and Little Andaman PYQ 1
- **Coco Channel:** Between Coco Islands (Myanmar) and North Andaman
- **Great Channel (6° Channel):** Between Great Nicobar and Sumatra (Indonesia)

Points to Note

1. **Highest peak of India: K2 (Godwin Austen, 8611 m)** — Karakoram Range of Trans-Himalaya; "highest in Indian Himalaya"/"highest undisputed in India" = **Kanchenjunga (8598 m)** [2]
2. **Highest peak of South India:** Not Doddabetta — **Anamudi (2695 m, Anaimalai)**; Doddabetta (2637 m) is second [1]
3. **Port Blair:** Not in Middle Andaman (largest island) — in **South Andaman**
4. **Meeting point of Eastern and Western Ghats:** Not Anaimalai or Cardamom — **Nilgiri** [1]
5. **Bhabar vs Terai:** Rivers disappear in **Bhabar**, re-emerge in **Terai** (not the reverse) [1]

Mnemonic (Trans and main Himalayan ranges, North → South): "Kal Ja Peer Dhaula Shiv" — Ka = Karakoram, L = Ladakh, Ja = Zanskar, Peer = Pir Panjal, Dhaula = Dhauladhar, Shiv = Shiwalik

Mnemonic (Arunachal hills, West → East): "DM AM" — D = Dafla, M = Miri, A = Abor, M = Mishmi

Mnemonic (South Indian hills, North → South): "Nalla Jave Shev Neela Anna Card" — Nalla = Nallamala, Jave = Javadi, Shev = Shevaroy, Neela = Nilgiri, Anna = Anaimalai, Card = Cardamom

Mnemonic (Western Ghats passes, North → South): "Thal Bhor Pal Sen" — Thal = Thalghat, Bhor = Bhorghat, Pal = Palghat, Sen = Shencottah

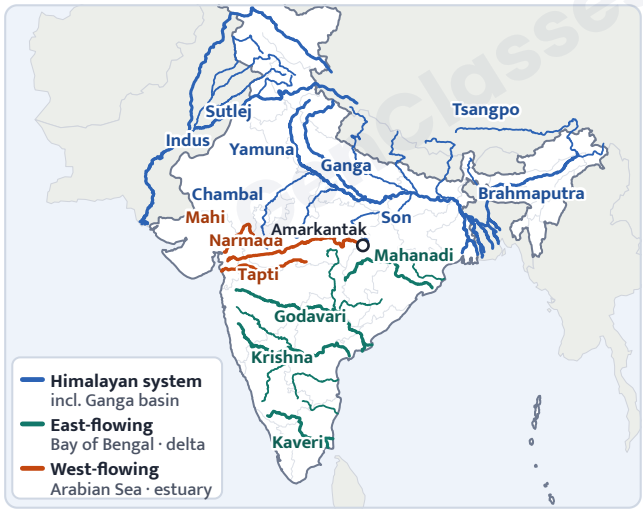
One-liner Facts

- **Oldest/youngest landmass:** **Peninsular Plateau** oldest • **Great Northern Plains (and Shiwalik)** youngest
- **Nelang Valley (Uttarkashi, Uttarakhand):** Near India-China border • Closed after 1962 war • Reopened to tourists **May 2015** PYQ 1
- **Meghalaya Plateau:** Garo, Khasi, Jaintia, Mikir-Rengma hills • Structurally not Himalayan — north-eastern extension of **Peninsular Plateau (Chotanagpur)** PYQ 1
- **Silent Valley, Kerala:** Western Ghats (Nilgiri Biosphere Reserve) • Highest biodiversity • **Kunthipuzha River** flows through PYQ 1
- **Wheeler Island:** Odisha, near Chandipur coast (at mouth of Brahmani) • Renamed **Abdul Kalam Island** PYQ 1
- **Sriharikota (Andhra Pradesh):** **Barrier island** separating Pulicat Lake from Bay of Bengal • Satish Dhawan Space Centre

10 Drainage System of India (Rivers, Lakes, Waterfalls and Projects)

- **Himalayan drainage:** Antecedent rivers (Indus, Ganga, Brahmaputra) • Perennial • Deep gorges and deltas
- **Peninsular drainage:** Consequent and superimposed rivers • Seasonal • Shallow valleys

Major Rivers of India — Amarkantak: Narmada west, Son north



Indus River System

- **Indus:** Total 2880 km (1114 km in India) • Source: **Bokhar Chu Glacier** near Mansarovar (Tibetan name **'Singi Khamban'** — Lion's Mouth) PYQ 2
- **Indus flow in India:** Only through **Ladakh** (Leh and Kargil districts) • Deep gorge at **Bunji** near Nanga Parbat • Enters Pakistan near Chilas
- **Indus right-bank tributaries:** Shyok • Gilgit • Kabul • Kurram • Gomal
- **Indus left-bank tributaries:** Zanskar • Suru • Dras • **Panchnad** (Jhelum, Chenab, Ravi, Beas, Sutlej) PYQ 2
- **Panchnad (North → South):** Jhelum → Chenab → Ravi → Beas → Sutlej
- **Jhelum (Vitasta):** Source **Verinag Spring** (Sheshnag Lake, Pir Panjal, J&K) • Flows through Srinagar and **Wular Lake** • Forms India-Pakistan border PYQ 3
- **Jhelum — others:** Tributary **Kishanganga** (Neelum in Pakistan) • Projects: Tulbul, Uri, Kishanganga • Only Himalayan river **meandering** in Kashmir Valley

- **Chenab (Askini):** Source **Baralacha-La** (HP) • **Largest tributary of Indus** • Formed by Chandra + Bhaga (Chandrabhaga) • Projects: **Salal, Baglihar, Dulhasti, Ratle, Pakal Dul** PYQ 3
- **Ravi (Parushni):** Source west of **Rohtang Pass** (Kullu Hills, HP) • Chamba Valley • Projects: **Thein Dam (Ranjit Sagar)**, Chamera
- **Beas (Vipasha):** Source **Beas Kund** (near Rohtang, HP) • **Only Panchnad river flowing entirely within India** • Pong Dam PYQ 2
- **Harike (Kapurthala, Punjab):** Beas-Sutlej confluence • Origin of India's longest canal, **Indira Gandhi Canal**
- **Sutlej (Shutudri):** Source **Rakas Tal** (near Mansarovar, Tibet) • Tibetan name **'Langchen Khambab'** • **Antecedent river** • Enters India via Shipki-La PYQ 3
- **Sutlej projects:** **Bhakra-Nangal (Gobind Sagar Lake)** • Nathpa Jhakri • Kol Dam
- **Indus Waters Treaty (1960):** Karachi • Nehru and Ayub Khan • World Bank mediation
- **Water sharing:** Western rivers (Indus, Jhelum, Chenab) to Pakistan (limited use for India) • Eastern rivers (Ravi, Beas, Sutlej) **entirely to India** • About **80% of total water to Pakistan** PYQ 1

Ganga River System

- **Ganga:** India's largest drainage system • 2525 km
- **Panch Prayag (Uttarakhand; North → South):** Vishnuprayag → Nandprayag → Karnaprayag → Rudraprayag → Devprayag PYQ 4
- **Vishnuprayag:** Alaknanda (from Satopanth Glacier) + Dhauliganga
- **Nandprayag:** Alaknanda + Nandakini
- **Karnaprayag:** Alaknanda + **Pindar**
- **Rudraprayag:** Alaknanda + **Mandakini** (from Kedarnath/Chorabari Tal)
- **Devprayag:** Alaknanda + **Bhagirathi** (from Gaumukh/Gangotri Glacier) • Combined stream called **'Ganga'** after this
- **Tehri Dam:** At confluence of Bhagirathi and Bhilangana
- **Course of Ganga:** Enters plains at Haridwar • Enters UP at **Bijnor**, leaves to Bihar from **Ballia** • 28 districts of UP • Longest stretch in **Budaun** PYQ 3
- **Ganga in Bangladesh:** After Farakka (WB) main stream enters Bangladesh as **'Padma'** (Bhagirathi-Hooghly branch stays in WB) • Joins Brahmaputra (**'Jamuna'**) • Finally **'Meghna'** • World's largest **Sundarbans Delta**
- **Ganga left-bank tributaries (West → East):** Ramganga → Gomti → Ghaghara/Saryu → Gandak → Kosi → Mahananda PYQ 5
- **Ramganga:** Joins Ganga at Kannauj
- **Gomti:** From **Fulhar Lake**, Pilibhit • **Only plains-origin tributary** • Joins at Kaithi, Ghazipur
- **Ghaghara/Saryu:** From Mapchachungo Glacier • Joins at Chhapra • Tributaries: Sharda/Kali, Rapti
- **Gandak:** Joins at Sonpur/Hajipur
- **Kosi:** From Gosainthan • **'Sorrow of Bihar'** • Notorious for shifting course • Joins at Kursela
- **Mahananda:** From Darjeeling Hills • **Last left-bank tributary** of Ganga
- **Yamuna (1376 km):** **Longest** and westernmost tributary of Ganga • From **Yamunotri Glacier**, Bandarpunch Range • Joins Ganga at Prayagraj PYQ 5
- **Yamuna right-bank tributaries (West → East):** Chambal → Sindh → Betwa → Ken
- **Chambal:** From Janapav Hill (Mhow) • Badlands/ravines and gharials • Joins near Etawah
- **Betwa:** From Vindhyan hills of Raisen • Joins at Hamirpur • Matatila and Rajghat dams
- **Ken:** From Kaimur Hills • Joins at Chilla, Banda
- **Yamuna left-bank tributaries:** Tons/Tamsa • Hindon • Sengar • Rind
- **Tons (Tamsa):** From Kaimur Hills • Joins Ganga **directly** at Prayagraj (Sirsa)
- **Son:** From Amarkantak Plateau • Flows north, joins Ganga **directly** near Patna (Danapur/Maner) • Tributaries: **Rihand** (Govind Ballabh Pant Sagar Dam), North Koel PYQ 2
- **Damodar:** From Chotanagpur Plateau (Palamu, Jharkhand) • **Rift valley** • Joins Hooghly • Was called 'Sorrow of Bengal' • Coal reserves make valley 'Ruhr of India' • Tributary: Barakar PYQ 2

Brahmaputra River System

- **Brahmaputra:** Total 2900 km (916 km in India) • Source: **Chemayungdung/Angsi Glacier** near Mansarovar
- **Different names:** **'Tsangpo'** in Tibet (the purifier) • **'Dihang'/'Siang'** in Arunachal • **'Brahmaputra'** from Sadiya to Dhubri (Assam) • **'Jamuna'** in Bangladesh PYQ 2
- **Majuli (Assam):** World's largest river island • Due to braided channel PYQ 1
- **Left-bank tributaries (from south):** Dibang • **Lohit** (river of blood; Bhupen Hazarika Setu) • Burhi Dihing • Dhansiri (South) • Kopili PYQ 4
- **Right-bank tributaries (from north):** **Subansiri** (largest tributary; antecedent) • **Kameng (Jia Bharali)** • Manas • **Sankosh** (Assam-WB border) • Teesta
- **Teesta:** From Zemu Glacier/Cholamu Lake, Sikkim • Joins Jamuna near Rangpur, Bangladesh

Peninsular River System

East-flowing vs West-flowing

Peninsular rivers

Eastward — Bay of Bengal	≠	Westward — Arabian Sea
Godavari 1465 km Trimbakeshwar · longest		**Narmada** 1312 km Amarkantak · rift valley
Krishna 1400 km Mahabaleshwar		**Tapti** 724 km Multai · rift valley
Mahanadi 851 km Sihawa (Dhamtari)		**Mahi** Tropic of Cancer twice
Kaveri 800 km Brahmagiri · only perennial		**Sabarmati** Ahmedabad
Form **deltas** East coast broad, emergent		**Estuary, not delta** West coast narrow, submergent

Trap: **Luni** never reaches the sea — disappears in Rann of Kachchh · Mnemonic "Samanta" = Sabarmati, Mahi, Narmada, Tapti

- **Godavari (Dakshin Ganga/Vriddha Ganga):** Source **Trimbakeshwar** (Nashik, Maharashtra) • 1465 km • Bay of Bengal • **Longest and largest river of Peninsular India** PYQ 3
- **Godavari tributaries:** Left — Purna, Pranhita (Penganga + Wardha + Wainganga), **Indravati**, Sabari • Right — **Manjira** (only major right-bank), Pravara, Maner • Projects: Polavaram, Pochampad

- **Krishna:** Source **Mahabaleshwar** (Maharashtra) • 1400 km • Bay of Bengal • Second longest in Peninsular India PYQ 3
- **Krishna tributaries:** Left — **Bhima**, Musi (Hyderabad on its bank), Muneru • Right — Koyna, Panchganga, Dudhganga, Ghataprabha, Malaprabha, **Tungabhadra** (largest)
- **Krishna dams:** **Almatti** (Karnataka) • **Srisailam** and **Nagarjuna Sagar** (Andhra-Telangana border)
- **Kaveri (Ganga of South India):** Source **Brahmagiri Hills** (Kodagu/Coorg, Karnataka) • 800 km • Bay of Bengal • **Only perennial Peninsular river** PYQ 3
- **Kaveri water source:** Upper basin — South-West Monsoon • Lower basin — North-East Monsoon
- **Kaveri tributaries:** Left — Hemavati, Shimsha, **Arkavathi** • Right — Lakshmana Tirtha, **Kabini**, Suvarnavati, Bhavani, **Amaravati**
- **Kaveri — others:** Shivasamudram Falls • Srirangapatna and Shivasamudram islands • **Mettur Dam** (Tamil Nadu) • Krishna Raja Sagar Dam (Karnataka)
- **Mahanadi:** Source **Sihawa Hills** (Dhamtari, Chhattisgarh) • 851 km • Bay of Bengal near Paradip • Makes Chhattisgarh basin the 'Rice Bowl' PYQ 2
- **Mahanadi tributaries and dam:** Left — Seonath, **Hasdeo**, Mand, Ib • Right — **Jonk, Ong, Tel** • **Hirakud**, world's longest main dam, at Sambalpur (Odisha)
- **Narmada:** Source **Amarkantak Plateau** (Anuppur, MP) • 1312 km • Gulf of Khambhat (Arabian Sea) • Flows west in **rift valley** between Vindhya and Satpura PYQ 4
- **Narmada — others:** **Forms estuary, not delta** • Longest west-flowing Peninsular river into Arabian Sea • **Dhuandhar Falls** in marble rocks, Jabalpur • Dams: Indira Sagar (MP), **Sardar Sarovar** (Gujarat)
- **Narmada tributaries:** Tawa (largest) • Hiran • Orsang • Barna • Kolar • Dudhi • Shakkar • Sher
- **Tapti (Tapi):** Source **Multai** (Betul, MP) • 724 km • Gulf of Khambhat near Surat • Rift valley between Satpura and Ajanta • Estuary PYQ 2
- **Tapti tributaries and projects:** Purna, Girna, Bori, Panjhra, Arunavati • **Ukai and Kakrapar** in Gujarat
- **Mahi:** From Vindhya Range (Mehad Lake, Dhar, MP) • MP-Rajasthan-Gujarat • Gulf of Khambhat • **Crosses Tropic of Cancer twice** PYQ 2
- **Sabarmati:** From Mewar Hills of Aravalli (Udaipur) • Gulf of Khambhat • Ahmedabad and Gandhinagar on its banks
- **Luni:** From Nag Hills (Aravalli), Ajmer • **Disappears in Rann of Kachchh (inland drainage)** • Does not reach sea • Sweet till Balotra, saline thereafter PYQ 2
- **Small west-flowing rivers from Western Ghats:** Shetrunji, Bhadar (Gujarat) • Vaitarna, Ulhas (Maharashtra) • **Mandovi, Zuari** (Goa) • Sharavati, Netravati (Karnataka) PYQ 2
- **Rivers of Kerala:** **Periyar** (longest in Kerala; lifeline) • Bharathapuzha (second longest) • Pamba
- **East-flowing rivers (North → South; UPPCS very important):** Subarnarekha → Baitarani → Brahmani → Mahanadi → Vamsadhara → Godavari → Krishna → Pennar → Palar → Ponnaiyar → Kaveri → **Vaigai** (Madurai; into Palk Bay) → Tamraparni PYQ 4

Major Waterfalls

- **Kunchikal (455 m):** **Varahi River** (Shimoga, Karnataka) • Highest waterfall of India by height (multi-tiered) PYQ 2
- **Barehipani (399 m):** Budhabalanga River • Simlipal National Park (Mayurbhanj, Odisha)
- **Nohkalikai (340 m):** Cherrapunji (East Khasi Hills, Meghalaya) • Highest rain-fed plunge waterfall
- **Jog/Gersoppa/Mahatma Gandhi (253 m):** **Sharavati** (Karnataka) • Largest by width • Highest single-drop PYQ 2
- **Shivasamudram:** Kaveri (Karnataka) • Largest in India by **volume** PYQ 1
- **Hundru:** **Subarnarekha** (Ranchi, Jharkhand) PYQ 2
- **Chitrakote:** **Indravati** (Bastar, Chhattisgarh) • Horseshoe shape • **'Niagara of India'** PYQ 2
- **Dhuandhar and Kapildhara:** Narmada (Jabalpur and Anuppur, MP) PYQ 1
- **Chulia:** Chambal (Kota/Chittorgarh, Rajasthan)
- **Gokak:** Ghataprabha (Belgaum, Karnataka)
- **Dudhsagar:** **Mandovi** (Goa-Karnataka border) PYQ 2
- **Duduma:** Machkund River (Odisha-Andhra Pradesh border)

Major Lakes

- **Chilika (Odisha):** South of Mahanadi delta • **Largest coastal lagoon and brackish-water lake** • Naval training centre • Irrawaddy dolphins PYQ 2
- **Sambhar (Rajasthan):** **Most saline inland salt lake** • Near Jaipur PYQ 1
- **Wular (J&K):** **Largest freshwater lake** • **Tectonic** lake • Fed by Jhelum PYQ 2
- **Loktak (Manipur):** **Largest freshwater lake of Northeast** • Floating vegetation islands **'Phumdis'** • World's only floating national park **Keibul Lamjao** (Sangai deer) PYQ 3
- **Pulicat:** **Andhra Pradesh (84%) and Tamil Nadu (16%) border** • Second largest lagoon • Sriharikota separates it from Bay of Bengal PYQ 3
- **Kolleru (Andhra Pradesh):** Between Godavari and Krishna deltas • Freshwater PYQ 1
- **Vembanad (Kerala):** **Longest lake** (96.5 km) • Willingdon Island • Boat race (Nehru Trophy) PYQ 1
- **Ashtamudi and Sasthamkotta:** Kerala
- **Lonar (Buldhana, Maharashtra):** Crater lake from volcanic activity or **meteorite impact** PYQ 2
- **Cholamu (Tso Lhamo) and Gurudongmar (Sikkim):** Highest-altitude lakes PYQ 1
- **Roopkund (Uttarakhand):** **'Mystery Lake'** • 'Skeleton Lake' PYQ 2
- **Other lakes:** Surajkund (Haryana) • Chapanala and Haflong (Assam) • Palak (Mizoram) • Renuka, Surajtal/Chandratal (HP) • Pangong Tso, Tso Moriri (Ladakh) PYQ 2

Points to Note

1. **Chambal, Betwa, Ken:** Not direct tributaries of Ganga — of **Yamuna**; **Son and Tons** join **Ganga directly** (right bank), not Yamuna [2]
2. **Dakshin Ganga vs Ganga of South India:** By size-extent **Godavari** = 'Dakshin Ganga'/'Vriddha Ganga'; by sanctity and perennial flow **Kaveri** = 'Ganga of South India'
3. **Teesta confluence:** Not Ganga — **Brahmaputra (Jamuna)** in Bangladesh (joined Ganga before the 1787 flood) [1]

4. **Rivers from Amarkantak: Narmada, Son** (and Johila) — radial drainage; **Tapti** from Multai (Betul), **Mahanadi** from Sihawa (Dhamtari) [2]
5. **Gomti:** Not from Himalaya — from **Fulhar Lake (Gomat Taal)**, Pilibhit; **only plains-origin tributary** [1]

Mnemonic (Panch Prayag, North → South): "VK R D" — Vishnu → Nand → Karna → Rudra → Dev

Mnemonic (Prayag confluences; Alaknanda common): "Vi-Dhau, Ka-Pin, Ru-Man (RUM), De-Bha" — Vishnu = Dhauliganga, Karna = Pindar, Rudra = Mandakini, Dev = Bhagirathi

Mnemonic (Yamuna right-bank tributaries, West → East): "CSBK" — C = Chambal, S = Sindh, B = Betwa, K = Ken

Mnemonic (Major rivers into Arabian Sea): "Samanta (Sa-Ma-Na-Ta)" — Sa = Sabarmati, Ma = Mahi, Na = Narmada, Ta = Tapti

One-liner Facts

- **Antecedent rivers:** Flowed before Himalayan uplift • Cut deep gorges through rising Himalaya • Indus, Sutlej, Ganga (Alaknanda), Kosi, Teesta, Brahmaputra PYQ 2
- **Rift valley rivers:** **Damodar, Narmada, Tapti** PYQ 2
- **Chambal dams:** **Gandhi Sagar** (Mandsaur, MP) • **Rana Pratap Sagar** (Chittorgarh) • **Jawahar Sagar and Kota Barrage** (Kota) — Rajasthan PYQ 2
- **Mullaperiyar Dam:** Periyar River (Idukki, Kerala) • Controlled and operated by **Tamil Nadu government** • Inter-state dispute PYQ 1
- **Ken-Betwa Link:** India's first river interlinking project • **Panna Tiger Reserve** submerged/affected PYQ 2

11 Climate of India (Monsoon, Seasons, Rainfall Distribution and Koppen's Classification)

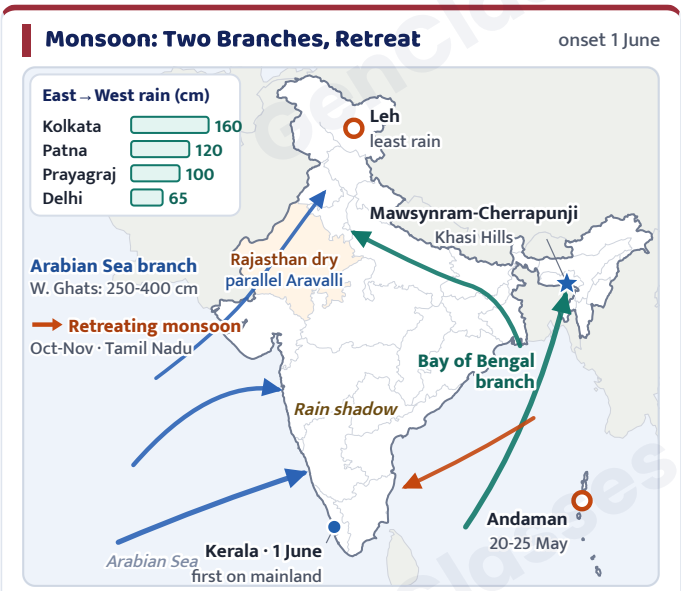
- **Climate type:** **Tropical Monsoon** PYQ 2
- **Tropic of Cancer:** 23°30′ N • passes through the middle of the country
- **Role of Himalayas:** blocks icy cold winds from Central Asia • obstructs South-West monsoon winds • gives the whole of India a **tropical character**

Mechanism of Indian Monsoon and Influencing Factors

- **ITCZ (Inter Tropical Convergence Zone):** in summer, with northward apparent movement of the Sun, ITCZ shifts north and settles over the Ganga plain (20° N to 25° N) • called **Monsoon Trough**
- **Effect of Monsoon Trough:** low-pressure zone • attracts trade winds of the Southern Hemisphere • after crossing the Equator, deflected to the right by **Ferrel's Law (Coriolis force)** → **South-West Monsoon** PYQ 2
- **Jet Stream Theory:** Koteswaram's theory
- **Sub-tropical Westerly Jet Stream:** in winter flows south of Himalayas • brings Western Disturbances to India • its shift north of Himalayas in summer causes the **Burst of Monsoon**
- **Tropical Easterly Jet Stream (TEJ):** originates in summer from **intense heating of the Tibetan Plateau** • flows over peninsular India (14° N) • pushes monsoon towards India PYQ 2
- **El-Nino (ENSO):** appearance of a **warm ocean current** off the west coast of Peru (South America) • Indian monsoon **weakens** • **drought** in India PYQ 3
- **La-Nina:** cold current strengthens off Peru coast • **favourable (good rainfall)** for Indian monsoon
- **Positive IOD (Indian Ocean Dipole):** western Indian Ocean (Arabian Sea) warmer than eastern Indian Ocean (near Indonesia) • **strengthens** monsoon • can also reduce negative effect of El-Nino PYQ 1

Onset, Branches and Retreat of Monsoon (Retreating Monsoon)

- **Onset (mainland):** first on **1 June** • **Kerala (Malabar Coast)** • in India as a whole, first around 20-25 May in Andaman-Nicobar PYQ 2
- **Share of South-West Monsoon:** about **75% to 80%** of India's total rainfall (June to September) PYQ 1
- **Arabian Sea branch:** strikes western slopes of Western Ghats • heavy **Orographic Rainfall** • 250-400 cm PYQ 2
- **Rain Shadow Area:** region east of Western Ghats (Deccan Plateau — Pune, Bengaluru, Hyderabad) • remains dry
- **Reason for drought in Rajasthan:** Aravalli lies **parallel** to the Arabian Sea branch • cannot obstruct moist winds • winds pass without rainfall PYQ 3



- **Bay of Bengal branch:** strikes Arakan Yoma and turns towards India • one stream trapped in **Khasi Hills of Meghalaya** (funnel-shaped Garo-Khasi-Jaintia) → world's highest rainfall at **Mawsynram and Cherrapunji** PYQ 2
- **Second stream:** moves **east to west** over Ganga plain • from east to west (Kolkata → Patna → Prayagraj → Delhi) and from north to south, rainfall **continuously decreases** PYQ 3

Winter Rainfall and Pre-monsoon Showers

- **Retreating Monsoon (North-East Monsoon):** **Tamil Nadu (Coromandel Coast)** and southern Andhra Pradesh • in October-November NE winds pick moisture over **Bay of Bengal** → maximum winter rainfall on **Tamil Nadu (Chennai)** coast PYQ 4
- **Western Disturbances:** Punjab, Haryana, western UP, Rajasthan and Himalayan states • winter (December-February) • temperate cyclones originating in **Mediterranean Sea** • called **Mawath** in Rajasthan • highly beneficial ("tonic") for **Rabi (especially wheat)** PYQ 4

- **Mango Showers:** **Kerala and coastal Karnataka** • pre-monsoon showers at end of summer (April-May) • help early ripening of mangoes PYQ 2
- **Cherry Blossom (Blossom Showers):** **Karnataka** (and Kerala) • pre-monsoon rain • helps flowering of **coffee** PYQ 1
- **Kalbaisakhi (Norwesters):** **West Bengal** (and Odisha) • destructive evening thunderstorms in Baisakh • useful for tea, jute, rice PYQ 1
- **Bardoli Chheerha (Tea Showers):** **Assam** • Assamese name of Kalbaisakhi • "Tea Shower" • beneficial for tea PYQ 1
- **Loo:** North and North-West India • May-June • extremely hot and dry local winds

Koppen's Climatic Classification of India

- **Importance:** very important • asked almost every alternate year PYQ 4
- **Amw:** monsoon type with short dry season • **west coast (Malabar and Konkan)** south of Goa • more than 300 cm rainfall
- **As:** monsoon type with dry summer • **Coromandel Coast of Tamil Nadu** and adjoining Andhra Pradesh • dry summer, rain in winter PYQ 2
- **Aw:** Tropical Savanna • **most of peninsular plateau** south of Tropic of Cancer (Maharashtra, Telangana, Chhattisgarh, Jharkhand, Odisha)
- **BShw:** semi-arid Steppe • north-western Gujarat, parts of western Rajasthan, Punjab, **rain shadow region** of peninsular India
- **BWhw:** hot desert • westernmost Rajasthan (**Thar Desert** — Jaisalmer, Bikaner)
- **Cwg:** monsoon type with dry winter • **Great Ganga Plain (Uttar Pradesh, Bihar)**, eastern Rajasthan, northern Madhya Pradesh, Assam PYQ 2
- **Dfc:** short summer and cold humid winter • **Arunachal Pradesh** and Sikkim
- **E / ET:** Polar / Tundra-like • **high Himalayan mountain regions** of Jammu-Kashmir, Ladakh, Himachal, Uttarakhand

Points to Note

Points to Note

1. **Winter rain in Tamil Nadu:** not SW Monsoon or Western Disturbances — **North-East (retreating) Monsoon** • winter rain in NW India (Punjab/UP) from **Western Disturbances** [2]
2. **As vs Amw:** 's' = Summer Dry • only Coromandel Coast is dry in summer, wet in winter • **As = Coromandel Coast (Tamil Nadu)** • **Amw = Malabar Coast (Kerala/Goa)** [1]
3. **Cherrapunji/Mawsynram:** not Garo or Jaintia — **Khasi Hills** (Meghalaya) [1]
4. **Lowest rainfall:** not Jaisalmer — **Leh (Ladakh)** • annual rainfall below 10 cm • cold desert [1]

Mnemonics

Mnemonic (Koppen code): "g in Cwg = Ganga Plain" — Cwg = Ganga plain (UP/Bihar)

Mnemonic (As): "s = summer dry" — As = Chennai/Coromandel (dry summer, winter rain)

Mnemonic (BWhw): "Big Warm Hot West Desert" — BWhw = Thar Desert

- **Decreasing rainfall east to west:** **Kolkata (160 cm) > Patna (120 cm) > Prayagraj (100 cm) > Delhi (65 cm)**

One-liner Facts

- **October Heat / Kwar ki Umas:** oppressive humid heat in October-November due to high temperature and high humidity during retreating monsoon PYQ 1
- **Highest diurnal range of temperature:** **desert areas of Rajasthan** • lowest diurnal and annual range: **coastal Kerala and Andaman-Nicobar** PYQ 1
- **Average annual rainfall:** about **118 cm (or 125 cm)** • high spatial and temporal variability PYQ 1
- **Break in Monsoon:** in July-August the monsoon trough (ITCZ) shifts to Himalayan foothills • no rain in plains for one or more weeks PYQ 1
- **Thornthwaite / Trewartha:** Thornthwaite — most of India in sub-humid and dry-winter regions • Trewartha — symbol **Caw** for Ganga plain

12 Soils of India (Classification, Characteristics, Problems and Conservation)

- **ICAR:** Indian Council of Agricultural Research • established 1929 • HQ New Delhi • **8 major soil groups**
- **Decreasing order by area:** **Alluvial (43.4%) > Red and Yellow (18.6%) > Black (15.2%) > Laterite (3.7%)** PYQ 2
- **Universal truth:** almost **all soils deficient in Nitrogen (N), Phosphorus (P) and Humus** • exception: humus in marshy/peat soil PYQ 3

Comparative Study of Major Soils of India

Area Share of Soils (ICAR) — decreasing order

Alluvial — N. plains, deltas	43.4%	rich in potash, lime · azonal · Khadar/Bangar
Red & Yellow — Tamil Nadu	18.6%	red = iron oxide · hydrated = yellow · acidic
Black (Regur) — Maharashtra	15.2%	best for cotton · self-ploughing · max water retention
Laterite — Kerala (Malabar)	3.7%	leaching · Fe, Al oxides remain · cashew, tea, rubber

Trap: almost all soils lack **N, P and humus** · black colour is not from humus but from **titaniferous magnetite**

- **Alluvial soil (Kanp/Kachhari):** deposition of sediments by Himalayan rivers • **Azonal Soil** • largest **43.4% area** (northern plains and coastal deltas)
- **Alluvial — elements:** rich in **Potash (K)** and **Lime** • deficient in Nitrogen, Phosphorus, Humus
- **Alluvial — crops and facts:** wheat, paddy, sugarcane, pulses, oilseeds • **Khadar** (new, more fertile) • **Bangar** (old, kankar-rich) • moderate water retention • highest fertility PYQ 3
- **Black soil (Regur, Cotton soil, Tropical Chernozem):** from **weathering of basaltic lava** of Deccan Trap • largest in **Maharashtra** • then Gujarat, MP (Malwa Plateau), Karnataka, Andhra Pradesh

- **Black — elements:** rich in **iron, lime, calcium, potash, aluminium, magnesium carbonate** • deficient in **Nitrogen, Phosphorus, organic matter (humus)**
- **Black — crops: best for cotton** • sugarcane, soybean, citrus fruits
- **Black — special facts:** black colour due to **titaniferous magnetite** and organic matter • **highest water retention** • sticky when wet, deep cracks when dry • **Self-ploughing Soil** • needs less irrigation PYQ 5
- **Red and Yellow soil:** from disintegration of Archaean crystalline igneous/metamorphic rocks (granite, gneiss) in low-rainfall areas • second largest group (18.6%) • **Tamil Nadu** (largest), Andhra Pradesh, Chhattisgarh, Odisha, Bundelkhand (UP)
- **Red-Yellow — elements:** rich in iron (ferric oxide), silica • deficient in Nitrogen, Phosphorus, Humus, Lime
- **Red-Yellow — crops and facts:** bajra, groundnut, tobacco, potato, coarse grains • **red colour: iron oxide** • **yellow when hydrated** • **acidic** in nature PYQ 3
- **Laterite soil (brick-like):** formed by intense **leaching** under **high temperature and heavy rainfall (over 200 cm)** • largest in **Kerala (Malabar Coast)**, Western Ghats, Meghalaya, Odisha, Rajmahal Hills
- **Laterite — elements:** rich in **iron oxide and aluminium oxide** (remain in top layer) • deficient in **Nitrogen, Potash, Lime, Silica** (lime-silica leach down), Humus
- **Laterite — crops and facts:** less fertile for food grains • suitable for plantation crops like **cashew, tapioca (cassava), tea, coffee, rubber, cinchona** • soft like butter when wet, hard like brick when dry PYQ 5
- **Desert (Arid) soil:** western Rajasthan, southern Punjab/Haryana, northern Gujarat (Kutch) • rich in **soluble salts and phosphate (phosphorus)** • deficient in nitrogen, **severe lack of humus**
- **Desert — crops and facts:** bajra, jowar, guar, mustard • wheat too with Indira Gandhi Canal irrigation • alkaline • lower layer has **kankar** (lime layer) that blocks water percolation PYQ 1
- **Mountain (Forest) soil:** Himalayan region (Jammu-Kashmir to Arunachal) and Western-Eastern Ghats • rich in **humus (organic matter)** • deficient in potash, phosphorus, lime
- **Mountain — facts:** **acidic** due to excess humus • slopes grow **tea, apple, pear, potato, spices** PYQ 1
- **Peat and Marshy soil:** coastal areas with heavy rain and high humidity • **Kari** in **Alleppey** (Kerala) • Sundarbans delta • Terai region
- **Peat — elements and facts:** rich in **humus/organic matter (up to 40-50%)** and soluble salts • deficient in potash, phosphate • heavy, black, highly acidic • suitable for mangroves and paddy PYQ 1

Acidic and Alkaline/Saline Soils and Soil Reclamation

- **Section weight:** saline-alkaline-acidic soils and reclamation PYQ 4
- **Saline-alkaline soil — local names:** Reh • Kallar • Usar • Thur • Rakar • Chopan PYQ 1
- **Cause of formation:** arid/semi-arid areas (Punjab, Haryana, western UP — Green Revolution areas) • **over-irrigation by canals** • **Capillary Action** brings sodium, calcium, magnesium salts to surface as white crust PYQ 3
- **Largest extent: Saline soil — Gujarat** • **Alkaline soil — Uttar Pradesh** PYQ 2
- **Alkaline soil reclamation:** pH > 7.5 • **Gypsum ($CaSO_4 \cdot 2H_2O$) or Pyrite** • green manure (Dhaincha/Sunhemp) PYQ 3
- **Acidic soil — cause:** leaching of basic elements in high-rainfall areas • decomposition of humus • good for tea gardens
- **Acidic soil reclamation:** pH < 6.5 • **Lime (limestone)** • suitable pH for normal crops **6.0 to 7.0** PYQ 2

Soil Erosion and Soil Conservation

- **Successive stages of erosion:** correct chronological order PYQ 3
- **1. Splash erosion:** scattering of soil particles by raindrop impact • initial stage
- **2. Sheet erosion:** uniform removal of most fertile top layer • not easily visible to farmers • **"Death of Farmers"**
- **3. Rill erosion:** small, shallow channels (finger-like) in fields
- **4. Gully erosion:** rills become large and deep • land unfit for farming • **Badland Topography / Ravines** PYQ 4
- **Gully — most affected: Chambal river valley** (Morena, Bhind, Etawah, Dholpur) • Yamuna region
- **5. Stream Channel erosion:** water cuts banks as a thick stream
- **Soil conservation measures:** Crop Rotation • Contour Ploughing • Terrace Farming • Strip Cropping • Mulching (to conserve moisture) • ban on overgrazing and jhum cultivation PYQ 2

Major Soils of Uttar Pradesh

- **Bhabar and Terai soil:** Saharanpur to Kushinagar • in Himalayan foothills
- **Bhur:** 10 to 20 feet high wind-deposited sand dunes in Ganga-Yamuna Doab PYQ 1
- **Mar and Kabar:** Bundelkhand (Jhansi, Lalitpur, Jalaun, Hamirpur) • clayey soils similar to **black soil (Regur)** • high water retention PYQ 2
- **Parwa and Rakar: red-yellow and sandy/gravelly** soils of Bundelkhand • Rakar on slopes • gram and sesame cultivation PYQ 1
- **Water erosion in UP:** most affected — Himalayan Terai region • gully erosion in Chambal-Yamuna region — **Etawah** district
- **Wind erosion in UP:** most affected — **Mathura and Agra** (south-western UP) PYQ 3

Points to Note

Points to Note

1. **Lime vs Gypsum: Acidic = Lime** (trick: "A-Lime") • **Alkaline (Usar) = Gypsum** [2]
2. **Humus in black soil:** assuming high humus from dark colour is wrong — colour from **titaniferous magnetite** (iron compound) • deficient in **all three**: nitrogen, phosphorus, humus [2]
3. **Laterite: lime and silica** leach down (deficient) • **iron and aluminium** oxides remain on top (abundant) [1]
4. **Top states:** most **saline = Gujarat** • most **alkaline (Usar) = Uttar Pradesh** [1]

Mnemonics

Mnemonic (soil area order): "Ja-La-Ka-Lai" — Ja = Jalodh/Alluvial (43%) > La = Lal/Red (18%) > Ka = Kali/Black (15%) > Lai = Laterite (3.7%)

Mnemonic (erosion order): "Boond ne parat banai, rill ne nali (gully) bahai" — Splash → Sheet → Rill → Gully → Stream Channel

Mnemonic (soil reclamation): "Amla ko choona lagao, kshar ko gypsum khilao" — Acid gets lime, alkali gets gypsum

One-liner Facts

- **Soil particle size (large → small):** **Gravel > Sand > Silt > Clay** • clay particle diameter below 0.002 mm PYQ 1
- **Loam / Sandy loam:** about 40% sand • 40% silt • 20% clay • most ideal for agriculture PYQ 1
- **Nitrogen fixation:** **leguminous crops** (gram, pea, arhar, moong) • **Rhizobium** in root nodules • in paddy fields **Blue-Green Algae (BGA) and Azolla** PYQ 3
- **Soil institutes:** **CSSRI** (Central Soil Salinity Research Institute) — **Karnal (Haryana)** • **IISS** (Indian Institute of Soil Science) — **Bhopal (MP)** • **CAZRI** (Central Arid Zone Research Institute) — **Jodhpur (Rajasthan)** PYQ 2
- **Bhojpatra (Betula utilis):** Himalayan alpine forests (high mountain soil, 3000-4500 m) • its bark used for writing texts in ancient times PYQ 2

Part 3: Economic Geography of India

13 Indian Agriculture (Cropping Pattern, Major Crops and Crop Varieties)

- **3 categories of agriculture questions in UPPCS:** specific crop varieties • major diseases (and micronutrient deficiencies) • agro-climatic and agro-ecological zones

Agro-climatic and Agro-ecological Zones and Jhum Cultivation

- **Agro-climatic zones:** Planning Commission • 1989 • **15 zones** PYQ 2
- **Agro-ecological zones:** National Bureau of Soil Survey and Land Use Planning (Nagpur) • **20 zones** • 60 sub-zones PYQ 2
- **Agro-climatic zones of Uttar Pradesh: 9** PYQ 2
- **Regional names of shifting (Jhum) cultivation:** PYQ 3
- **Jhum:** North-East India (Assam, Meghalaya, Nagaland, Mizoram)
- **Podu / Penda:** Andhra Pradesh and Telangana
- **Bewar / Dahiya:** Madhya Pradesh (Bundelkhand and Baghelkhand)
- **Kumari:** Western Ghats (hilly region of Kerala)
- **Walre / Waltre:** South-eastern Rajasthan
- **Khil:** Himalayan belt (Uttarakhand and Himachal Pradesh)
- **Kuruwa:** Jharkhand
- **Pamlou:** Manipur
- **Dipa:** Bastar district (Chhattisgarh) • Andaman-Nicobar Islands
- **Ponam:** Kerala
- **Koman, Bringa, Dhavi:** Odisha

Paddy (Rice)

- **Climate:** tropical • average temperature above 24°C • annual rainfall above 100 cm
- **Top producers:** 1. West Bengal • 2. Uttar Pradesh • 3. Punjab PYQ 2
- **Rice Bowl:** **Krishna-Godavari Delta** • "Rice Bowl" as a state: **Chhattisgarh** PYQ 1

Top Producing States of Crops (1 → 2 → 3)

	1	2	3
Paddy	West Bengal	Uttar Pradesh	Punjab
Wheat	Uttar Pradesh	Madhya Pradesh	Punjab
Sugarcane	Uttar Pradesh	Maharashtra	Karnataka
Tea	Assam	Coffee	Karnataka (about 70%)

Trap: **Punjab** tops wheat yield per hectare · Rice Bowl: **Krishna-Godavari Delta** · "Rice Bowl" state: **Chhattisgarh**

- **Major varieties:** Jaya • Padma • Pusa Sugandha-5 • Mahi Sugandha • Barani Deep • Govind • Kaveri • Ratna • Karuna • Lunishree • Saket-4 • Swarna Sub-1 • Sahbhagi (drought tolerant) PYQ 4
- **Kala Namak:** cultivated since Buddhist period • Siddharthnagar-Gorakhpur region • Geographical Indication (GI)
- **Hybrid variety:** **Pusa RH-10** • first hybrid Basmati PYQ 1
- **Golden Rice:** genetically modified • rich in **Beta-carotene (precursor of Vitamin A)** PYQ 2
- **Bio-fertilisers:** **Blue-Green Algae (Cyanobacteria)** • **Azolla** • **Azospirillum** — nitrogen fixation • **Anabaena** algae symbiotic with Azolla PYQ 3
- **Khaira disease:** **Zinc deficiency** PYQ 2
- **Other diseases:** False Smut (Ustilaginoidea virens fungus) • Blast (Pyricularia oryzae fungus) • Tungro virus (spread by green leafhopper)

Wheat

- **Climate:** temperate (Rabi) • 10-15°C at sowing • 20-25°C at ripening • 50-75 cm rainfall
- **Top producers:** 1. Uttar Pradesh • 2. Madhya Pradesh • 3. Punjab • **Punjab** tops per-hectare productivity PYQ 2
- **Dwarf wheat gene:** **Norin-10** • from Japan • developed in Mexico by Norman Borlaug PYQ 2
- **Major varieties:** Kalyan Sona • Sonalika • Lerma Rojo • Sonora-64 • Raj-3077 • UP-308 • Pusa Sindhu-Ganga • Pusa Baker • HD-2967 • HD-3226 (Pusa Yashasvi) • Karan Vandana (DBW-187) PYQ 3
- **Macaroni wheat:** most suitable for unirrigated conditions
- **Triticale:** **wheat × rye** • first man-made hybrid cereal PYQ 2
- **Karnal Bunt:** Neovossia indica fungus PYQ 2
- **Rust disease:** brown, yellow, black rust • Puccinia fungus
- **Ear Cockle / Tundu disease:** nematode and bacteria
- **Mandusi / Gulli-danda (Phalaris minor):** mimic weed of wheat • control: **Isoproturon** or **Sulfosulfuron** PYQ 2

Pulses (Gram, Arhar, Pea, Lentil)

- **Nitrogen fixation:** **Rhizobium** in root nodules • **Cobalt** and **Molybdenum** essential • Molybdenum = main component of nitrogenase enzyme PYQ 2
- **Rajma (French Bean):** only major pulse without root nodules • **does not fix atmospheric nitrogen** PYQ 2
- **Varieties (section):** arhar, pea, gram, lentil PYQ 4
- **Arhar (Tur):** Bahar • Amar • Azad • Malviya Chamatkar • Paras • Pusa-9 • ICPL-87 (Pragati) • UPAS-120
- **Pea:** Aparna (leafless) • Sapna • Rachna • Malviya Matar-15 • Prakash • Pant Uphar

- **Gram:** Pusa Chamatkar • Radhey • Vishal • Uday • Avrodhi • Samrat • Pant G-114 • Gaurav
- **Lentil (Masoor):** Pant L-406 • Garima • Priya • Narendra Masoor-1 • Mallika
- **Wilt disease:** arhar and gram • Fusarium oxysporum fungus PYQ 1

Oilseeds (Mustard, Groundnut, Soybean, Sunflower)

- **Mustard varieties:** Varuna • Pusa Bold • **Pusa Jai Kisan** (first variety developed through biotechnology) • Pitambari (yellow mustard) • Rohini • Kranti • **Dhara Mustard Hybrid-11 (DMH-11)** — genetically modified PYQ 3
- **Groundnut varieties:** Kaushal • Chitra • Chandra • Jyoti • Amber • Vikram • TG-37A PYQ 2
- **Groundnut — Pegging:** high requirement of **gypsum (calcium and sulphur)** • major disease: **Tikka disease** (Cercospora fungus) PYQ 2
- **Soybean varieties:** Ahilya • Alankar • Pratikar • Shilajeet • Bragg • Clark-63 • Gaurav PYQ 1

Cash and Plantation Crops (Sugarcane, Cotton, Jute, Coffee, Tea, Potato, Mango, Guava, Amla)

- **Sugarcane — top producers:** 1. Uttar Pradesh • 2. Maharashtra • 3. Karnataka PYQ 2
- **Sugarcane institutes:** Sugarcane Breeding Institute — **Coimbatore (Tamil Nadu)**, 1912 • Indian Institute of Sugarcane Research — **Lucknow**, 1952 PYQ 2
- **Adsali crop:** Maharashtra and South India • **18-month** sugarcane crop PYQ 1
- **Sugarcane disease:** **Red Rot** • Colletotrichum falcatum fungus PYQ 2
- **Cotton — climate:** 210 frost-free days • black soil (Regur) • clear sunny sky at ripening PYQ 1
- **Bt Cotton:** **Cry1Ac** gene from soil bacterium **Bacillus thuringiensis** • resistant to pink bollworm PYQ 2
- **Bale weight:** 1 cotton bale = **170 kg** • 1 jute bale = **180 kg** PYQ 1
- **Tea:** Assam top producer • **Darjeeling Tea** — India's first GI tag in 2004 PYQ 1
- **Coffee:** **Karnataka** top (about 70%) • **Arabica** and **Robusta** • first seeds sown in **Baba Budan Hills** of Karnataka PYQ 3
- **Horticulture varieties (section):** mango, guava, amla, potato, papaya PYQ 5
- **Mango:** **Sindhu** — world's only **seedless** variety (Ratna × Alphonso) • **Amrapali** — Dashehari × Neelum, for high-density planting • **Mallika** — Neelum × Dashehari
- **Mango (others):** Arka Anmol • Ratna • Chausa • Dashehari • Alphonso (Hapus)
- **Guava:** **Lalit** (pink pulp, best for jelly) • Allahabad Safeda • Sardar (L-49) • Arka Mridula • Shweta
- **Amla:** Kanchan • Krishna • Banarasi • Chakaiya • Narendra Amla-7 • Hathijhool (Francis)
- **Potato varieties:** Kufri Chipsona-1 and 2 (for chips) • Kufri Badshah • Kufri Bahar • Kufri Chandramukhi • Kufri Jyoti • Kufri Neelima
- **Potato diseases:** **Late Blight** — Phytophthora infestans fungus • **Black Heart** — lack of oxygen in storage PYQ 2
- **Papaya:** **Pusa Nanha** (dwarf variety for pots/terraces) • Pusa Majesty • Coorg Honey Dew

Agricultural Revolutions, Major Boards and Research Institutes

- **Agricultural revolutions and their fathers:** PYQ 4
- **Green Revolution:** food grains (mainly wheat and rice) • **M. S. Swaminathan** (world: Norman Borlaug)
- **White Revolution (Operation Flood):** milk production • **Dr. Verghese Kurien** • started 1970
- **Yellow Revolution:** oilseeds (especially mustard and sunflower) • **Sam Pitroda**
- **Blue Revolution:** fish production • Dr. Arun Krishnan and Hiralal Chaudhary
- **Pink Revolution:** onion and prawn production • Durgesh Patel
- **Red Revolution:** meat and tomato production • Vishal Tiwari
- **Silver Revolution:** egg and poultry production • Indira Gandhi
- **Silver Fibre Revolution:** cotton production
- **Golden Revolution:** fruits, honey and overall horticulture • Nirpakh Tutej
- **Golden Fibre Revolution:** jute production
- **Grey Revolution:** fertiliser production
- **Black Revolution:** petroleum / biodiesel (Jatropha) production
- **Round Revolution:** potato production
- **Evergreen Revolution:** organic farming and overall agricultural development • M. S. Swaminathan
- **Rainbow Revolution:** integrated form of all agricultural revolutions • proposed in National Agriculture Policy (2000)
- **Major agricultural boards and HQs:** PYQ 4
- **Coffee Board:** Bengaluru (Karnataka) • 1942
- **Rubber Board:** Kottayam (Kerala) • 1947
- **Tea Board:** Kolkata (West Bengal) • 1954
- **Tobacco Board:** Guntur (Andhra Pradesh) • 1976
- **Spices Board:** Kochi (Kerala) • 1987
- **National Jute Board:** Kolkata (West Bengal) • 2009
- **National Fisheries Development Board:** Hyderabad (Telangana) • 2006
- **National Grape Processing Board:** Pune (Maharashtra) • 2009

Points to Note

Points to Note

1. **Variety confusion:** **Varuna** = mustard • **Aruna** = mutant variety of castor • **Karuna** = dwarf paddy variety • **Kaushal** = groundnut • **Kanchan and Krishna** = amla [2]
2. **Pea vs Mustard:** **Aparna** = leafless pea • **Pusa Jai Kisan** (Bio-902) = mustard [1]
3. **Sugarcane institutes:** Indian Institute of Sugarcane Research (IISR) — **Lucknow** • National Sugar Institute (NSI) — **Kanpur** • Sugarcane Breeding Institute — **Coimbatore** [2]
4. **Spice production:** as per latest data **Madhya Pradesh** tops (ahead of Rajasthan and Gujarat) [1]

Mnemonics

Mnemonic (Amrapali): "DNA" — D = Dashehari × N = Neelum = A = Amrapali

Mnemonic (Mallika): "NDM" — N = Neelum × D = Dashehari = M = Mallika

Mnemonic (Sindhu): "RAS" — R = Ratna × A = Alphonso = S = Sindhu (seedless)

Mnemonic (Amla varieties): "Kanchan aur Krishna ne Banaras ke Chakaiya me hathi dekha" — Kanchan, Krishna, Banarasi, Chakaiya, Hathijhool

One-liner Facts

- **Impact of Green Revolution:** most positive on wheat and rice • area under coarse grains and pulses declined PYQ 1
- **Fully organic: Sikkim** — first 100% organic state in 2016 • **Lakshadweep** — first fully organic Union Territory PYQ 2
- **Karnal Bunt vs Khaira:** Karnal Bunt — fungal disease of wheat • Khaira — zinc deficiency in paddy PYQ 1
- **Institutes:** Indian Institute of Pulses Research — **Kanpur** • Indian Institute of Vegetable Research — **Varanasi** PYQ 2
- **CACP (Commission for Agricultural Costs and Prices):** established 1965 • recommends MSP for **23 crops** • Fair and Remunerative Price (FRP) for sugarcane • final approval: Cabinet Committee on Economic Affairs (**CCEA**) PYQ 2

14 Irrigation, Multipurpose River Valley Projects and Water Resources

- **Exam trend:** 'Dam ↔ river/state' matching • direct questions on tube-well and canal irrigation data

Means and Pattern of Irrigation in India

Sources of Irrigation (share) — decreasing order

Tube-wells	46-48%	UP: 74.9% by tube-wells
Canals	24%	
Wells	16%	wells + tube-wells > 62%
Tanks	3%	top: Tamil Nadu (hard rocks)

Project class (Culturable Command Area)

Minor	Medium	Major
< 2,000 ha	2,000-10,000 ha	> 10,000 ha

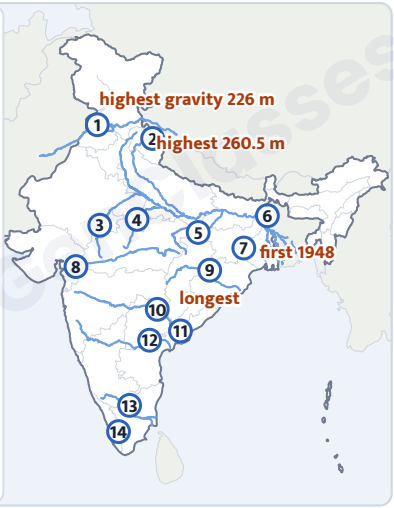
Trap: highest irrigated by percentage: **Punjab** (over 98%) · top in total irrigated area: **Uttar Pradesh**

- **Sources in decreasing order: Tube-wells** (about 46-48%) > Canals (about 24%) > Wells (about 16%) > Tanks (about 3%) > Other sources PYQ 2
- **Wells + tube-wells combined:** share **over 62%**
- **State most irrigated by tube-wells: Uttar Pradesh** • about 74.9% of total irrigated area by tube-wells PYQ 3
- **State most irrigated by tanks: Tamil Nadu** • reason: hard rocks of the Peninsular Plateau
- **Highest irrigated state by percentage: Punjab** • over about 98% of total cropped area irrigated
- **Top in total irrigated area:** Uttar Pradesh
- **Top in micro-irrigation (drip and sprinkler):** Karnataka • Andhra Pradesh • Maharashtra • Gujarat
- **Basis of project classification:** Culturable Command Area (CCA) PYQ 2
- **Minor irrigation project:** CCA **less than 2,000 hectares**
- **Medium irrigation project:** CCA 2,000 to 10,000 hectares
- **Major irrigation project:** CCA **more than 10,000 hectares**

Major Multipurpose River Valley Projects of India

Major Multipurpose Projects and Rivers

1 Bhakra-Nangal · Sutlej
2 Tehri · Bhagirathi
3 Gandhi Sagar · Chambal
4 Matatila · Betwa
5 Rihand · Rihand river
6 Farakka · Ganga
7 Panchet (DVC) · Damodar
8 Sardar Sarovar · Narmada
9 Hirakud · Mahanadi
10 Kaleshwaram · Godavari
11 Polavaram · Godavari
12 Nagarjuna Sagar · Krishna
13 Mettur · Kaveri
14 Idukki · Periyar



- **Bhakra-Nangal:** Sutlej • Punjab, Haryana, Rajasthan (joint venture) • India's highest **gravity dam** (226 m) • **Gobind Sagar** reservoir behind it (Himachal Pradesh) PYQ 1
- **Tehri Dam:** confluence of Bhagirathi and Bhilangana • Uttarakhand, Uttar Pradesh • India's **highest dam** (260.5 m; rock-fill and earth-fill) PYQ 2
- **Hirakud Dam:** Mahanadi • Odisha • **longest main dam** of India and the world (main length 4.8 km, total 25.8 km) PYQ 1
- **Damodar Valley Corporation (DVC):** Damodar and tributaries • Jharkhand and West Bengal • independent India's **first multipurpose project** (1948) • modelled on the **Tennessee Valley Authority** of the USA PYQ 2
- **DVC dams:** Tilaiya and Maithon (Barakar river) • Panchet Hill (Damodar river) • Konar (Konar river)
- **Chambal Project:** Chambal • Rajasthan and Madhya Pradesh • 3 dams: **Gandhi Sagar** (Mandsaur, MP) • **Rana Pratap Sagar** (Rawatbhata, Rajasthan) • **Jawahar Sagar** (Kota, Rajasthan) PYQ 3
- **Indira Gandhi Canal (Rajasthan Canal):** Sutlej-Beas confluence (Harike Barrage) • Rajasthan (Ganganagar, Bikaner, Jaisalmer, Barmer) • India's **longest irrigation canal system** PYQ 1
- **Nagarjuna Sagar:** Krishna • Telangana-Andhra Pradesh border • among the world's largest **masonry** dams PYQ 2
- **Almatti and Srisailam:** Krishna • Almatti (Karnataka), Srisailam (Andhra Pradesh-Telangana) • main focus of the **Krishna water dispute** PYQ 2
- **Pochampad (Sriram Sagar):** Godavari • Telangana • major irrigation project of South India PYQ 1
- **Kaleshwaram Lift Irrigation:** Godavari (Pranhita confluence) • Telangana • world's **largest multi-stage lift irrigation** project PYQ 1
- **Polavaram:** Godavari • Andhra Pradesh • **national project** linked to joining the Godavari and Krishna PYQ 1
- **Idukki:** Periyar • Kerala • major **arch dam** • between Kuravan and Kurathi hills PYQ 2
- **Mullaperiyar:** Periyar • located in Kerala, operated by **Tamil Nadu** • Kerala-Tamil Nadu dispute over safety and water level PYQ 1

- **Mettur and Sivasamudram:** Kaveri • Mettur (Tamil Nadu), Sivasamudram (Karnataka) • Sivasamudram (1902) among the oldest hydropower projects • first: **Sidrapong, Darjeeling (1897)** PYQ 2
- **Sardar Sarovar:** Narmada • Gujarat, Madhya Pradesh, Maharashtra, Rajasthan • most water to **MP**, most irrigation benefit to **Gujarat** • Narmada Bachao Andolan (Medha Patkar) PYQ 1
- **Indira Sagar and Omkareshwar:** Narmada • Madhya Pradesh • Indira Sagar: India's reservoir with the **largest water storage capacity** PYQ 1
- **Ukai and Kakrapar:** Tapti (Tapi) • Gujarat • near Surat PYQ 2
- **Mahi Bajaj Sagar and Kadana:** Mahi • Bajaj Sagar (Rajasthan), Kadana (Gujarat) • Mahi: river crossing the **Tropic of Cancer twice** PYQ 1
- **Rihand Dam (Govind Ballabh Pant Sagar):** Rihand (tributary of Son) • Uttar Pradesh (Sonbhadra - Pipri) • among India's largest artificial lakes PYQ 2
- **Matatila and Rajghat (Rani Laxmibai):** Betwa • Uttar Pradesh and Madhya Pradesh • lifeline of the Lalitpur-Jhansi region PYQ 3
- **Bansagar:** Son • Madhya Pradesh, Uttar Pradesh and Bihar • dam in **Shahdol (MP)** PYQ 2
- **Mayurakshi (Canada Dam/Massanjore):** Mayurakshi • Jharkhand and West Bengal • built with **Canadian** assistance PYQ 1
- **Farakka Barrage:** Ganga • West Bengal • diverts water into the Hooghly to keep Kolkata port silt-free (1975) PYQ 1
- **Chukha, Tala and Mangdechhu:** Wangchu and Mangdechhu • India and Bhutan • hydropower projects built in **Bhutan** with Indian cooperation PYQ 2

Strategic Projects of Jammu-Kashmir and Ladakh (Indus River System)

- **Context:** **Indus Waters Treaty (1960)** • project ↔ river matching asked repeatedly PYQ 5
- **On Chenab:** **Dulhasti** • **Baglihar** • **Salal** • **Ratle** • **Pakal Dul** (Marusudar river - Chenab tributary) • **Kiru** • **Kwar**
- **On Jhelum and tributary:** **Uri** (Jhelum) • **Tulbul Navigation Project** (Jhelum - at the mouth of Wular Lake) • **Kishanganga** (Kishanganga/Neelum - Jhelum tributary)
- **On Indus and Suru:** **Nimoo Bazgo** (Indus - Leh) • **Chutak** (Suru - Kargil)
- **On Ravi:** **Thein Dam (Maharaja Ranjit Singh Dam)** • **Shahpurkandi Dam** (Punjab-Jammu-Kashmir border)
- **On Beas:** **Pong Dam** (Maharana Pratap Sagar) • **Pandoh Dam** (Himachal Pradesh)
- **On Sutlej:** **Nathpa Jhakri** • **Kol Dam** (Himachal Pradesh)

Points to Note

1. **Gobind Sagar vs Govind Ballabh Pant Sagar:** Gobind Sagar — Bilaspur (Himachal Pradesh), Sutlej (Bhakra Dam); Govind Ballabh Pant Sagar — **Sonbhadra (UP), Rihand river** [2]
2. **Chambal dams:** only **Gandhi Sagar is in MP (Mandsaur)**; Rana Pratap Sagar and Jawahar Sagar both in **Rajasthan** [1]
3. **Damodar Valley:** Tilaiya and Maithon not on the main Damodar — on the **Barakar river**; Panchet on the main **Damodar** [2]
4. **Kishanganga vs Baglihar:** Kishanganga is not on the Chenab — it is a **Jhelum tributary**; Baglihar, Salal, Dulhasti are on the Chenab [1]

Mnemonic (Chenab projects): "Chenab ka **Dulha Bagal** me **Sala** ko lekar **Raat** me bhaga" — Dulha = Dulhasti, Bagal = Baglihar, Sala = Salal, Raat = Ratle

Mnemonic (Betwa projects): "Betwa ki **Mata** ka **Raj Pari** jaisa hai" — Mata = Matatila, Raj = Rajghat, Pari = Parichha

One-liner Facts

- **Ken-Betwa Link:** **first** project under the National River Linking Project • part of **Panna Tiger Reserve** (MP) to be submerged • irrigation benefit to Bundelkhand PYQ 2
- **Sharda Canal:** UP's **largest** canal system • built 1920-1928 • on the Sharda (Kali) river from Banbasa (Champawat border) PYQ 1
- **Eastern Yamuna Canal:** UP's **oldest** canal • built by Shah Jahan • reorganised by the British in 1830 PYQ 1
- **Tanks vs tube-wells:** tanks used most in **Tamil Nadu** • tube-wells used most in the alluvial plains of North India (**UP**) PYQ 1
- **Polavaram vs Kaleshwaram:** Polavaram — Godavari, **Andhra Pradesh** • Kaleshwaram — Godavari, **Telangana** PYQ 1

15 Mineral and Energy Resources (India and State-wise Distribution)

- **Exam trend:** most frequent is **'mine ↔ mineral/state'** matching • at least 2-3 questions every year directly on mine names

Major Mineral Belts of India

- **North-Eastern Plateau Belt:** Chota Nagpur Plateau, Odisha Plateau, Chhattisgarh • **'Ruhr of India'** and 'mineral heartland of India' • coal, iron ore, manganese, mica, bauxite PYQ 2
- **South-Western Plateau Belt:** Karnataka, Goa, Tamil Nadu • high-grade iron ore (magnetite) • manganese • gold • lignite coal
- **North-Western Belt:** Aravalli (Rajasthan and Gujarat) • non-ferrous minerals (copper, lead, zinc, silver) • limestone • marble • gypsum • petroleum PYQ 1

Major Metallic and Non-metallic Minerals: Mines, Reserves and Production

Top Mineral-Producing States — not reserves

Odisha	Madhya Pradesh	Rajasthan
• Iron ore (over 50%) • Bauxite (over 70%) • Chromite-nickel (100%)	• Manganese • Copper • Diamond (100%)	• Lead-zinc (about 100%) • Silver (99%) • Tungsten · Gypsum
Karnataka	**Andhra Pradesh**	**Chhattisgarh**
Gold (about 99%, Hutti)	Mica	Tin (only producer)

Trap (reserves ≠ production): Manganese — reserves **Odisha**, production **MP** · Copper — reserves **Rajasthan**, production **MP** · Gold — primary reserves **Bihar**, production **Karnataka**

- **Iron ore — Odisha:** Badampahar, Gurumahisani, Sulaipat (Mayurbhanj) • Keonjhar • Bonai (Sundargarh) PYQ 2
- **Iron ore — Chhattisgarh:** **Bailadila** (Dantewada - exported to Japan via Visakhapatnam) • **Dalli-Rajhara** (Durg - supplies Bhilai) PYQ 2

- **Iron ore — Jharkhand:** Noamundi • Gua • Kiriburu (West Singhbhum)
- **Iron ore — Karnataka:** Bababudan Hills • Kudremukh (Chikmagalur) • Sandur-Hospet (Bellary) • Kemmangundi PYQ 2
- **Iron ore — Maharashtra:** Ratnagiri • Chandrapur
- **Iron ore reserves:** Haematite — **Odisha** • Magnetite — **Karnataka** PYQ 1
- **Iron ore production:** 1. **Odisha** (over 50%) • 2. Chhattisgarh • 3. Karnataka PYQ 2
- **Manganese — Madhya Pradesh:** Balaghat (**Bharweli** - Asia's largest underground manganese mine) • Chhindwara PYQ 2
- **Manganese — Maharashtra / Odisha:** Nagpur, Bhandara, Ratnagiri • Keonjhar, Sundargarh, Koraput, Kalahandi
- **Manganese reserves:** 1. **Odisha** • 2. Karnataka • 3. Madhya Pradesh PYQ 1
- **Manganese production:** 1. **Madhya Pradesh** • 2. Maharashtra • 3. Odisha PYQ 2
- **Copper — Madhya Pradesh: Malanjkhand** (Balaghat - country's largest open-cast copper mine) PYQ 3
- **Copper — Rajasthan: Khetri-Singhana** (Jhunjhunu) • Kho-Dariba (Alwar) • Chandrapura PYQ 2
- **Copper — Jharkhand:** Mosabani • Rakha • Ghatshila (East Singhbhum)
- **Copper reserves:** 1. **Rajasthan** • 2. Jharkhand • 3. Madhya Pradesh PYQ 1
- **Copper production:** 1. **Madhya Pradesh** • 2. Rajasthan • 3. Jharkhand PYQ 1
- **Bauxite — Odisha: Panchpatmali** (Koraput - largest deposit) • Kalahandi • Bolangir • Sambalpur PYQ 2
- **Bauxite — others:** Jharkhand (Lohardaga, Palamu, Ranchi) • MP (Amarkantak Plateau, Katni, Shahdol) • Chhattisgarh (Phutka Pahar, Bastar)
- **Bauxite reserves:** 1. **Odisha** (over 50%) • 2. Andhra Pradesh • 3. Gujarat
- **Bauxite production:** 1. **Odisha** (over 70%) • 2. Gujarat • 3. Jharkhand PYQ 2
- **Lead and zinc (twin minerals) — Rajasthan:** Zawar (Udaipur) • **Rampura-Agucha** (Bhilwara - world-class deposit) • Rajpura-Dariba (Rajsamand) • Chauth ka Barwara (Sawai Madhopur) PYQ 3
- **Lead-zinc reserves / production:** Rajasthan (about **89%** monopoly) • production Rajasthan (about **100%**) PYQ 1
- **Silver — mines:** Rajasthan (Zawar, Sindri, Chittorgarh - by-product of lead-zinc) • Karnataka (Kolar and Hutti - by-product with gold)
- **Silver reserves / production:** reserves 1. Rajasthan (87%) 2. Jharkhand • production 1. **Rajasthan (99%)** 2. Karnataka PYQ 1
- **Gold — mines:** Karnataka (**Kolar** and **Hutti**, Raichur - metallic gold) • Andhra Pradesh (Ramagiri, Anantapur) • Jharkhand (sands of Subarnarekha river - placer deposits) PYQ 2
- **Gold reserves / production:** primary gold ore reserves — **Bihar (Jamui, about 44%)** [1] • production — **Karnataka** (about 99%, Hutti mine)
- **Diamond — mines:** MP (**Panna** - Majhgawan and Hinota mines • Chhatarpur - Bunder Project) • Andhra Pradesh (Golconda, Vajrakarur) PYQ 2
- **Diamond reserves / production:** reserves 1. Madhya Pradesh (90%) 2. Andhra Pradesh • production **Madhya Pradesh (100%)**
- **Mica — Andhra Pradesh:** Nellore (**Gudur** mine - green mica) PYQ 2
- **Mica — Jharkhand: Koderma** ('Mica Capital of India') • Giridih • Hazaribagh (ruby mica) PYQ 2
- **Mica — Rajasthan:** Bhilwara • Ajmer
- **Mica reserves / production:** reserves 1. Andhra Pradesh 2. Rajasthan 3. Odisha • production 1. **Andhra Pradesh** 2. Rajasthan PYQ 2
- **Chromite and nickel:** Odisha — **Sukinda Valley** (Jajpur - over 96% of India's chromite and nickel) • Cuttack • Keonjhar PYQ 2
- **Chromite-nickel reserves / production:** Odisha (96%) • production Odisha (100%)
- **Tungsten: Degana** (Nagaur, Rajasthan) • reserves and production Rajasthan PYQ 1
- **Gypsum:** Hanumangarh • Bikaner • Nagaur • Jaisalmer (Rajasthan) • reserves and production Rajasthan
- **Tin (cassiterite):** Chhattisgarh — Dantewada and Bastar Plateau PYQ 2
- **Tin reserves / production:** reserves Haryana, Chhattisgarh • production **Chhattisgarh** (India's only tin-producing state)

Coal (Black Gold)

- **Types (decreasing carbon content):** Anthracite > Bituminous > Lignite > Peat PYQ 3
- **Anthracite: 80-95%** carbon • best quality, smokeless • in India only in **Jammu-Kashmir (Kalakot)** in limited quantity
- **Bituminous:** 55-80% carbon • **most common** in India • in Gondwana rocks (about 98% of total coal) PYQ 2
- **Lignite:** 40-55% carbon • brown coal • **Tertiary** period • **Neyveli (Tamil Nadu - largest reserves and production)** • Palana and Barmer (Rajasthan) • Kutch (Gujarat) • Makum (Assam) PYQ 2
- **Peat:** less than 40% carbon • lowest grade • highest moisture and smoke
- **Top reserve states:** 1. **Jharkhand** • 2. Odisha • 3. Chhattisgarh PYQ 2
- **Top producing states (latest):** 1. Odisha / Chhattisgarh (by quantity **Odisha**, by value **Chhattisgarh**) • 3. Jharkhand • 4. Madhya Pradesh PYQ 2
- **Coking coal production: Jharkhand** tops
- **Coalfield ↔ state (matching):** PYQ 5
- **Jharkhand: Jharia** (India's largest coking coal field) • Bokaro • Giridih • Karanpura • Ramgarh • Daltonganj
- **West Bengal: Raniganj** • start of modern coal mining in India (**1774**)
- **Odisha:** Talcher (Brahmani valley) • Ib river valley (Sambalpur-Jharsuguda)
- **Chhattisgarh: Korba** (Hasdeo valley) • Tatapani-Ramkola • Bishrampur • Jhilmili • Chirimiri
- **Madhya Pradesh: Singrauli** (spread over Sidhi-Singrauli of MP and Sonbhadra of UP) • Sohagpur • Umaria • Pench-Kanhan valley
- **Telangana: Singareni** (Godavari valley)
- **Maharashtra:** Wardha valley • Kamptee • Chandrapur
- **Meghalaya and Assam (Tertiary coal):** Makum, Nazira (Assam) • Cherrapunji, Langrin (Meghalaya)
- **Arunachal Pradesh:** Namchik-Namphuk PYQ 2

Petroleum (Crude Oil) and Natural Gas

- **Occurrence:** **Tertiary** sedimentary rocks (offshore and onshore basins) • **anticlines** and fault traps PYQ 1
- **Major oil fields:** PYQ 4
- **Assam (Brahmaputra valley):** **Digboi** (oldest oil well of Asia and India - 1889/1901) • Naharkatiya • Moran-Hugrijan • Rudrasagar • Lakwa • Surma valley
- **Gujarat (Khambhat and Ankleshwar basins):** **Ankleshwar** • Kalol • Lunej • Navagam • Kosamba • Mehsana • Sanand
- **Maharashtra (western offshore):** **Mumbai High** (176 km from Mumbai, Arabian Sea • discovered 1974, production 1976 • drilling platform **'Sagar Samrat'** • country's largest oil-producing area) • Bassein • Aliabet PYQ 2
- **Rajasthan (Barmer-Sanchore basin):** oil — **Mangala** • Bhagyam • Aishwariya • Saraswati • Rajeshwari • gas — **Raageshwari** PYQ 2
- **Eastern offshore:** **Krishna-Godavari basin** (Ravva offshore field) • huge natural gas reserves (**D-6 block**)
- **Crude oil production order:** 1. Offshore (Mumbai High) • 2. Rajasthan • 3. Gujarat • 4. Assam PYQ 2
- **Natural gas production order:** 1. Offshore • 2. Assam • 3. Rajasthan PYQ 1
- **HBJ (Hazira-Vijaipur-Jagdishpur) pipeline:** **GAIL** (founded 1984) • **1,750 km** • first cross-country gas pipeline • Hazira (Gujarat) → Vijaipur (MP) → Jagdishpur and Aonla (UP) PYQ 2

Atomic Minerals and Nuclear Power Plants

- **Uranium — Jharkhand:** Singhbhum Thrust Belt • **Jaduguda** (India's first uranium mine - 1967) • Narwapahar • Turamdih • Bagjata • Bhatin PYQ 3
- **Uranium — Andhra Pradesh (largest reserves):** **Tummalapalle** (Kadapa - among the world's largest uranium reserves) • Lambapur PYQ 2
- **Uranium — Meghalaya / Rajasthan:** **Domiasiat** (Wahchil hills) • Rohil (Sikar)
- **Thorium:** India has the world's **largest reserves** • from **monazite sands** • coastal areas of Kerala (Palakkad, Kollam), Tamil Nadu, Odisha PYQ 3
- **Major operational nuclear plants:** PYQ 4
- **Tarapur (TAPS):** Maharashtra • India's **first atomic power station** (1969) • **USA** assistance
- **Rawatbhata (RAPS):** Rajasthan (near Kota) • **Canada** cooperation (1973) • start of heavy water reactors
- **Kalpakkam (Madras - MAPS):** Tamil Nadu • India's first plant on fully **indigenous technology** (1984) • 'KAMINI' and Fast Breeder Reactor here
- **Narora (NAPS):** Uttar Pradesh (Bulandshahr) • safe indigenous design in seismic zone-4 (1991)
- **Kakrapar (KAPS):** Gujarat (near Surat) • **700 MW** indigenous Pressurised Heavy Water Reactor
- **Kaiga (KGS):** Karnataka (Uttara Kannada) • amid forests on the Kali river (2000)
- **Kudankulam (KKNPP):** Tamil Nadu (Tirunelveli) • **Russian** cooperation • India's **highest-capacity** nuclear plant
- **Jaitapur and Gorakhpur (proposed):** Jaitapur (Maharashtra) — proposed with **French** assistance • Gorakhpur — under construction in **Haryana (Fatehabad)** PYQ 1

Mineral Resources of Uttar Pradesh

- **Exam trend:** 1-2 questions every year on UP mineral-district matching PYQ 6
- **Sonbhadra:** coal (Singrauli area) • limestone (Kajrahat and Rohtas areas) • dolomite (Bari) • marble • gold (Hardi area and Son Pahari)
- **Lalitpur:** **uranium** (only district in UP) • **copper** (Sonrai) • **rock phosphate** • **iron ore** • **gold** (Bhikhampur) • **platinum**
- **Mirzapur:** **andalusite** • **pyrophyllite** • **asbestos** • sandstone • marble • diamond (forest area) • Bansi and Makri-Khoh (fire clay)
- **Prayagraj (Shankargarh), Chandauli (Chakia) and Banda:** **glass sand (silica sand)** • UP **leads** the country in production
- **Banda and Chitrakoot:** bauxite (Rajgawan) • ochre (geru) • diamond (Kalinjar) • dolomite
- **Hamirpur and Mahoba:** **diaspore** (UP among top producing states) • **pyrophyllite** • **gypsum** • soapstone (talc)
- **Jhansi:** potash salts • soapstone • glass sand • diaspore

Points to Note

1. **Singrauli vs Singareni:** Singrauli — MP and UP (Sonbhadra); Singareni — **Telangana (Godavari valley)** [2]
2. **Dariba mines: Kho-Dariba** (Alwar) — **copper**; **Rajpura-Dariba** (Rajsamand) — **lead and zinc** [1]
3. **Two mines of Balaghat: Malanjkhand** — copper; **Bharweli** — manganese (both Balaghat, MP) [1]
4. **Gorakhpur nuclear plant:** not in Gorakhpur, UP — in Gorakhpur village of **Fatehabad district, Haryana** [1]

Mnemonic (Lalitpur minerals): "Lalit ne **taambe** ke bartan me **uranium** aur **rock phosphate** se **sona** banaya" — copper, uranium, rock phosphate, gold, platinum = Lalitpur only

Mnemonic (Odisha iron ore): "Mayur ne **badaam** khakar **guru** ko **sulai** aur **kyon Bonai** gaya?" — Mayur = Mayurbhanj, Badaam = Badampahar, Guru = Gurumahisani, Sulai = Sulaipat, Kyon = Keonjhar, Bonai = Bonai

One-liner Facts

- **Gondwana coal:** about **98%** of India's coal reserves • Damodar, Mahanadi, Son, Godavari valleys • mainly **bituminous** PYQ 1
- **Bailadila vs Kudremukh:** Bailadila (Chhattisgarh) — high-grade **haematite**, exported to Japan via Visakhapatnam • Kudremukh (Karnataka) — was exported via **New Mangalore** port PYQ 2
- **Barmer basin:** Mangala, Bhagyam, Aishwariya — among India's largest **onshore** crude oil producing fields PYQ 1
- **Geothermal energy sites:** **Puga valley** and Chumathang (Ladakh) • **Manikaran** (Parvati valley, HP) • Tattapani (Odisha) • Anhoni (MP) • Tatapani (Balrampur, Chhattisgarh) PYQ 3
- **Wind vs solar capacity:** wind installed capacity — **Gujarat** and Tamil Nadu top • solar — **Rajasthan** and Gujarat top PYQ 2

16 Major Industries and Industrial Regions of India

- **Commission's focus:** **partner countries** of iron-steel plants • locations of aluminium companies • fertilizer plants • **nicknames** of industrial cities

Iron and Steel Industry (Basic Industry)

- **1874 — Bengal Iron Works:** on Barakar river at **Kulti (West Bengal)** • beginning of modern iron-steel industry PYQ 1
- **1907 — TISCO (Tata Iron and Steel Company):** Jamsetji Tata • **Sakchi (Jamshedpur, Jharkhand)** • confluence of Subarnarekha and Kharkai rivers PYQ 2
- **1923 — Visvesvaraya Iron and Steel Plant (VISL): Bhadravati (Karnataka)** • iron ore: Bababudan Hills • hydropower: Sharavati project • initially **charcoal** used due to lack of coal PYQ 2
- **Public steel plants of 2nd and 3rd Five Year Plans:** order — plant • state/river • foreign collaboration • iron ore • coal/power PYQ 5
- **Rourkela (1959):** Odisha (Brahmani river) • collaboration **Germany** (Krupps and Demag) • ore: Keonjhar and Bonai (Sundargarh) • coal: Jharia and Talcher • power: Hirakud project
- **Bhilai (1959):** Chhattisgarh (Durg district) • collaboration **former Soviet Union (Russia)** • ore: **Dalli-Rajhara** mines [1] • coal: Korba and Jharia • Korba thermal power station
- **Durgapur (1962):** West Bengal (Damodar river) • collaboration **Britain (United Kingdom)** • ore: Noamundi and Bolani • coal: Raniganj and Jharia • power: Damodar Valley Corporation (DVC)
- **Bokaro (1964 — 3rd Plan):** Jharkhand (Damodar-Bokaro confluence) • collaboration **former Soviet Union (Russia)** • ore: Kiriburu (Odisha) • coal: Jharia — based on **least transport cost** principle PYQ 1
- **Visakhapatnam (1992):** Andhra Pradesh • India's **first shore-based (coastal)** steel plant [1] • ore: Bailadila (Chhattisgarh) • coal: Damodar Valley and coking coal imported from Australia
- **Salem (1982):** Tamil Nadu • famous for **stainless steel** • ore: local (Kanjamalai Hills) • coal: Neyveli lignite
- **SAIL (Steel Authority of India Limited):** established **January 1973** • manages most public-sector steel plants

Aluminium Industry (Power-intensive Industry)

Foreign Partners: Steel, Aluminium

matching

Steel plants		Aluminium	
Rourkela 1959 · Odisha	Germany	BALCO Korba	Russia
Bhilai 1959 · Chhattisgarh	Russia	NALCO Damanjodi, Angul	France
Durgapur 1962 · W. Bengal	Britain	HINDALCO Renukoot (UP)	USA
Bokaro 1964 · Jharkhand	Russia	INDALCO Muri, Hirakud	Canada
Visakhapatnam (1992) first shore-based		MALCO Mettur	Italy
Salem (1982) stainless steel		Vedanta Jharsuguda	Germany

Trap: Bhilai and Bokaro = **Russia** · Rourkela = **Germany** · Durgapur = **Britain**

- **Location factor:** where **cheap electricity** and **bauxite** are available • Commission asks matching of location and partner country PYQ 4
- **BALCO (Bharat Aluminium Company):** **Korba** (Chhattisgarh) and Ratnagiri (Maharashtra) • collaboration **former Soviet Union (Russia)**
- **NALCO (National Aluminium Company — 1981):** **Damanjodi** (Koraput) and **Angul** (Odisha) • collaboration **France** • India's largest integrated aluminium company
- **HINDALCO (Hindustan Aluminium Corporation):** **Renukoot** (Sonbhadra, Uttar Pradesh) • power: Rihand Dam • collaboration **United States of America** (Birla Group) PYQ 2
- **INDALCO (Indian Aluminium Company):** **JK Nagar** (West Bengal) • **Muri** (Jharkhand) • **Hirakud** (Odisha) • **Alwaye** (Kerala) • collaboration **Canada**
- **MALCO (Madras Aluminium Company):** **Mettur** (Salem district, Tamil Nadu) • collaboration **Italy**
- **Vedanta Aluminium:** **Jharsuguda** and Lanjigarh (Odisha) • collaboration **Germany**

Cotton Textile Industry (India's largest and oldest organised industry)

- **First modern cotton mill: 1818** • **Fort Gloster** near Kolkata • failed
- **First successful factory: 1854 Mumbai** • **Cowasji Nanabhai Davar** PYQ 2
- **Major centres and nicknames:** Mumbai • Ahmedabad • Coimbatore • Kanpur PYQ 3
- **Mumbai: 'Cotton textile capital of India'** • 'Cottonopolis of India'
- **Ahmedabad (on Sabarmati): 'Manchester of India'** • **'Boston of the East'** PYQ 2
- **Coimbatore (Tamil Nadu): 'Manchester of South India'** • Tamil Nadu has the **most spinning mills** in the country
- **Kanpur (Uttar Pradesh): 'Manchester of North India'**

Jute Industry

- **First factory: 1855** • **Rishra** near Kolkata (on Hooghly bank) • George Auckland PYQ 2
- **Hooghly industrial region:** about **more than 80%** of India's jute mills • both banks of Hooghly in West Bengal (Bansberia to Birlanagar)
- **Major Hooghly centres:** Titagarh • Jagatdal • Budge Budge • Howrah • Rishra
- **Jute mills in Uttar Pradesh: Sahjanwa (Gorakhpur)** • Kanpur PYQ 1
- **International Jute Organisation:** established 1984 • HQ **Dhaka (Bangladesh)** PYQ 1

Sugar and Paper Industry

- **Sugar industry: weight-losing** and seasonal industry
- **First modern sugar mill: 1903** • **Pratappur (Deoria, Uttar Pradesh)** PYQ 2
- **Reasons for shift towards South India:** analytical question PYQ 2
- **Reason 1 — Sucrose:** tropical climate • southern sugarcane has **higher sucrose content** (10.5 to 12%)
- **Reason 2 — Crushing season:** longer in South (**7-8 months**) • only 4-5 months in North
- **Reason 3 — Cooperatives:** more efficient management of cooperative sugar mills in South (especially **Maharashtra**)
- **Paper industry — first successful factory: 1879 Lucknow** • 1881 Titagarh (West Bengal) PYQ 3
- **Newsprint: Nepanagar** (Burhanpur district, Madhya Pradesh — 1947)
- **Currency note paper (Security Paper Mill): Hoshangabad (Narmadapuram, Madhya Pradesh)**

Fertilizer, Cement, Heavy Engineering and Petrochemical Centres

- **First superphosphate fertilizer factory: 1906 • Ranipet (Tamil Nadu)**
- **First large public fertilizer plant after Independence: 1951 • Sindri (Dhanbad, Jharkhand)** PYQ 2
- **Major fertilizer centres:** Aonla (Bareilly) • Phulpur (Prayagraj — IFFCO) • Jagdishpur (Amethi) • Gorakhpur • Talcher (Odisha — **coal-based**) • Namrup (Assam) • Ramagundam (Telangana) • Vijaipur (Madhya Pradesh) • Thal-Vaishet and Trombay (Maharashtra) PYQ 2
- **First modern cement factory: 1904 • Chennai**
- **Cement centres:** raw material limestone and gypsum • Vindhyan Range (Satna • Katni • Kaimur • Churk • Dala — Sonbhadra) • Rajasthan (Chittorgarh • Sawai Madhopur • Lakheri) PYQ 2
- **Specific industrial centres and products:** direct matching questions PYQ 5
- **Rail locomotives:** Chittaranjan (West Bengal) • Varanasi (DLW/BLW)
- **Rail coaches:** Perambur (Chennai) • Kapurthala (Punjab) • Rae Bareli (Uttar Pradesh)
- **HAL (Hindustan Aeronautics Limited — aircraft):** Bengaluru • Kanpur • Koraput (Odisha) • Nashik
- **Shipbuilding:** Visakhapatnam (Hindustan Shipyard) • Kochi (Cochin Shipyard) • Mazagon (Mumbai) • Garden Reach (Kolkata)
- **Antibiotics (Hindustan Antibiotics Limited):** Pimpri (Pune) • Rishikesh (Uttarakhand)
- **Chemical fertilizers:** Sindri (Jharkhand) • Nangal (Punjab)
- **Uranium processing: Jaduguda (Jharkhand)**
- **Petrochemicals and oil refineries:** Hazira • Jamnagar • Koyali (Gujarat) • Haldia (West Bengal)
- **Jamnagar:** Reliance's **world's largest single-location oil refinery**
- **Silk textile industry:** Kanchipuram (Tamil Nadu) • Mysuru (Karnataka) • Varanasi (Uttar Pradesh) • Bhagalpur (Bihar — **Tasar**) • Sualkuchi (Assam — **Muga**) PYQ 2
- **Muga silk:** India's world **monopoly** • only in **Assam**

Points to Note

1. **Bhilai and Bokaro = Russia • Rourkela = Germany • Durgapur = Britain** (confusion over Rourkela-Durgapur partners) [2]
2. **Nepanagar = newsprint • Hoshangabad (Narmadapuram) = currency note paper (security paper)** — both in Madhya Pradesh • Bank Note Press: **Dewas (Madhya Pradesh)** and **Nashik (Maharashtra)** [2]
3. **Churk and Dala (Sonbhadra) = cement** • Sonbhadra's **Renukoot = aluminium (HINDALCO) • Obra and Anpara = thermal power** [2]
4. **Mathura refinery:** the **only** refinery of Uttar Pradesh (Indian Oil Corporation) • crude via pipeline from **Salaya** (Gulf of Kutch, Gujarat) • SO2 gas → 'acid rain' on Taj Mahal (**stone leprosy**) [2]

Mnemonic (steel plants and partners): "Bhiru, Boru, Dubri, Raj" — Bhiru = Bhilai-Russia, Boru = Bokaro-Russia, Dubri = Durgapur-Britain, Raj = Rourkela-Germany

Mnemonic (HINDALCO): "Hind ke Renukoot me America" — HINDALCO = Renukoot = USA

Mnemonic (BALCO): "Balak Korba me Rus gaya" — BALCO = Korba = Russia

Mnemonic (NALCO): "Naal Daman-Angul me Phans gayi" — NALCO = Damanjodi/Angul = Phans (France)

One-liner Facts

- **Textile industry: second largest** employment provider after agriculture PYQ 1
- **Core sector (eight core industries) weightage, descending: Refinery products (highest — 28.04%)** > Electricity > Steel > Coal > Crude oil > Natural gas > Cement > **Fertilizers (lowest — 2.63%)** PYQ 2
- **Pithampur (Dhar district, Madhya Pradesh): 'Detroit of India'** (automobile industry) • globally **Chennai** = Detroit of South Asia PYQ 1
- **All four commercial silk varieties:** Mulberry • Tasar • Eri • Muga — all produced in India • **Mulberry** highest (**Karnataka** top) PYQ 1
- **World-famous centres of Uttar Pradesh:** Bhadohi (Sant Ravidas Nagar) and Mirzapur — carpets • Firozabad — glass bangles • Aligarh — locks • Moradabad — brassware • Meerut — sports goods PYQ 2

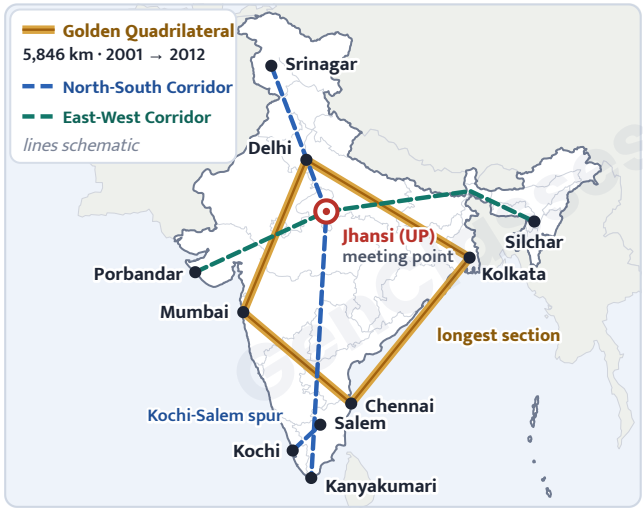
17 Transport and Communication (Road, Rail, Water and Air Transport)

- **Direct questions:** National Waterways (especially 1 to 6) • port features (deepest, tidal, artificial) • railway zone headquarters • matching of Golden Quadrilateral/corridors

Road Transport (National Highways and Major Projects)

Golden Quadrilateral and Corridors

NHAI



- **India's road network: second largest in world** after USA • total length: **Maharashtra** first, **Uttar Pradesh** second • Maharashtra and Uttar Pradesh top in National Highway length too PYQ 2
- **Major National Highways (new numbering system):** PYQ 3
- **NH-44 (longest): Srinagar to Kanyakumari** • about 4,112 km • formed by merging old NH-1A, 1, 2, 3, 75, 26 and 7 • 11 states and 1 Union Territory
- **NH-27 (second longest — East-West Corridor): Porbandar (Gujarat) to Silchar (Assam)** • 3,507 km

- **NH-48:** Delhi to Chennai • via Mumbai and Bengaluru • old NH-8 and NH-4
- **NH-19:** Delhi to Kolkata • via Agra, Kanpur, Prayagraj, Varanasi • major part of historic **Grand Trunk Road** • old NH-2 PYQ 1
- **NH-16:** Kolkata to Chennai • parallel to east coast • old NH-5
- **NH-66:** Panvel (Mumbai) to Kanyakumari • parallel to west coast • old NH-17
- **Shortest National Highway:** **NH-548** (Maharashtra, Kalamboli to Karanja — only 5 km) • **NH-118** (Jharkhand, Asanbani to Jamshedpur — 5 km) PYQ 2
- **Earlier shortest:** **NH-47A** on Willingdon Island, Kerala (new name NH-966B — 6 km) • if 47A appears in Commission's options, choose it
- **Golden Quadrilateral and Corridor project:** operated by NHAI PYQ 3
- **Golden Quadrilateral (5,846 km):** 4/6 lane project linking **Delhi** • **Mumbai** • **Chennai** • **Kolkata** • started by Atal Bihari Vajpayee in **2001** • completed **2012** • longest stretch **Chennai-Kolkata**
- **North-South Corridor:** Srinagar (Jammu and Kashmir) to Kanyakumari (Tamil Nadu) • including Kochi-Salem spur
- **East-West Corridor:** Silchar (Assam) to Porbandar (Gujarat)
- **Intersection point:** North-South and East-West corridors cross at **Jhansi**, Uttar Pradesh PYQ 3
- **Border Roads Organisation (BRO):** established **May 1960** • under Ministry of Defence PYQ 2
- **Atal Tunnel (Rohtang Tunnel — 9.02 km):** links Manali to Lahaul-Spiti (Leh) • in **Pir Panjal** Range
- **Bharatmala Pariyojana (2017):** **umbrella** project linking border, coastal and backward areas and ports with highways PYQ 1

Rail Transport (Railway Zones, Konkan Railway and Dedicated Freight Corridor)

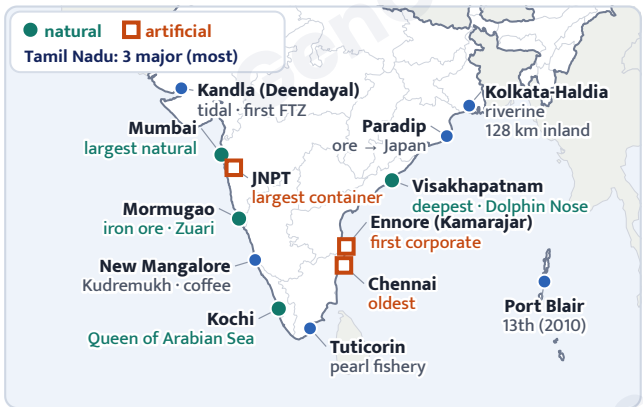
- **Beginning:** **16 April 1853** • Lord Dalhousie's period • Mumbai (Bori Bunder) to Thane (34 km)
- **18 zones of Indian Railways and headquarters:** very important matching question PYQ 4
- **2 headquarters in Uttar Pradesh:** **North Central Railway (Prayagraj)** • **North Eastern Railway (Gorakhpur)** PYQ 2
- **3 headquarters in Kolkata:** Eastern Railway • South Eastern Railway • Kolkata Metro Railway
- **2 headquarters in Mumbai:** Central Railway (Chhatrapati Shivaji Maharaj Terminus) • Western Railway (Churchgate)
- **Northeast Frontier Railway:** **Maligaon (Guwahati)**
- **East Central Railway:** **Hajipur (Bihar)**
- **West Central Railway:** **Jabalpur (Madhya Pradesh)**
- **South East Central Railway:** **Bilaspur (Chhattisgarh)**
- **East Coast Railway:** **Bhubaneswar (Odisha)**
- **South Western Railway:** **Hubballi (Karnataka)**
- **North Western Railway:** **Jaipur (Rajasthan)**
- **South Central Railway:** **Secunderabad (Telangana)**
- **South Coast Railway (18th and newest zone):** **Visakhapatnam (Andhra Pradesh)** PYQ 1
- **Konkan Railway (established 1998):** **Roha (Maharashtra) to Mangaluru (Karnataka)** • 741 km • parallel to Western Ghats (Sahyadri) PYQ 3
- **4 states benefiting from Konkan Railway:** Maharashtra • Goa • Karnataka • **Kerala**
- **UNESCO World Heritage Mountain Railways:** Darjeeling Himalayan Railway (1999) • Nilgiri Mountain Railway (2005) • Kalka-Shimla Railway (2008) PYQ 1
- **Chhatrapati Shivaji Maharaj Terminus (Mumbai):** also a UNESCO heritage site
- **Eastern Dedicated Freight Corridor (EDFC):** Ludhiana (Sahnewal, Punjab) to Dankuni (West Bengal) • passes through Dadri, Khurja, Kanpur, Prayagraj, Pt. Deen Dayal Upadhyaya Junction in Uttar Pradesh
- **Western Dedicated Freight Corridor (WDFC):** Dadri (Gautam Buddh Nagar, Uttar Pradesh) to Jawaharlal Nehru Port (Mumbai) PYQ 1
- **Junction of both corridors:** **Dadri/Khurja**, Uttar Pradesh

National Waterways (Commission's favourite topic)

- **Inland Waterways Authority of India (IWAI):** established **1986** • HQ **Noida (Uttar Pradesh)** PYQ 2
- **National Waterways Act, 2016:** total **111 National Waterways** declared
- **Major waterways:** asked in previous exams PYQ 5
- **NW-1:** **Prayagraj to Haldia** • **first** waterway declared in 1986 • Ganga-Bhagirathi-Hooghly system • **1,620 km** • India's **longest** operational waterway • Uttar Pradesh, Bihar, Jharkhand, West Bengal PYQ 2
- **NW-2:** **Sadiya to Dhubri** (1988) • Brahmaputra river • 891 km • Assam PYQ 2
- **NW-3:** **Kollam to Kottapuram** (1993) • West Coast Canal, Champakara and Udyogmandal canals • 205 km • Kerala • India's **first 24-hour navigable** waterway PYQ 1
- **NW-4:** **Kakinada to Puducherry** (with Krishna-Godavari extension) • Godavari and Krishna rivers and Kakinada-Puducherry canal • 1,078 km (extended 2,890 km) • Andhra Pradesh, Tamil Nadu, Telangana, Puducherry PYQ 1
- **NW-5:** **Talcher to Dhamra** (up to Paradip) • Brahmani-Mahanadi delta and East Coast Canal • 623 km • Odisha and West Bengal
- **NW-6:** **Lakhipur to Bhanga** • Barak river • 121 km • Assam PYQ 2
- **NW-110:** Yamuna river • Delhi, Haryana, Uttar Pradesh
- **NW-40:** Ghaghara (Saryu) river • Uttar Pradesh and Bihar

India's 13 Major Ports (West Coast vs East Coast)

13 Major Ports: West vs East Coast



- **Number:** 13 major • more than 200 minor/intermediate
- **Control:** major ports — Union Government (Ministry of Ports, Shipping and Waterways) • minor — State Governments (Concurrent List)

- **Sagarmala Project (2015):** port-led development and modernisation PYQ 2
- **Port facts:** asked from 2010 to 2025
- **1. Kandla (Deendayal Port):** Gujarat (Gulf of Kutch — west coast) • **tidal port** • developed after Partition to compensate loss of Karachi • **first Free Trade Zone (FTZ — 1965)** PYQ 3
- **2. Mumbai Port:** Maharashtra (west coast — Salsette Island) • India's **largest natural port** PYQ 1
- **3. Jawaharlal Nehru Port (Nhava Sheva):** Maharashtra (Navi Mumbai — west coast) • built to decongest Mumbai Port • **largest container port** • fully mechanised modern artificial port PYQ 2
- **4. Mormugao:** Goa (west coast) • on **estuary of Zuari river** • famous for Goa's **iron ore** export PYQ 2
- **5. New Mangalore:** Karnataka (west coast) • iron ore from **Kudremukh** • export of Karnataka's **coffee** PYQ 1
- **6. Kochi (Cochin):** Kerala (west coast) • natural port on **Willingdon Island** at mouth of **Vembanad Kayal (lagoon)** • **'Queen of the Arabian Sea'** • spices and tea export PYQ 3
- **7. Tuticorin (V.O. Chidambaranar Port):** Tamil Nadu (east coast — Gulf of Mannar) • shallow sea, big ships anchor 5 miles away • **pearl fishery** PYQ 1
- **8. Chennai Port:** Tamil Nadu (east coast) • **oldest** and **second largest artificial** open-sea port • all-weather PYQ 2
- **9. Ennore (Kamarajar Port):** Tamil Nadu (north of Chennai — east coast) • India's **first corporatised** port (registered under Companies Act) PYQ 2
- **10. Visakhapatnam:** Andhra Pradesh (east coast) • **deepest** and best **landlocked** natural port • **'Dolphin's Nose'** hill protects from cyclones and strong waves PYQ 3
- **11. Paradip:** Odisha (east coast — Mahanadi delta) • deep port to export iron ore and coal of Odisha and Jharkhand to **Japan** PYQ 1
- **12. Kolkata-Haldia (Syama Prasad Mookerjee Port):** West Bengal (east coast) • only major **riverine** port • on Hooghly **128 km** inland from sea • ships come with tidal help • Haldia is subsidiary port PYQ 3
- **13. Port Blair:** Andaman and Nicobar • declared 13th major port in **June 2010**
- **Vadhavan Port:** Palghar (Maharashtra) • being developed as new major **deep-water** port
- **Major private and non-major ports:** asked in exams PYQ 3
- **Mundra (Kutch, Gujarat):** India's **largest private** commercial port (Adani Group) • surpassed Kandla in total cargo handling
- **Dahej (Gujarat):** **'India's first Chemical Port'** PYQ 2
- **Pipavav (Gujarat):** India's **first private-sector** port PYQ 1
- **Alang (near Bhavnagar, Gujarat):** world's **largest ship-breaking yard**
- **Dhamra and Gopalpur:** Odisha
- **Karaikal:** Puducherry
- **Gangavaram and Krishnapatnam:** Andhra Pradesh • **Gangavaram** = India's **deepest non-major** port PYQ 1

Points to Note

1. **Natural:** Mumbai, Kochi, Mormugao, Visakhapatnam • **Artificial:** Chennai, Ennore, Nhava Sheva (JNPT) [2]
2. **Kandla** = **tidal** (Gulf of Kutch) • **Kolkata** = **riverine and tidal** (Hooghly) [1]
3. **Most major ports: Tamil Nadu — 3** (Chennai, Ennore, Tuticorin) • Maharashtra — 2 (Mumbai, JNPT) • most minor/intermediate ports in **Maharashtra** [1]
4. **Uttar Pradesh — first state with 5 international airports:** Chaudhary Charan Singh (Amausi, Lucknow) • Lal Bahadur Shastri (Babatpur, Varanasi) • Kushinagar • Maharishi Valmiki (Ayodhya) • Noida International Airport (Jewar, Gautam Buddh Nagar — among Asia's largest) [3]

Mnemonic (river systems of NW 1 to 6): "Ganga aur Brahm ne Kerala me Godavari-Krishna aur Brahmani ke saath Barak paar kiya" — Ganga = NW-1, Brahm = NW-2, Kerala = NW-3, Godavari-Krishna = NW-4, Brahmani = NW-5, Barak = NW-6

Mnemonic (east coast ports, north to south): "Kallu ne Paara peekar Vishakha ko Anna aur Chennai ke saath Tota dikhaya" — Kallu = Kolkata-Haldia, Paara = Paradip, Vishakha = Visakhapatnam, Anna = Ennore, Chennai = Chennai, Tota = Tuticorin

One-liner Facts

- **Total navigable length of inland waterways:** **Uttar Pradesh** first • West Bengal second PYQ 1
- **Beginning of air transport:** **1911** • Allahabad (Prayagraj) to Naini • world's **first airmail service** PYQ 2
- **Cochin International Airport (Kerala):** world's **first fully solar-powered** airport PYQ 2
- **Longest railway length:** **Uttar Pradesh** • borders 9 states and 1 Union Territory • major National Highways and freight corridors pass through PYQ 1
- **Sethusamudram Project:** proposed shipping canal linking **Palk Bay** and **Gulf of Mannar** • reduces sea distance between east and west coasts (no circumnavigation of Sri Lanka) PYQ 1

Part 4: Geography of Uttar Pradesh

18 Geographical Location, Extent and Boundaries of Uttar Pradesh PYQ 28

Location and Extent

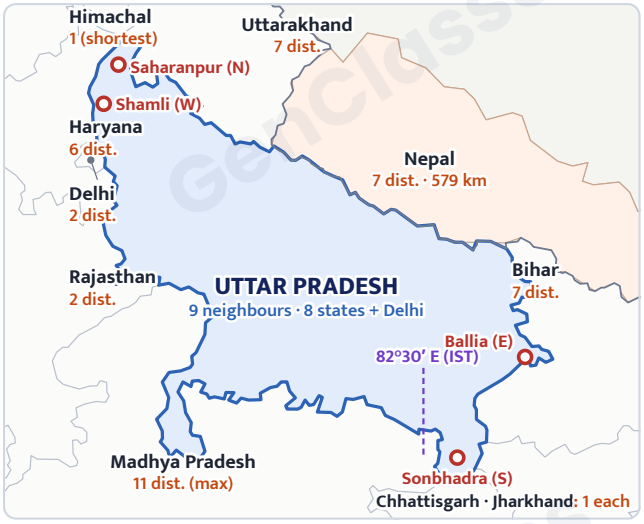
- **Latitudinal extent:** 23°52′ N to 30°24′ N • total **6°32′**
- **Longitudinal extent:** 77°05′ E to 84°38′ E • total **7°33′**
- **Total area:** 2,40,928 sq km • **7.33%** of India (32,87,263 sq km)
- **Rank by area:** **Fourth** • 1. Rajasthan • 2. Madhya Pradesh • 3. Maharashtra • 4. Uttar Pradesh
- **Length-breadth:** East-West **650 km** • North-South **240 km** • elongated shape (like a long corridor)

Boundaries

- **International border:** only country **Nepal** • about **579 km**
- **7 districts bordering Nepal (East → West):** Maharajganj • Siddharthnagar • Balrampur • Shravasti • Bahraich • Lakhimpur Kheri • Pilibhit PYQ 9
- **States/UTs touching UP:** total **9** • 8 states + 1 UT (Delhi)
- **Bordering Uttarakhand (7 districts):** Saharanpur • Muzaffarnagar • Bijnor • Moradabad • Rampur • Bareilly • Pilibhit
- **Bordering Himachal Pradesh (1 district):** Saharanpur • **shortest** state boundary

- **Bordering Haryana (6 districts):** Saharanpur • Shamli • Baghpat • Gautam Buddh Nagar • Aligarh • Mathura
- **Bordering Delhi (UT) (2 districts):** Ghaziabad • Gautam Buddh Nagar

UP: 9 neighbours + Nepal number = bordering districts



- **Bordering Rajasthan (2 districts):** Mathura • Agra
- **Bordering Madhya Pradesh (11 districts, maximum):** Agra • Etawah • Jalaun • Jhansi • Lalitpur • Mahoba • Banda • Chitrakoot • Prayagraj • Mirzapur • Sonbhadra PYQ 7
- **Bordering Chhattisgarh (1 district):** Sonbhadra
- **Bordering Jharkhand (1 district):** Sonbhadra
- **Bordering Bihar (7 districts):** Sonbhadra • Chandauli • Ghazipur • Ballia • Deoria • Kushinagar • Maharajganj

Four Extreme Districts and Area PYQ 12

- **Easternmost district: Ballia** • border with Bihar • near Ganga-Ghaghara confluence region
- **Westernmost district: Shamli** • natural boundary with Haryana formed by Yamuna
- **Northernmost district: Saharanpur** • touches 3 states (Uttarakhand, Himachal Pradesh, Haryana) • in Shivalik foothills
- **Southernmost district: Sonbhadra** • only district in India touching 4 states (Madhya Pradesh, Chhattisgarh, Jharkhand, Bihar)
- **4 largest districts by area:** 1. **Lakhimpur Kheri** (7,680 sq km, largest) • 2. Sonbhadra • 3. Hardoi • 4. Sitapur
- **4 smallest districts by area:** 1. **Hapur** (660 sq km, smallest) • 2. Ghaziabad • 3. Bhadohi (Sant Ravidas Nagar) • 4. Shamli

Mnemonic (7 districts bordering Nepal): "Peeli Kheer Bahraich se Shravasti Balram Siddharth Maharaj laaye" — Peeli = Pilibhit, Kheer = Lakhimpur Kheri, Bahraich, Shravasti, Balram = Balrampur, Siddharth = Siddharthnagar, Maharaj = Maharajganj

Mnemonic (extreme districts in four directions): "Saharan ke bete (Son) ne Ballia ki shaam dekhi" — North = Saharanpur, South = Sonbhadra, East = Ballia, West = Shamli

Points to Note

1. **Lalitpur vs Sonbhadra:** Lalitpur surrounded on three sides by **Madhya Pradesh** • touches only one UP district (**Jhansi**); Sonbhadra touches **4 states**
2. **District touching most districts:** earlier Budaun (9 districts) — after formation of Sambhal, **Hardoi and Budaun 8 each** • Lucknow **5 districts**; beware options of Sonbhadra (4 states) and Lalitpur (minimum, 1 district — Jhansi)
3. **8 UP districts in NCR:** Meerut • Ghaziabad • Gautam Buddh Nagar • Bulandshahr • Hapur • Baghpat • Muzaffarnagar • Shamli (8th, 2017) — **Mathura, Aligarh, Hathras, Saharanpur** only proposed, not included

One-liner Facts

- **Indian Standard Time line (82°30′ E):** passes through **Prayagraj (Naini/Mirzapur)** • cuts 5 districts — Prayagraj, Mirzapur, Sonbhadra, Bhadohi, Kaushambi
- **Longest/shortest boundary:** longest with **Madhya Pradesh (11 districts)** • shortest with **Himachal Pradesh (1 district, Saharanpur)**
- **7 districts each:** **7** with Uttarakhand • **7** with Bihar • **7** with Nepal
- **Saharanpur:** borders 3 states (Himachal Pradesh, Uttarakhand, Haryana)
- **7 districts of Bundelkhand (UP):** Jhansi • Lalitpur • Jalaun • Hamirpur • Mahoba • Banda • Chitrakoot

19 Geological Structure and Physiographic Divisions PYQ 19

Geological Structure

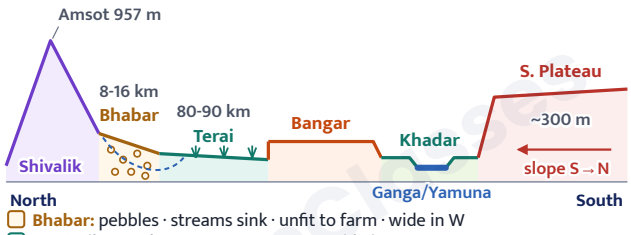
- **Southern plateau region:** extension of the oldest **Gondwana Land** • Archaean/Pre-Cambrian gneiss and granite rocks
- **Bundelkhand Gneiss:** Archaean rocks • abundance of red loam and granite in Lalitpur, Jhansi, Mahoba
- **Vindhyan System rocks:** Kaimur and Sonbhadra region • abundant limestone, sandstone, dolomite • **absence of fossils**
- **Tertiary rock group:** Shivalik foothills (Saharanpur to Kushinagar) • formation of **Bhabar and Terai**
- **Quaternary group:** vast Ganga plain • deposition of new and old alluvium (**Khadar and Bangar**)

Major Physiographic Regions of UP PYQ 14

- **Bhabar — extent:** Saharanpur to Kushinagar, just south of Shivaliks • width **8-16 km** (wide in west, narrow in east)
- **Bhabar — features:** pebbles, boulders and coarse sand deposits • highly porous (permeable) • small streams disappear underground • **unsuitable** for agriculture • thorny bushes
- **Terai — extent:** just south of Bhabar, Saharanpur to Deoria/Kushinagar • width **80-90 km** (wider in east)
- **Terai — features:** fine silt, clay, marshy land • high water table and moisture • Bhabar's lost streams re-emerge • intensive **sugarcane, rice, jute** • malaria-prone
- **Ganga-Yamuna plain — extent:** south of Terai and north of plateau • slope **north-west to south-east**

UP physiography: north to south

not to scale



- **Bhabar:** pebbles · streams sink · unfit to farm · wide in W
- **Terai:** silt, marsh · streams re-emerge · wide in E
- **Bangar:** old alluvium · kankar/reh · flood-safe
- **Khadar:** new alluvium · yearly floods · most fertile
- **Plateau:** Bundelkhand (W) · Baghelkhand (E) · ravines

- **Ganga-Yamuna plain — soil:** **Bangar** (old alluvium, kankar/reh, safe from floods) • **Khadar** (new alluvium, renewed by annual floods, most fertile)
- **Ganga-Yamuna plain — features:** Khol (high mounds of upper Ganga plain) • Bhur (wind-blown sand heaps) • bowl of wheat, rice, pulses
- **Southern plateau — extent:** south of Yamuna and Ganga up to Vindhyan scarp • area about **45,200 sq km**
- **Southern plateau — relief:** **Bundelkhand plateau** (west) • **Baghelkhand plateau** (east) • average height 300 m • slope **south to north**
- **Southern plateau — features:** ravines by **Chambal and Betwa** • **Kaimur and Sonakar** hills • gram, arhar, mustard, jowar
- **Highest peak of southern plateau:** **Sonakar and Kaimur** hills • extend over Mirzapur and Sonbhadra
- **Highest peak of UP:** **Amsot peak** • Shivalik range, Saharanpur • about **957 m**
- **Bhur:** 10-20 ft high **sandy mounds** along Ganga-Yamuna and Ramganga banks • formed by wind action in Pleistocene
- **Typical grasses of Bundelkhand:** **Sewan and Lampa**

Points to Note

1. **Bhabar vs Terai width:** Bhabar wide in west (Saharanpur), narrow in east (Kushinagar) — Terai exactly opposite: widest in east (Maharajganj, Kushinagar, Deoria) (80-90 km), narrow in west (Saharanpur); reversed in assertion-reason questions
2. **Plain slope vs plateau slope:** Ganga plain **NW → SE** • southern plateau (Bundelkhand) **SW → NE** — hence Chambal, Betwa, Ken flow north to join Yamuna

One-liner Facts

- **Red soil of Bundelkhand:** from weathering of granite and gneiss • **Rakar and Parwa**
- **Sonbhadra-Mirzapur plateau:** **Vindhyan rock system** • famous for limestone and cement industry
- **Khadar soil:** rich in potash, lime, fossils • deficient in **nitrogen, phosphorus, humus**
- **Alluvial fans and cones:** upper Ganga plain — Saharanpur, Muzaffarnagar, Meerut (western UP)
- **Vindhyan range:** formed by deposition at **end of Pre-Cambrian (Palaeozoic era)**

20 Climate, Seasons and Rainfall Distribution PYQ 21

Climate and Classification

- **Climate type:** tropical monsoon (humid subtropical)
- **Koppen:** most plains **Cwg** — monsoon type with dry winter • western and southern dry plateau (Agra, Mathura, Jhansi) **BShw** — semi-arid steppe PYQ 6
- **Thornthwaite:** **CB'w** — mesothermal sub-humid • eastern Terai **CA'w**
- **Prof. Stamp and Dr. Kazi Syed Ahmad:** **2 main climatic regions** (1. humid and hot, 2. moderately humid and hot) • 5 sub-regions

Three Major Seasons PYQ 11

- **Summer — period:** March to mid-June • sharp temperature rise as sun moves north • average maximum 36-39°C
- **Summer — Loo:** hot dry winds blowing west to east in May-June called **Loo** • maximum heat in Bundelkhand (near Tropic of Cancer and rocky surface)
- **Summer — records:** hottest **Agra and Jhansi** • least hot **Bareilly**
- **Rainy season — period:** mid-June to October • **80-85%** of total rainfall • from **Bay of Bengal branch of SW monsoon** (locally **Purva**)
- **Rainy season — distribution:** rainfall decreases **east to west** and **north (Terai) to south (plateau)**
- **Rainy season — records:** highest **Gorakhpur** (184.7 cm, avg 56 days) • lowest **Mathura** (54.4 cm, avg 32 days)
- **Winter — period:** November to February • temperature decreases south to north • **January** coldest
- **Winter — Mawath:** **Western Disturbances (Mediterranean cyclones)** give up to **10 cm** winter rain in western UP • beneficial for **Rabi (wheat)**
- **Winter — records:** coldest **Bareilly** • southern plateau average minimum 13.3°C

Agro-Climatic and Agro-Ecological Regions

- **Agro-climatic zones:** on basis of rainfall, terrain and soil, UP has **9** (India 15) • UP falls in India's **Middle Gangetic Plain, Upper Gangetic Plain, Lower Gangetic Plain, Central Plateau** regions PYQ 10
- **9 agro-climatic zones:** 1. Bhabar and Terai • 2. Western Plain • 3. Mid-Western Plain • 4. South-Western Semi-Arid Plain • 5. Central Plain
- **9 agro-climatic zones (contd.):** 6. Bundelkhand • 7. North-Eastern Plain • 8. Eastern Plain • 9. Vindhya Region
- **Agro-ecological zones:** **20** • soil group zones **8**

Points to Note

1. **Agro-climatic vs agro-ecological:** agro-climatic zones = **9** • agro-ecological zones = **20** • soil group zones = **8** — 9 and 20 commonly confused
2. **Monsoon branch:** most rainfall not from Arabian Sea branch — **Bay of Bengal branch**; Arabian Sea branch gives little rain only in southern Bundelkhand

One-liner Facts

- **Annual average rainfall:** about **95 cm** • eastern plain 112 cm • central plain 94 cm • western plain 84 cm • southern plateau 91 cm

- **Monsoon duration:** maximum Gorakhpur (**56 days**) • minimum western UP/Mathura (**32 days**)
- **Dust storms (May-June):** highest frequency **Agra and Mathura**
- **Flood-affected:** most **eastern plains and Terai** • total flood-prone area about 73.06 lakh hectares
- **Frost:** in northern parts mainly **December and January**

21 Soils, Erosion and Problem Lands PYQ 22

Alluvial Soil

- **Alluvial soil (Kanp/loam/clayey):** largest area (Ganga-Yamuna plain) • rich in **potash and lime** • deficient in **nitrogen, phosphorus, humus**

Local Soils of Bundelkhand (Southern Plateau) PYQ 15

- **Parwa (Padwa/Paduwa) — red group:** light red-brown sandy loam • humus-deficient • good yield with irrigation • Hamirpur, Jalaun, Yamuna ravines • **jowar (chari) and gram**
- **Rakar (Rankad) — red group:** red-brown coarse gravelly soil • found on slopes • low fertility • Jhansi, Lalitpur, Mirzapur • with manure **sesame (Kharif) and gram (Rabi)**
- **Mar (Mad) — black group:** black like Kabar • **60% silicate, 15% iron, 25% aluminium** • becomes sticky (gum-like) when wet
- **Mar — extent:** Jhansi, Lalitpur, Jalaun, Hamirpur (western Bundelkhand) • hard to cultivate but excellent for **cotton and wheat**
- **Kabar (Kabad) — black group:** deep black/brown clayey soil • very high water-holding capacity • large cracks when dry • Jalaun, Hamirpur, Banda, Chitrakoot • **pulses (gram, lentil) and soybean**
- **Bhonta — Vindhyan group:** gravelly-stony red soil from breaking of Vindhyan rocks • upper plateau of Sonbhadra and Mirzapur • **coarse grains (Kodo, Sawan)**

Saline and Alkaline Soils (Usar, Reh, Kallar, Banjar)

- **Reh/Kallar:** white crust of sodium, calcium, magnesium salts on surface due to excessive canal irrigation, waterlogging and capillary action
- **Districts with most Usar/alkaline land:** Raebareli • Aligarh • Mainpuri • Etah • Etawah • Kanpur • Unnao • Pratapgarh • Sultanpur
- **Reclaiming alkaline/Usar land: gypsum, pyrite** • green manure (**Dhaincha**) • **paddy** crop
- **Reclaiming acidic soil: lime**

Soil Erosion

- **Water erosion (gully erosion): biggest** cause of soil erosion in UP • deep **ravines** in catchments of **Chambal, Yamuna, Betwa** PYQ 11
- **Areas affected by gully erosion:** most **Etawah** • then Agra, Jalaun, Hamirpur
- **Sheet erosion:** invisible washing away of fine fertile top layer from flat fields • **"death of the farmer"**
- **Wind erosion:** in summer, south-western UP — **Agra and Mathura** (adjoining Rajasthan border)

Mnemonic (Bundelkhand soils): "Kaali Mar-Kabar, Laal Parwa-Rakar" — black soil = Mar and Kabar, red soil = Parwa and Rakar

Points to Note

1. **Beehad vs Bhur:** Beehad = deep gullied land by gully erosion of Chambal-Yamuna (Etawah, Agra) • Bhur = 10-20 ft high wind-deposited sandy mounds in Ganga plain (Bijnor, Meerut, Budaun)
2. **Wind vs water erosion:** "area most affected by soil erosion" → **Chambal/Yamuna ravines (Etawah)** • if "wind erosion" is specified → **Agra and Mathura**

One-liner Facts

- **Mar (black soil):** western UP and Bundelkhand • locally equivalent to **Regur/Karail**
- **Karail-Kachhar:** clayey soil of low-lying areas of Azamgarh, Jaunpur, Mau, Ballia
- **Bhat and Manjh:** lime-free loam of eastern UP (Gorakhpur, Basti, Maharajganj)
- **Total problem land (Usar/ravine/waterlogged):** about **12-15 lakh hectares**
- **UP Bhumi Sudhar Nigam:** established **1978** • to make Usar and barren land cultivable

22 Drainage System, Major Rivers and Lakes PYQ 34

3 Groups of Rivers by Origin

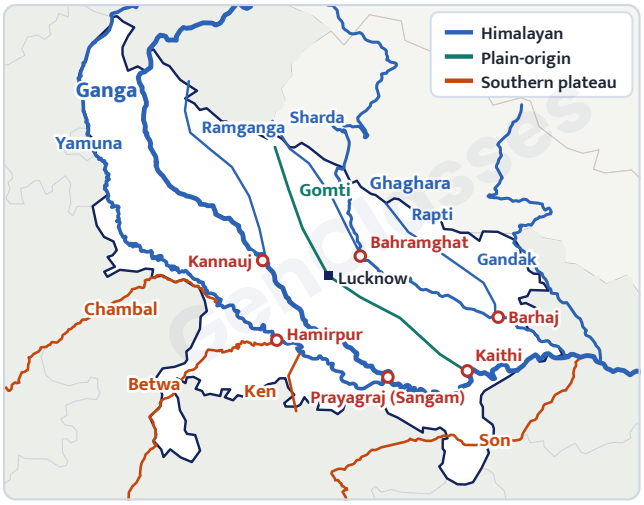
- **From Himalayas (perennial):** Ganga • Yamuna • Ramganga • Kali (Sharda) • Ghaghara (Saryu) • Gandak • Rapti
- **From Ganga plain (lakes/marshes):** Gomti • Varuna • Sai • Pando • Isan • Rind (Arind) PYQ 8
- **From southern plateau (Vindhya/Satpura):** Chambal • Betwa • Ken • Son • Tons (Tamsa) • Rihand • Kanhar • Karmanasa

Major Rivers: Origin, Entry, Confluence PYQ 18

- **Ganga — origin:** Gangotri glacier (Gaumukh, as Bhagirathi) • joins Alaknanda at Devprayag
- **Ganga — course:** enters **Bijnor** • exits **Ballia** • through 28 districts • into Bay of Bengal (as Padma/Meghna)
- **Ganga — length and cities: 1,450 km** in UP • maximum length in **Budaun** • banks: Kannauj, Kanpur, Prayagraj, Varanasi, Mirzapur, Ghazipur, Ballia
- **Yamuna — origin:** Yamunotri glacier (western slope of Bandarpunch, Uttarkashi)
- **Yamuna — course:** enters **Saharanpur** (near Faizabad village) • through **19 districts** • confluence with Ganga at Prayagraj
- **Yamuna — bank towns:** Mathura • Vrindavan • Agra • Bateshwar • Etawah • Kalpi (Jalaun) • Hamirpur • Kaushambi • forms a large arc
- **Ramganga:** origin Doodhatoli range (Pauri Garhwal, Uttarakhand) • enters **Bijnor** (descends to plains near Kalagarh fort) • joins Ganga from left bank near **Kannauj**
- **Ramganga — special:** flows through Corbett National Park • Moradabad, Bareilly, Shahjahanpur, Hardoi • tributary **Koh**
- **Gomti:** origin **Fulhar Jheel (Gomat Tal)**, Madhotanda (Pilibhit) • entirely in UP (plains origin) • joins Ganga from left bank at **Kaithi (Ghazipur)**

UP rivers and confluences

colour = origin class



- **Gomti — special:** banks: **Lucknow, Sultanpur, Jaunpur** • **Sai** (via Pratapgarh, Raebareli) joins Gomti at **Rajepur, Jaunpur**
- **Ghaghara (Saryu):** origin Mapchachungo glacier (Tibetan plateau) • **Karnali** in Nepal • enters at Lakhimpur Kheri-Bahraich border (Sheeshpani gorge) • exits Deoria/Ballia • joins Ganga near **Chhapra (Bihar)**
- **Ghaghara — special:** called **Saryu** at Ayodhya • **Rapti** (Gorakhpur on its bank) joins Ghaghara near **Barhaj (Deoria)** • Rapti's tributary **Rohini**
- **Sharda (Kali):** origin Milam glacier (Kumaon Himalaya, Pithoragarh) • enters **Pilibhit** (Nepal-UP border) • joins Ghaghara (Karnali) at **Bahramghat (near Barabanki/Sitapur)** • snake-like meandering course
- **Gandak:** Nepal Himalaya (called Shaligrami and Narayani) • enters at Maharajganj-Kushinagar border • joins Ganga near Hajipur/Patna (Bihar) • **Triveni Barrage** at Bhainsalotan (Valmikinagar)
- **Chambal:** origin Janapav hill (Mhow, Indore, MP) • enters **Agra** • forms UP-MP-Rajasthan boundary • joins Yamuna at **Muradganj (near Etawah/Auraiya)**
- **Chambal — special:** notorious for gully erosion and ravines • tributaries Kali Sindh, Parvati, Banas
- **Betwa (Vetravati):** origin Kumhara village, Raisen (Vindhya range, MP) • enters **Lalitpur** (via Jhansi, Jalaun) • joins Yamuna near **Hamirpur** • tributaries **Dhasan and Jamni** • Rajghat and Matatila dams
- **Ken (Karnavati):** origin Kaimur hills (Katni, MP) • enters **Banda** (near Kalinjar) • joins Yamuna at **Bhojha/Chillatara (Banda)** • Ken-Betwa Link Project • Banda town on its bank
- **Son (Swarna river):** origin Amarkantak plateau (Anuppur, MP) • enters **Sonbhadra** (exits to Bihar south of Mirzapur) • joins Ganga from right bank near Patna (Maner)
- **Son — tributaries (in UP):** **Rihand (Renuka)** • Kanhar • North Koel

Rivers Joining Ganga (West→East)

- **Left bank:** Ramganga • Gomti • Ghaghara • Gandak
- **Right bank:** Yamuna • Tons (Tamsa — at Sirsa, Prayagraj) • Karmanasa (near Chausa) • Son

Major Lakes and Tals (District-wise) PYQ 16

- **Keetham Lake (Sur Sarovar):** Agra • Ramsar site • bear rescue centre
- **Fulhar Jheel (Gomat Tal):** Pilibhit • source of Gomti (near Madhotanda)
- **Bakhira Lake:** Sant Kabir Nagar • state's **largest natural floodplain wetland** • Ramsar site
- **Surha Tal:** Ballia • oxbow lake in Ganga-Saryu doab • Jayaprakash Narayan Bird Sanctuary
- **Ramgarh Tal and Chilua Tal:** Gorakhpur • major tourist and aquatic lakes of eastern UP
- **Govind Ballabh Pant Sagar:** Sonbhadra (Pipri) • **largest artificial lake of India** • reservoir of Rihand dam
- **Parvati Arga Tal:** Gonda • pair of two oxbow lakes • Ramsar site
- **Benti, Naiya and Ajgara lakes:** Pratapgarh • Dr. Bhimrao Ambedkar Bird Sanctuary at Benti
- **Samaspur and Vishaitha Tal:** Raebareli • Samaspur Bird Sanctuary and Ramsar site
- **Nawabganj Lake (Kundra Samudra):** Unnao • Chandrashekhar Azad Bird Sanctuary
- **Bada Tal:** Shahjahanpur • famous **oxbow** lake
- **Barua Sagar and Laxmi Tal:** Jhansi • historic reservoirs of Bundelkhand
- **Madan Sagar and Kirat Sagar:** Mahoba • built by **Chandela** rulers
- **Radha Kund, Shyam Kund, Mansi Ganga, Nauh Lake:** Mathura • religious-cultural ponds
- **Sheikha Lake:** Aligarh • **smallest** sanctuary developed as national bird sanctuary
- **Darwan Lake:** Ambedkar Nagar • Tamsa basin
- **Chittaura Lake (Ashtavakra Lake):** Bahraich • source of **Tedhi river** (tributary of Saryu)
- **Moti Jheel and Balahapara:** Kanpur city • major urban artificial lakes
- **Aundhi Tal:** Varanasi • famous local reservoir
- **Raja ka Bandh, Laundhi and Bhojpur Tal:** Sultanpur • major lakes of middle Ganga plain
- **Dahar Lake and Bhijwan Lake:** Hardoi • **Sai river** originates from Bhijwan
- **Alwara Lake:** Kaushambi • famous for migratory birds
- **Shukratal:** Muzaffarnagar • historical-mythological importance

Mnemonic (southern tributaries of Yamuna, West → East): "Chambal ne Sindh ke bete (Betwa) ko Ken (chhadi) se maara" — 1. Chambal = Etawah, 2. Sindh = Jalaun, 3. Betwa = Hamirpur, 4. Ken = Banda PYQ 9

Points to Note

1. **Tons confusion:** two Tons — Uttarakhand/Himachal one is the largest tributary of **Yamuna** near Kalsi (Dehradun) • **Tamsa (Tons)** from Kaimur (Maihar, MP) joins **Ganga** directly from right bank at **Sirsa**, Prayagraj (tributary Belan)
2. **Sharda vs Rapti confluence:** Sharda (Kali) joins Ghaghara (Karnali) near **Bahramghat (Barabanki)** • Rapti joins Ghaghara near **Barhaj (Deoria)**
3. **Origin of Sai:** not Pilibhit — **Bhijwan Lake, Hardoi** • major tributary of Gomti • joins Gomti at **Rajepur, Jaunpur**

One-liner Facts

- **Ganga entry and maximum length:** enters **Bijnor** • maximum length in **Budaun** (about 133 km)
- **National River:** Ganga • **4 November 2008** • National Waterway-1 (**Prayagraj to Haldia, 1620 km**) on Ganga
- **Gorakhpur and Chitrakoot:** Gorakhpur near **Rapti-Rohini** confluence • Chitrakoot on **Mandakini (Payasvini)**
- **Rivers of Sonbhadra:** Belan • Rihand • **Kanhar** • **Pandu**
- **Karmanasa:** rises in Vindhyas • natural boundary between UP (Chandauli, Ghazipur) and Bihar

23 Means of Irrigation, Canal Network and Multipurpose Projects PYQ 29

Percentage Contribution of Means of Irrigation (Descending Order) PYQ 10

- **Tubewells: 73.4%** (highest) • UP ranks first in India in tubewell-irrigated area • first tubewell 1930, Meerut
- **Canals: 15.1%** • maximum canal-irrigated area: western UP and Raebareli/Lakhimpur
- **Wells: 8.6%** • central plain region
- **Tanks, lakes and ponds: 0.6%** • most common in Bundelkhand/southern plateau region
- **Other means:** remaining share

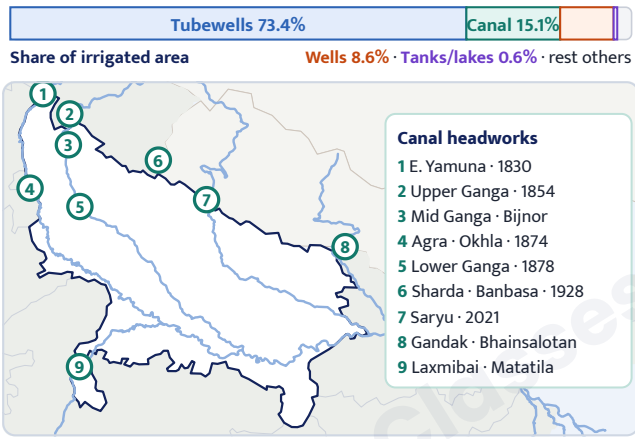
Major Canal Network of Uttar Pradesh PYQ 12

- **Eastern Yamuna Canal:** left bank of Yamuna (Faizabad, Saharanpur) • **1830 AD** • **oldest canal** of the state • built by Shah Jahan, restored by the British • Saharanpur, Shamli, Muzaffarnagar, Meerut, Ghaziabad
- **Upper Ganga Canal:** right bank of Ganga (**Haridwar, Uttarakhand**) • **1854 AD** • father: **Sir Proby Cautley** • main length 340 km (6,496 km with branches)
- **Upper Ganga Canal route:** Haridwar → Saharanpur, Muzaffarnagar, Meerut, Bulandshahr, Aligarh, Etah, Mathura, Firozabad, Mainpuri → up to Kanpur
- **Middle Ganga Canal:** Ganga (**Bijnor Barrage**) • purpose: remove water shortage in Upper Ganga Canal • Bijnor, Amroha, Sambhal, Bulandshahr, Aligarh, Hathras, Mathura
- **Lower Ganga Canal:** Ganga (**Narora, Bulandshahr**) • **1878 AD** • Bulandshahr, Aligarh, Etah, Firozabad, Mainpuri, Farrukhabad, Kannauj, Kanpur, Fatehpur, Prayagraj
- **Sharda Canal:** Sharda (Kali) river (**Banbasa, Champawat** • UP, Uttarakhand and Nepal border) • **1928 AD** (built 1920-1928) • **largest canal system of UP** • total length with branches **12,368 km**
- **Sharda Canal districts:** Pilibhit, Bareilly, Shahjahanpur, Lakhimpur, Hardoi, Sitapur, Barabanki, Lucknow, Unnao, Raebareli, Sultanpur, Pratapgarh, Prayagraj
- **Saryu Canal Project:** Ghaghara river (**Katarniaghat, Bahraich**) • started 1978 • dedicated to the nation from Balrampur in **December 2021** • **links 5 rivers:** Ghaghara, Saryu, Rapti, Banganga, Rohini
- **Saryu Canal's 9 eastern districts:** Bahraich, Shravasti, Balrampur, Gonda, Siddharthnagar, Basti, Sant Kabir Nagar, Gorakhpur, Maharajganj
- **Agra Canal:** right bank of Yamuna (**Okhla, Delhi**) • **1874 AD** • inter-state canal: Delhi, Haryana, Rajasthan and Mathura & Agra of UP
- **Gandak Canal:** Gandak river (**Bhainsalotan/Valmikinagar**) • joint project of UP, Bihar and Nepal • irrigation in Gorakhpur, Maharajganj, Kushinagar, Deoria
- **Rani Laxmibai Dam Canal:** Betwa river (**Matatila, Lalitpur**) • joint project of MP and UP • two branches: Gursarai and Mandar • Jhansi, Lalitpur, Jalaun, Hamirpur benefited

Major Dams, Reservoirs and Hydel Projects PYQ 17

- **Rihand Dam (Govind Ballabh Pant Sagar):** Rihand (Renuka) river • **Pipri (Sonbhadra)** • 1962 • concrete gravity dam (length 934 m, height 91.46 m) • 300 MW hydropower

Irrigation sources and canal headworks



Trap: oldest **E. Yamuna (1830)** · largest **Sharda (12,368 km)**

- **Obra Dam:** Rihand river • Robertsganj (Sonbhadra) • 25 km north of Rihand Dam • thermal and hydel power station
- **Matatila Dam:** Betwa river • Lalitpur/Jhansi border • **1958** • with Madhya Pradesh's cooperation
- **Rajghat Dam Project:** Betwa river • Lalitpur • equal (50:50) UP-MP partnership • reservoir: **Rani Laxmibai Sagar**
- **Pariksha Dam and Sukma-Dukma Dam:** Betwa river • Jhansi • Bundelkhand's irrigation and thermal power water supply projects
- **Govind Sagar Dam and Shahzad Dam:** Shahzad river • Lalitpur • Govind Sagar Dam (1952) • Rohini Dam (Rohini river) also in Lalitpur
- **Sajnam and Jamni Dams:** Lalitpur • on Sajnam and Jamni rivers respectively
- **Maudaha Dam (Swami Brahmanand Dam):** Birma river • Hamirpur • major irrigation project of Hamirpur in Bundelkhand
- **Arjun Dam and Urmil Dam:** Arjun and Urmil rivers • Mahoba • Kabrai Dam also in Mahoba
- **Pathrai Dam and Lakheri Dam:** Pathrai and Sukhnai rivers • Jhansi • regional irrigation projects
- **Meja Reservoir Dam:** Belan river • Mirzapur • famous earthen dam • water supply to Mirzapur and Prayagraj
- **Ahraura and Jargo Dams:** Garai and Jargo rivers • Mirzapur • Sirsi Dam (Bakher river) also in Mirzapur
- **Chandraprabha and Musakhand Dams:** Chandraprabha and Karmanasa rivers • Chandauli • Musakhand: joint UP-Bihar project
- **Kanhar Dam:** Kanhar river • Duddhi (Sonbhadra) • irrigation project of Sonbhadra's tribal area

- **Kalagarh Dam (Ramganga Project):** Ramganga river • Bijnor • **128 m high** • **highest dam of UP** (earth and rock) • Kathi Dam also on Ramganga in Bijnor
- **Bansagar Dam Project:** Son river • Shahdol (MP) • **3 states (UP, MP, Bihar)** • cost and water ratio **1:2:1** • benefits **Mirzapur and Prayagraj** of UP
- **Hathnikund Barrage (New Tajewala):** Yamuna river • Saharanpur border • **5 states (Haryana, Himachal Pradesh, UP, Delhi, Rajasthan)** • completed 1999
- **Gokul Barrage and Luv-Kush Barrage:** Yamuna and Ganga • Mathura and Kanpur • **drinking water projects:** Gokul (Mathura-Agra), Luv-Kush (Kanpur), **Agra Barrage** (Yamuna, Agra)
- **Ken-Betwa Link Project:** water transfer from Ken to Betwa • Panna/Chhatarpur to Jhansi • country's first river-interlinking project under Amrit Kranti
- **Ken-Betwa route:** surplus Ken water via **Daudhan Dam** through a 221 km canal into Betwa

Points to Note

1. **Govind Ballabh Pant Sagar:** Sonbhadra, **Rihand river** • **Govind Sagar Dam:** Lalitpur, **Shahzad river** • Bhakra Nangal (Himachal) reservoir is also Govind Sagar — UPPCS's favourite confusion point
2. **Oldest canal: Eastern Yamuna Canal (1830)** • **largest (longest) system: Sharda Canal (1928)**
3. Saryu Canal's 5 rivers do not include Gomti or Sharda — **Ghaghara, Saryu, Rapti, Banganga, Rohini**

One-liner Facts

- **Net irrigated area:** **western region first** • **Bundelkhand last**
- **Khara hydel project:** **Asan river (Yamuna canal)** • **Saharanpur**
- **Adwa and Meja Dams:** both in Mirzapur • on Adwa and Belan rivers respectively
- **'District of Dams':** **Lalitpur** • Matatila, Rajghat, Govind Sagar, Shahzad, Sajnam, Jamni, Rohini dams
- **Free Boring Scheme:** UP government • small and marginal farmers • **1985**

24 Natural Vegetation, Forest Report (ISFR) and Wildlife Conservation PYQ 31

Forest Policy and Types of Forests

- **Forest policy:** first **1952 AD** • second 1998 • current **State Forest Policy-2017**
- **Tropical moist deciduous forests:** Terai and Bhabar (100-150 cm rainfall) • **Sal, Teak, Shisham, Mahua, Semal, Amla, Jamun, Bamboo**
- **Tropical dry deciduous forests:** eastern, central and western plains (50-100 cm rainfall) • **Neem, Peepal, Shisham, Mango, Jamun, Babul, Amaltas, Palash (Dhak • State flower)**
- **Tropical thorny scrub:** southern plateau and western dry region (below 50 cm • Agra, Mathura, Jhansi, Banda) • **Acacia, Babul, Katha (Khair), Kakora, Thor, Dhaura, Tendu**

India State of Forest Report (ISFR 2021) — Uttar Pradesh PYQ 14

- **Total forest cover:** **14,817.89 sq km** • **6.15%** of geographical area
- **Top 3 forest cover (area):** 1. Sonbhadra (2,436 sq km) • 2. Lakhimpur Kheri • 3. Mirzapur
- **Top 3 forest cover (percentage):** 1. Sonbhadra (35.29%) • 2. Chandauli (21.78%) • 3. Chitrakoot
- **Lowest 3 forest cover (area):** 1. Bhadohi (Sant Ravidas Nagar • only 3.71 sq km) • 2. Mau • 3. Mainpuri
- **Lowest 3 forest cover (percentage):** 1. Bhadohi (0.37%) • 2. Mainpuri (0.49%) • 3. Deoria
- **Total tree cover:** **7,421 sq km** • **3.08%**
- **Very dense forest:** 2,626.61 sq km (1.09%) • highest in **Lakhimpur Kheri**
- **Moderately dense forest:** 4,029 sq km (1.67%) • **open forest:** 8,161 sq km (3.39%) • both highest in **Sonbhadra**
- **Forest + tree cover:** **22,238.89 sq km** • **9.23%**
- **Change in forest cover:** maximum increase Saharanpur (+49 sq km) • maximum decrease Sonbhadra (−103 sq km)

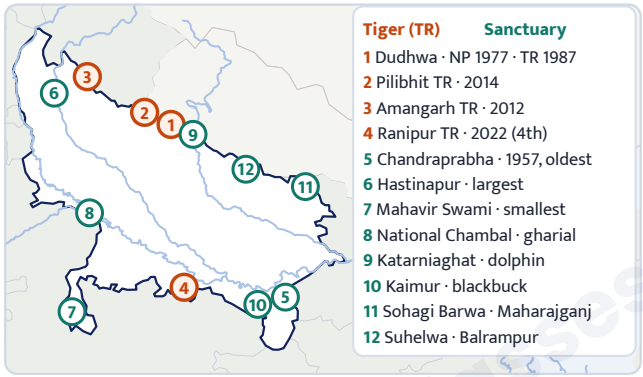
Major Forest-Produce Based Industries PYQ 9

- **Katha (catechu):** sap of **Khair** tree • Bareilly, Najibabad
- **Bidi:** **Tendu** leaves • Mirzapur, Sonbhadra, Prayagraj, Jhansi, Gursahaiganj
- **Matches and packing boxes:** **Semal and Gurel** wood • Bareilly, Saharanpur, Prayagraj
- **Cane, furniture and paper:** bamboo, cane and Bhabar grass • Bareilly, Saharanpur, Gorakhpur
- **Lisa (rosin and turpentine oil):** resin of **Chir pine**

National Park, Tiger Reserves and Zoological Parks PYQ 15

- **Only National Park:** **Dudhwa** • Lakhimpur Kheri • 490 sq km
- **Dudhwa history:** 1968 wildlife sanctuary • **1 February 1977** National Park • **1987-88** merged with Kishanpur and Katarniaghat sanctuaries into **Project Tiger (Dudhwa Tiger Reserve)**
- **Dudhwa famous for:** **Barasingha (swamp deer)** and tigers
- **Dudhwa Tiger Reserve (1987):** Lakhimpur Kheri and Bahraich
- **Pilibhit Tiger Reserve (June 2014):** Pilibhit and Shahjahanpur • 2020 international **TX2 Award** for doubling tiger numbers
- **Amangarh Tiger Reserve (2012):** Bijnor • buffer zone of Jim Corbett Tiger Reserve
- **Ranipur Tiger Reserve (October 2022):** Chitrakoot • UP's **4th** • India's **53rd** • first in Bundelkhand
- **Elephant Reserves (2):** 1. Uttar Pradesh Elephant Reserve (Saharanpur-Bijnor, 2009) • 2. **Terai Elephant Reserve (2022)** • Dudhwa + Pilibhit • India's 33rd

Tiger reserves and sanctuaries — only NP: Dudhwa



- **Recognised zoos (3):** Nawab Wajid Ali Shah (Lucknow, 1921) • Kanpur (Allen Forest, 1974) • **Shaheed Ashfaq Ulla Khan (Gorakhpur, 2021)**
- **Night Safari: Kukrail forest area (Lucknow)** • country's first (under development) • earlier proposed in Greater Noida
- **Lion Safari and Lion Breeding Centre:** Etawah • **Peacock Conservation Centre:** Vrindavan (Mathura)
- **Jatayu (Vulture) Conservation and Breeding Centre: Bhari-Baisi (Maharajganj)** • world's first Asian King Vulture conservation centre
- **Dolphin Sanctuary:** Chambal river area (Etawah) • **Turtle Wildlife Sanctuary:** Varanasi (now shifted to Prayagraj-Mirzapur-Bhadohi Ganga stretch)

Major Wildlife Sanctuaries of Uttar Pradesh PYQ 16

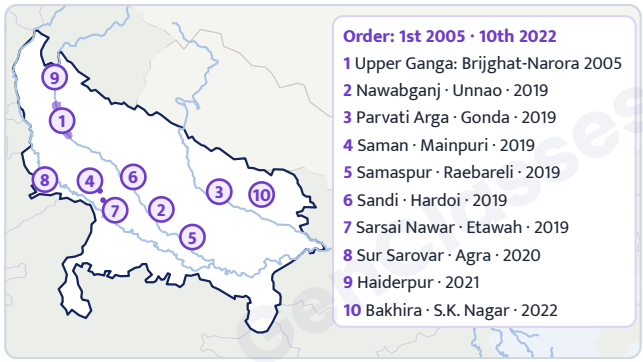
- **Chandraprabha: 1957** • Chandauli • **oldest (first) in the state** • 78 sq km
- **Kishanpur:** 1972 • Lakhimpur Kheri and Shahjahanpur • part of Dudhwa Tiger Reserve
- **Katarniaghat:** 1976 • Bahraich • gharial and Gangetic dolphin • Girwa river flows
- **Ranipur:** 1977 • Chitrakoot and Banda • 4th Tiger Reserve in 2022
- **Mahavir Swami:** 1977 • Lalitpur • **smallest in the state** (5.41 sq km)
- **National Chambal:** 1979 • Agra and Etawah • joint UP, MP and Rajasthan • **gharial breeding**
- **Kaimur:** 1982 • Mirzapur and Sonbhadra • blackbuck and chinkara • 501 sq km
- **Hastinapur: 1986** • Meerut, Muzaffarnagar, Bijnor, Amroha, Hapur • **largest in the state** (2,073 sq km) • Barasingha conservation
- **Sohagi Barwa:** 1987 • Maharajganj • adjoining Nepal border • 428 sq km
- **Suhelwa:** 1988 • Balrampur, Gonda and Shravasti • major Terai wildlife area • 452 sq km
- **Pilibhit:** 2008 • Pilibhit • Tiger Reserve in 2014

Major Bird Sanctuaries and 10 Ramsar Sites PYQ 20

- **Shaheed Chandrashekhar Azad (Nawabganj): 1984** • Unnao • **oldest (first) in the state** • 2.25 sq km • Ramsar (2019)
- **Lakh Bahosi: 1988** • Kannauj • **largest in the state** (80.24 sq km)
- **Samaspur:** 1987 • Raebareli • centre of migratory birds • Ramsar (2019)
- **Sandi:** 1990 • Hardoi • near Garra river • Ramsar (2019)
- **Bakhira:** 1990 • Sant Kabir Nagar • declared **10th Ramsar site** on World Wetlands Day **2022**
- **Okhla:** 1990 • Gautam Buddh Nagar • at Okhla Barrage on Yamuna
- **Saman:** 1990 • Mainpuri • cranes (Sarus) • Ramsar (2019)
- **Parvati Arga:** 1990 • Gonda • formed by two oxbow lakes • Ramsar (2019)
- **Vijay Sagar:** 1990 • Mahoba • major bird sanctuary of Bundelkhand
- **Patna:** 1990 • Etah (Jalesar) • formerly the smallest (1.09 sq km)
- **Suraha Tal (Jayaprakash Narayan):** 1991 • Ballia • oxbow lake • 34.32 sq km
- **Sur Sarovar (Keetham):** 1991 • Agra • Ramsar in **2020** (8th)
- **Dr. Bhimrao Ambedkar:** 2003 • Benti (Kunda, Pratapgarh) • 4.27 sq km

10 Ramsar sites of UP

2nd after Tamil Nadu



- **Sheikha Jheel: 2016** • Aligarh • **smallest in the state** (only **0.40 sq km**)
- **Total 10 Ramsar sites:** second in the country after Tamil Nadu by number
- **Ramsar 1: Upper Ganga River (Brijghat to Narora stretch)** • 2005 • **UP's first Ramsar site**
- **Ramsar 2-6 (2019):** Nawabganj (Unnao) • Parvati Arga (Gonda) • Saman (Mainpuri) • Samaspur (Raebareli) • Sandi (Hardoi)
- **Ramsar 7: Sarsai Nawar Jheel (Etawah)** • 2019
- **Ramsar 8:** Sur Sarovar/Keetham Lake (Agra) • 2020
- **Ramsar 9: Haiderpur Wetland (Muzaffarnagar-Bijnor border)** • December 2021
- **Ramsar 10:** Bakhira Wildlife Sanctuary (Sant Kabir Nagar) • February 2022

Points to Note

1. **Patna Bird Sanctuary:** not in Patna, capital of Bihar — in **Etah district (Jalesar tehsil)** of UP
2. **Smallest bird sanctuary:** if **Sheikha Jheel (Aligarh • 0.40 sq km)** is an option, that; otherwise **Patna (Etah • 1.09 sq km)**
3. **Total forest cover (area and percentage): Sonbhadra first** • but in **very dense forest** only, **Lakhimpur Kheri first**

One-liner Facts

- **Forests declared state property:** first in **1935 AD**
- **Social Forestry (1976): Eucalyptus** • called **'ecological terrorist'** for excessive water use • **Subabul** • provides both soil fertility and fodder
- **Operation Green Scheme:** to promote afforestation in UP • **1 July 2001**
- **Country's first Monkey Forest (Vanar Van):** Lucknow (near Kukrail/Indira Dam) • first **Cow Safari:** Jalaun/Lalitpur (under development)
- **State symbols:** tree **Ashoka (Sita Ashoka)** • animal **Barasingha** • bird **Sarus Crane (Kraunch)** • flower **Palash (Tesu)** (adopted 4 January 2011)

25 Agriculture, Animal Husbandry, Major Crops and Production Rank PYQ 27

Land Holdings and Land Use

- **Working population in agriculture:** about **59.3%** • cultivators 30.3% • agricultural labourers 29.0%
- **Average holding:** UP **0.73 hectare** • India 1.08 hectare

- **Farmers with marginal holdings (below 1 hectare):** about **80%**
- **Cropping intensity:** about **162.4%**

3 Main Crop Seasons

- **Rabi (sowing October-November • harvest March-April):** wheat, barley, gram, pea, lentil (masoor), mustard (rai), linseed, potato, tobacco
- **Kharif (sowing June-July • harvest September-October):** paddy (rice), maize, jowar, bajra, arhar (tur), groundnut, sesame, cotton, sugarcane, jute, sunn hemp
- **Zaid (sowing March-April • harvest June):** watermelon, muskmelon, kakdi, cucumber, moong, urad, cowpea, pumpkin

UP's Rank in the Country and Top Producing Districts PYQ 19

- **Wheat:** **first** (~32%) • **Hardoi** (highest area and production) • Gorakhpur (leads in per-hectare productivity) • sown on **24%** of cultivable land • Ganga-Ghaghara Doab best
- **Sugarcane and sugar:** **first** (~45%) • Muzaffarnagar, Lakhimpur Kheri, Shamli, Bijnor, Meerut • **most important cash crop** (13% of farm land)
- **First sugar mill:** 1903 • **Pratappur (Deoria)** • India's first
- **Potato:** **first** • Agra, Hathras, Kannauj, Farrukhabad, Firozabad • Agra potato exported as **'Taj Brand'** • Saharanpur mango as **'Nawab Brand'**
- **Regional station of Central Potato Research Institute:** **Modipuram (Meerut)**
- **Total foodgrains:** **first** (~18-19% of country) • Ganga-Yamuna plains • also among top states in total vegetables and fruits (especially mango and guava)
- **Milk and total livestock:** **first** (close competition with Rajasthan) • **first** in buffalo population • Budaun, Agra, Meerut • Gokul Award: milk production
- **Pt. Deendayal Upadhyay Veterinary Science University and Cattle Research Institute:** Mathura
- **Amla:** **first** • **Pratapgarh** (first in country and state) • processing centres: Kunda and Sadar
- **Mango:** **first** • **Lucknow (Malihabadi Dussehri)** • Saharanpur (Safeda and Chausa) • Varanasi (Langra) • Meerut/Baghpat (Rataul)
- **Guava:** **first** • **Prayagraj (Allahabadi Safeda and Surkha)**, Kaushambi, Budaun • main centre: Khusro Bagh, Prayagraj
- **Mentha oil (peppermint):** **first** (~85% of country) • Barabanki, Rampur, Budaun, Sambhal, Chandauli • intensive cultivation: Shivala Patti
- **Green pea, lentil and bajra:** **first/second** • **gram and lentil:** Bundelkhand (Banda, Hamirpur, Jhansi) • **bajra:** Agra, Aligarh, Budaun
- **Pulses:** **gram** highest in **Jhansi/Banda** • **arhar** Hamirpur/Chitrakoot
- **Rice (paddy):** **second** (West Bengal first) • Shahjahanpur, Lakhimpur Kheri, Pilibhit, Bareilly, Chandauli • Kranti region: Terai and eastern UP
- **Rice Bowl:** **Chandauli** • Siddharthnagar's **Kala Namak rice** • GI tag
- **Opium:** major producer • **Barabanki** (highest) • **Ghazipur** • only **Opium Factory at Ghazipur** (1820)
- **Groundnut and sesame:** major regional crops • **groundnut:** Jhansi, Hardoi, Lalitpur • **mustard:** Agra, Mathura
- **Turmeric and ginger:** highest in **Bundelkhand (Lalitpur and Jhansi)**
- **Betel (paan):** **Mahoba** (training centre too), Banda, Lalitpur • **orange and litchi:** Saharanpur
- **Lemon:** Saharanpur and Agra • **Jute:** Saryu-Ghaghara Doab (Deoria, Gorakhpur)

3 Agri Export Zones PYQ 11

- **Agra:** potato • with Hathras, Farrukhabad, Kannauj, Meerut
- **Lucknow:** mango • with Unnao, Hardoi, Sitapur, Barabanki
- **Saharanpur:** mango • with Meerut, Muzaffarnagar, Baghpat, Bulandshahr, Bijnor

Points to Note

1. **Opium:** highest production **Barabanki** • but country's oldest and state's only **Government Opium and Alkaloid Factory at Ghazipur**
2. **Amla** → **Pratapgarh** • **orange and litchi** → **Saharanpur** (humid climate of Shivalik foothills)
3. **Jute variety:** mainly **white jute (Corchorus capsularis)** • ropes and sacks

One-liner Facts

- **Farmer schemes:** **Kisan Mitra 18 June 2001** • **Kisan Bahi 1992** • **Kisan Credit Card 1998-99**
- **Consolidation of holdings (Chakbandi):** **1954 AD** • from Kairana tehsil (Muzaffarnagar) and Musafirkhana tehsil (Sultanpur) • statewide from **1958**
- **Research institutes:** **Indian Institute of Pulses Research (IIPR) Kanpur** • **Indian Institute of Vegetable Research (IIVR) Varanasi** • **Indian Grassland and Fodder Research Institute (IGFRI) Jhansi**
- **Sugarcane research:** **Indian Institute of Sugarcane Research (IISR) Lucknow** • **UP Council of Sugarcane Research (UPCSR) Shahjahanpur**
- **Agriculture Policy-2013:** agricultural growth rate target **5.1%**

26 Mineral Resources, Energy and Power Projects PYQ 32

Mineral Policy and Administration

- **UP State Mineral Development Corporation:** established **23 March 1974**
- **First Mineral Policy:** **December 1998** • mineral development given **'Industry'** status • **12 districts** declared 'Mineral-rich areas'
- **12 mineral-rich districts:** Prayagraj • Jhansi • Saharanpur • Mirzapur • Lalitpur • Jalaun • Mahoba • Sonbhadra • Chitrakoot • Chandauli • Banda • Hamirpur
- **Mineral wealth category:** a **lower-middle** category state • most minerals in the **southern plateau region (Vindhyan and Bundelkhand rock groups)**

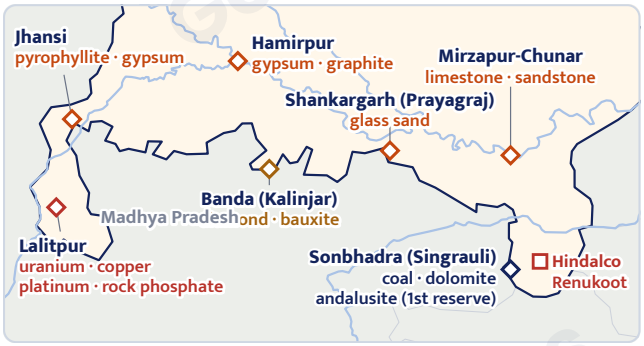
Major Minerals and Their Locations (District-wise Matching) PYQ 22

- **Limestone:** **Mirzapur (Gurma-Kanach-Bapuhari and Rohtas area)** • **Sonbhadra (Kajrahat)** • main raw material for **Churk, Dalla, Chunar** cement factories • UP 2nd in India in reserves
- **Coal:** **Singrauli area (Sonbhadra)** • Gondwana-period coal • used in Obra and Singrauli thermal power stations • UP **8th** in India in reserves

- **Bauxite: Banda • Chitrakoot (Ranipur)** • Chandauli (Chakia) • Lalitpur • main ore of aluminium • aluminium extracted at **Hindalco (Renukoot, Sonbhadra)**
- **Dolomite: Sonbhadra (Bari village and Kajrahat)** • Banda • used as flux and refractory in steel industry
- **Glass sand (Silica sand): Prayagraj (Shankargarh and Lohgarh)** • Banda • Chitrakoot (Bargarh) • Chandauli (Chakia) • Jhansi (Mundari Balabehat) • UP **first/leading** in India in production (obtained from Ganga-Yamuna)

Southern plateau: mineral belt

Vindhyan · Bundelkhand



Trap: uranium, copper only **Lalitpur** · coal, limestone **Sonbhadra-Mirzapur**

- **Potash salts:** Prayagraj • Chandauli • Banda • Jhansi • Sonbhadra • used in fertilizer manufacturing
- **Andalusite:** Sonbhadra and Mirzapur • UP **first** in India in reserves
- **Pyrophyllite and Diaspore:** Jhansi • Lalitpur • Mahoba • Hamirpur • UP **leading** in production • used in ceramics, pesticides, refractory bricks
- **Uranium: Lalitpur** (only district) • limited quantity
- **Copper and Platinum: Lalitpur (Sonrai area)** • in veins of igneous and sedimentary rocks
- **Gold: sands of Sharda and Ramganga • Sonbhadra (Bhikhampur, Hardi)** • Lalitpur (Girar) • small quantities in river sand and rock veins
- **Diamond: Banda (Kalinjar area)** • forest area of Mirzapur • in conglomerate rocks of the Vindhyan group
- **Rock Phosphate: Lalitpur (Durmala, Kimai)** • Banda • fertilizer industry and acidic soil treatment
- **Marble and Sandstone:** Mirzapur • Sonbhadra • **Chunar sandstone** world-famous for building construction (building stone: Mirzapur and Chunar)
- **Gypsum and Soapstone (Talc): Hamirpur • Jhansi** • cement, ammonium sulphate fertilizer, cosmetics and paper industry
- **Asbestos and Conglomerate:** Mirzapur • Jhansi (Badagaon) • heat insulators and cement sheets
- **Graphite:** Hamirpur • Chunar (Mirzapur) • pencil lead
- **Red ochre (Geru):** Banda • dyeing industry
- **Sillimanite and China Clay:** Sonbhadra • Mirzapur • porcelain/ceramic ware and electrical insulators
- **Feldspar:** Jhansi • **Pyrites:** Sonbhadra and Mirzapur • glass and ceramic industry

Mnemonic (Lalitpur minerals): "Lalit ne taambe ke rock (patthar) par uranium aur platinum se diaspore likha" — Copper • Rock Phosphate • Uranium • Platinum • Diaspore • Pyrophyllite • Iron ore — all found in Lalitpur

Energy Resources and Major Power Plants PYQ 14

- **UP State Electricity Board:** formed **1959** • **UP Electricity Regulatory Commission: 1998**
- **Contribution to power generation (order): Thermal** highest → Hydro → Nuclear → Renewable (solar/biomass)
- **Narora Atomic Power Station: Narora (Bulandshahr)** • **UP's only nuclear power centre** • two units (1st on 1 January 1991, 2nd on 1 July 1992)
- **Narora capacity:** total **440 MW** (220 × 2) • Seismic Zone-4 • indigenous Pressurised Heavy Water Reactor
- **Obra Thermal Power Station:** Sonbhadra • **1967-1971** • with help of **former Soviet Union (Russia)**
- **Anpara Thermal Power Station:** Sonbhadra • units A, B, C, D • Anpara 'C' by **Lanco** (private sector) • Anpara 'B' with Japanese (OECF) assistance
- **Harduaganj Thermal Power House: Kasimpur (Aligarh)** • **1942** • **oldest thermal power house in the state** • revived in 1968 with Russian collaboration
- **Pariksha Thermal Power Station:** Jhansi • on the Betwa bank • run by State Power Generation Corporation
- **Panki Thermal Power Station:** Kanpur • major state-sector plant
- **Ghatampur (Kanpur Nagar):** joint project of Neyveli Lignite and UPRVUNL • **1980 MW**
- **Meja (Prayagraj):** NTPC joint project
- **Rosa Thermal Power Project: Shahjahanpur** • private sector • acquired from Aditya Birla by **Reliance Group** • 1200 MW
- **Bara (Shankargarh, Prayagraj):** JP Group (now Tata/others) • private sector
- **Lalitpur Thermal Power: Bajaj Group** • private sector
- **8 NTPC plants (UP) — districts:** Sonbhadra • Gautam Buddh Nagar • Raebareli • Ambedkar Nagar • Auraiya • Prayagraj • Jhansi
- **NTPC coal-based (5): Singrauli** (Sonbhadra — NTPC's oldest, 1982) • Rihand (Sonbhadra) • Dadri (Gautam Buddh Nagar) • Unchahar (Raebareli) • Tanda (Ambedkar Nagar)
- **NTPC gas-based (2): Auraiya** (Dibiyapur) • **Dadri** (Gautam Buddh Nagar — both coal and gas)
- **NTPC solar/joint:** Meja (Prayagraj) • Bilhaur (Kanpur — solar)

Points to Note

1. **NTPC gas-based:** only **two in UP — Auraiya and Dadri** (Dadri has both coal + gas) • Singrauli, Rihand, Tanda, Unchahar are purely **coal-based**
2. **Uranium and Copper** only in **Lalitpur** • **Coal and Limestone** in **Sonbhadra and Mirzapur** — coal against Lalitpur is a wrong match

One-liner Facts

- **NEDA (UPNEDA):** Non-conventional Energy Development Agency • established **1983** • autonomous body
- **Waste-to-energy plant:** municipal solid waste based • with Centre's help • **Lucknow and Kanpur**
- **Aonla (Bareilly) and Jagdishpur (Amethi):** fertilizer and power plants based on Hazira-Vijaipur-Jagdishpur (HVJ) gas pipeline
- **Ethanol and bio-energy from molasses:** UP **first** in India
- **First Solar Expressway: Bundelkhand Expressway** (being developed) • **First Solar City: Ayodhya**

27 Industries, Industrial Corridors, ODOP and Special Economic Zones PYQ 35

Major Traditional and Modern Industries

- **Highest employment (after agriculture):** **Handloom and cotton textile industry**
- **Kanpur:** **'Manchester of North India'** • 1861 'Elgin Mill' • 1876 woollen mill **'Lal Imli' (British India Corporation)**
- **Sugar industry:** India's first sugar mill **1903, Pratappur (Deoria)** • UP first in sugarcane and ethanol production
- **Cement industry:** near limestone areas • **Churk (Sonbhadra)** • **Dalla (Sonbhadra)** • **Chunar (Mirzapur)** • Kajrahat • Dadri (Gautam Buddh Nagar) • Tanda (Ambedkar Nagar) • Raebareli • Jhansi • Aligarh • Sikandrabad

Major Industries, Industrial Centres and Locations PYQ 24

- **Glass and bangles:** **Firozabad (Suhag Nagari)** • Bahjoi (Sambhal) • Naini (Prayagraj) • Sasni (Hathras) • Balawali (Bijnor)
- **Bulbs and glassware:** **Makhanpur (Firozabad)** • **Shikohabad** • Bharat Yantra Nigam at Naini
- **Aluminium (Hindalco):** **Renukoot (Sonbhadra)** • **1958** • Birla Group + Kaiser Company (USA) • based on bauxite (Banda/Jharkhand) and Rihand power
- **Fertilizers (IFFCO and KRIBHCO):** **Aonla (Bareilly)** • **Phulpur (Prayagraj)** • **Gorakhpur** • **Jagdishpur (Amethi)** • **Shahjahanpur** • IFFCO plants at Phulpur and Aonla
- **Phosphate fertilizer factories:** **Chakeri (Kanpur)** • **Magarwara (Unnao)**
- **Rail engines and coaches:** **Banaras Locomotive Works (BLW — Manduadih, Varanasi)** — diesel and electric engines • **Modern Coach Factory (Lalganj, Raebareli)** — India's **third rail coach factory**
- **Oil refinery:** **Mathura** • UP's **only** • Indian Oil Corporation
- **Leather industry:** **Kanpur (main)** • **Agra (footwear)** • Unnao (Banthar Leather Technology Park) • India's first **Mega Leather Cluster Park — Ramaipur (Kanpur)** (under development)
- **Brass, copper and bells:** **Brass City — Moradabad** • brass idols — Mathura • bells (ghungroo) — Jalesar (Etah) • utensils — Mirzapur, Hapur, Shamli
- **Moradabad:** **leading** in India's total metal handicraft exports
- **Locks:** Aligarh • **Scissors and sports goods:** Meerut • **Knives:** Rampur • sports goods in both Agra and Meerut, but **Meerut** world-famous
- **Carpets (durries):** **Bhadohi (Sant Ravidas Nagar — Carpet City)** • Mirzapur • Khamaria • Gopiganj • Agra • Shahjahanpur • **~80%** of India's carpet exports from Bhadohi-Mirzapur belt
- **Silk/Banarasi sarees:** Varanasi • Mau • Mubarakpur (Azamgarh) • **Chikankari and Zardozi:** Lucknow • Bareilly • Farrukhabad (printing)
- **Terracotta (baked clay figures):** **Gorakhpur** • **Black Pottery:** **Nizamabad (Azamgarh)**
- **Ceramics:** **Khurja (Bulandshahr — Ceramic City, hub of pottery)** • Chinhat (Lucknow) • Ghaziabad
- **Wood carving:** Saharanpur • Nagina (Bijnor) • **Wooden toys:** Varanasi • Chitrakoot • **Wooden sandals (khadaun):** Pilibhit
- **Cane sticks:** Bareilly • **Flutes:** Pilibhit
- **Perfume (Ittar):** **Kannauj (Perfume City)** — International Perfume Park • rose water — Ghazipur • jasmine oil — Jaunpur
- **Paper and pulp:** **Saharanpur (most paper mills)** — **'School of Paper Technology'** • Muzaffarnagar • Meerut • Naini • Gorakhpur
- **Matches:** Bareilly (WIMCO factory) • Saharanpur • Prayagraj • **Synthetic rubber:** Modinagar (Ghaziabad) • **Polyfibre:** Barabanki
- **Churru, granular jaggery and khandsari:** Muzaffarnagar • **Opium factory:** Ghazipur
- **Defence and aeronautics:** **HAL** — Lucknow, Kanpur, Korwa (Amethi) • **BEL** — Ghaziabad • **Bharat Dynamics** — Jhansi
- **UP Defence Industrial Corridor (6 nodes):** **Aligarh** • **Agra** • **Jhansi** • **Chitrakoot** • **Kanpur** • **Lucknow** • **AK-203 rifle** factory at Korwa (with Russia)
- **Artificial Limbs Manufacturing Corporation (ALIMCO):** Kanpur • **Transformer factory:** Jhansi and Meerut
- **Triveni Structurals Ltd.:** Naini (Prayagraj) • **Diesel engines/pumpsets:** Ghaziabad and Agra

One District One Product (ODOP) — Confusing Districts PYQ 15

- **ODOP launched:** **24 January 2018**
- **Siddharthnagar:** Kala Namak rice • **Chandauli:** Zari-Zardozi and black rice
- **Amroha:** Dholak (musical instrument) • **Baghpat:** home furnishing
- **Balrampur:** pulses (food processing) • **Chitrakoot:** wooden toys
- **Etah:** ghungroo, bells and brass products • **Hathras:** asafoetida (hing) processing
- **Jalaun:** handmade paper art (Kalpi) • **Kaushambi:** banana (food processing)
- **Kushinagar:** banana fibre products • **Lalitpur:** zari silk sarees and food processing
- **Maharajganj:** furniture • **Mahoba:** Gaura stone handicraft
- **Mainpuri:** Tarkashi art (brass wire inlay on wood) • **Pratapgarh:** amla products
- **Sant Kabir Nagar:** brass utensils (Bakhira) • **Shravasti, Sultanpur and Amethi:** Moonj products
- **Sonbhadra:** carpets and durries • **Banda:** Shajar stone craft
- **Sambhal:** horn and bone products • **Pilibhit:** flutes

Industrial Development Authorities and Year of Establishment (Chronological) PYQ 8

- **UPSIDA (Kanpur):** 1961 (formerly UPSIDC)
- **NOIDA (New Okhla Industrial Development Authority):** **1976** (Gautam Buddh Nagar)
- **BIDA (Bhadohi Industrial Development Authority):** 1981
- **GIDA (Gorakhpur):** **1989** • **SIDA (Satharia, Jaunpur):** **1989**
- **Greater Noida:** **1991**
- **YEIDA (Yamuna Expressway Industrial Development Authority):** 2001
- **LIDA (Lucknow Industrial Development Authority):** 2005
- **BIDA (Bundelkhand Industrial Development Authority — New Noida):** **2023** • on land of 33 villages of Jhansi

> **Mnemonic (6 Defence Corridor districts):** "Ali aur Agra wale ne Jhansi ke chitra ko Kaan aur Lakhan ko dikhaya" — Ali = Aligarh, Agra, Jhansi, Chitra = Chitrakoot, Kaan = Kanpur, Lakhan = Lucknow

Points to Note

1. **SIDA (Satharia):** not Sonbhadra/Saharanpur — **Jaunpur**
2. **Gaura stone craft — Mahoba • Shajar stone craft — Banda** (stone from the Ken river)
3. **Kaushambi — banana (fruit/food processing) • Kushinagar — banana stem fibre** products

One-liner Facts

- **Asia's largest gold market: Meerut** • India's first **handicraft complex (Deendayal Hastkala Sankul): Varanasi**
- **First Software Technology Park: Noida** • **Biotechnology Park: Lucknow (Bakshi Ka Talab)**
- **Agro Parks:** Barabanki (Kursi Road) • Varanasi (Karkhiyaon) • Lucknow • Hapur
- **Apparel and Hosiery Park:** Kanpur • **Export Promotion Industrial Park (EPIP):** Greater Noida and Agra • **Mega Food Park:** Baheri (Bareilly)
- **Medical Device Park and Film City: YEIDA (Gautam Buddh Nagar)** • **Mega Textile Park (PM MITRA): Lucknow-Hardoi border**

28 Transport System (Road, Expressway, Rail, Water and Air) PYQ 24

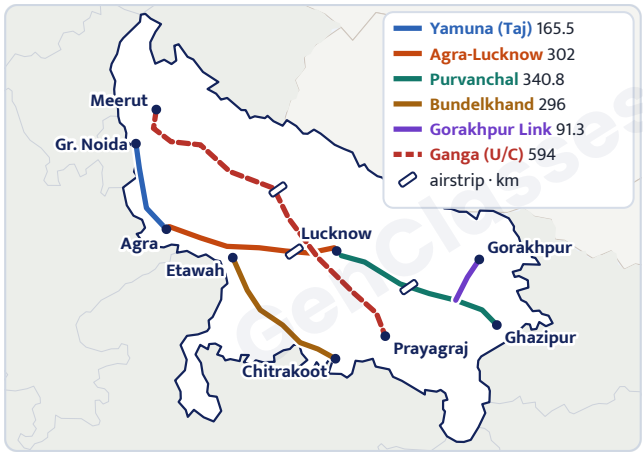
Road and Expressway Network ("Expressway State")

- **National Highway length:** UP **second** in India (Maharashtra first)
- **Longest National Highway (through UP): NH-34** (old numbering **NH-2 / NH-19** — Delhi-Kolkata Grand Trunk Road)
- **North-South and East-West Corridors:** intersect at **Jhansi** {11}

Major Expressways (Operated and Under Construction by UPEIDA) PYQ 14

- **Yamuna (Taj) Expressway: 165.5 km** • Greater Noida → Agra • operational **2012** • 6 lanes
- **Yamuna — 5 districts:** Gautam Buddh Nagar • Aligarh • Hathras • Mathura • Agra
- **Agra-Lucknow Expressway: 302 km** • Agra (Etmadpur) → Lucknow (Sarosa Bharosa) • **3.3 km airstrip** at Bangarmau (Unnao)
- **Agra-Lucknow — 10 districts:** Agra • Firozabad • Mainpuri • Etawah • Auraiya • Kannauj • Kanpur Nagar • Unnao • Hardoi • Lucknow
- **Purvanchal Expressway: 340.8 km** • Chandsarai (Lucknow) → Haidaria (Ghazipur) • inaugurated **November 2021** • **3.2 km airstrip** at Sultanpur (Kurebhar)
- **Purvanchal — 9 districts:** Lucknow • Barabanki • Amethi • Sultanpur • Ayodhya • Ambedkar Nagar • Azamgarh • Mau • Ghazipur
- **Bundelkhand Expressway: 296 km** • Kudrail (Etawah) → Gonda village, Bharatkoop (Chitrakoot) • inaugurated **July 2022** from Jalaun (Kaithari) • crosses Ken, Betwa, Yamuna
- **Bundelkhand — 7 districts:** Chitrakoot • Banda • Mahoba • Hamirpur • Jalaun • Auraiya • Etawah
- **Gorakhpur Link Expressway: 91.3 km** • Jaitpur (Gorakhpur) → Salarpur (Azamgarh — Purvanchal Expressway) • 4 districts: Gorakhpur • Sant Kabir Nagar • Ambedkar Nagar • Azamgarh

Expressway Pradesh — routes schematic · longest Ganga



- **Ganga Expressway (under construction): 594 km** • Bijauli village (Meerut) → Judapur Dandu (Prayagraj) • **3.5 km airstrip in Shahjahanpur** • UP's **longest**
- **Ganga — 12 districts:** Meerut • Hapur • Bulandshahr • Amroha • Sambhal • Budaun • Shahjahanpur • Hardoi • Unnao • Raebareli • Pratapgarh • Prayagraj

Rail Transport

- **First train in UP: March 1859** • Prayagraj to Kanpur
- **Longest rail network:** UP (~**9,167 km**) • first in India
- **2 Railway Zone headquarters: North Eastern Railway (NER) — Gorakhpur** (1952) • **North Central Railway (NCR) — Prayagraj** (2003) {9}
- **Gorakhpur platform: 1,366.33 m** • world's second longest (world's first before Hubballi, Karnataka)
- **First Vande Bharat Express (semi-high speed):** New Delhi–Varanasi • **First Namo Bharat (RRTS) rapid rail:** Sahibabad (Ghaziabad)–Duhai

Air and Water Transport

- **Only state with the highest 5 international airports:** Uttar Pradesh
- **Chaudhary Charan Singh International:** Amausi (Lucknow) • **Lal Bahadur Shastri International:** Babatpur (Varanasi)
- **Kushinagar International:** inaugurated 2021 • **Maharishi Valmiki International:** Ayodhya Dham (inaugurated December 2023)
- **Noida International Greenfield Airport: Jewar (Gautam Buddh Nagar)** • Asia's largest under-construction airport • by Zurich Airport International
- **Domestic airports: Chakeri** (Kanpur) • **Kheria/Pandit Deendayal** (Agra) • **Bamrauli** (Prayagraj) • **Hindon** (Ghaziabad) • **Sarsawa** (Saharanpur)
- **Fursatganj (Raebareli):** Indira Gandhi Rashtriya Uran Akademi • **Civil Aviation Training Centre:** Bamrauli (Prayagraj) • **National Parachute Training College:** Agra
- **National Waterway-1 (NW-1):** on the Ganga **Prayagraj–Haldia (1,620 km)** • **India's longest National Waterway**
- **First Multi-Modal Terminal: Ramnagar (Varanasi)**

Points to Note

1. **Bundelkhand Expressway: Jhansi and Lalitpur not included** • 7 districts — Chitrakoot, Banda, Mahoba, Hamirpur, Jalaun, Auraiya, Etawah (Auraiya and Etawah outside Bundelkhand)
2. **Ganga Expressway:** does not touch **Kanpur, Kannauj, Farrukhabad, Fatehpur** (or Varanasi) — goes via Hardoi-Unnao-Raebareli-Pratapgarh to Prayagraj

One-liner Facts

- **India's first airmail service: 18 February 1911** • Prayagraj (Allahabad) to Naini
- **UPSRTC:** established **1 June 1972** • first state roadways bus **15 May 1947** • Lucknow–Barabanki
- **Taj International Airport:** Firozabad/Agra region • **Trishul Airbase:** Bareilly
- **Metro rail operations:** Lucknow • Kanpur • Noida • Ghaziabad • Agra
- **Varanasi (Ganga):** India's first **water taxi service** and **Tent City** • **MV Ganga Vilas** — world's longest river cruise, Varanasi–Dibrugarh (Assam)

29 Census 2011, Urbanisation and Tribes

PYQ 38

Census 2011 — Basic Data

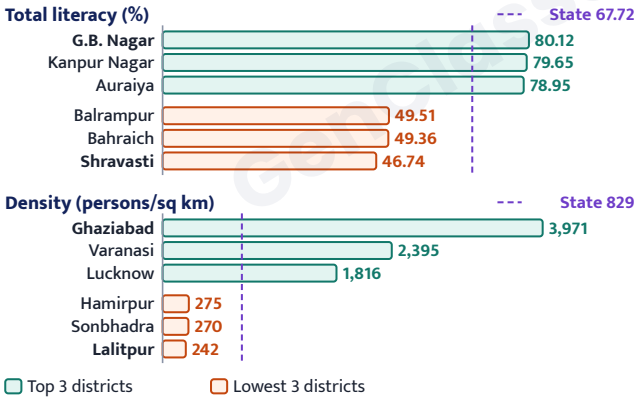
- **Total population: 19,98,12,341** (~20 crore) • **16.51%** of India • **first** in India
- **World comparison:** only 4 countries (China, India, USA, Indonesia) ahead of UP • more than Brazil
- **Male:** 52.29% • **Female:** 47.71%
- **Decadal growth (2001-2011): 20.22%** • above national 17.7% • 15th in India
- **Population density: 829 persons/sq km** • 447 above national 382 • **fourth** among states after Bihar, West Bengal, Kerala • 9th among states + UTs
- **Sex ratio: 912** • 31 below national 943 • up 14 points from 2001
- **Child sex ratio (0-6): 902** • 916 in 2001 • down 14 points
- **Total literacy: 67.72%** • male **77.3%** • female **57.2%** • gap **20.1%**

Census 2011: Top 3 and Bottom 3 Districts

PYQ 26

- **Total population — highest: Prayagraj** (59.54 lakh) → Moradabad → Ghaziabad (4. Azamgarh, 5. Lucknow)
- **Total population — lowest: Mahoba** (8.75 lakh) → Chitrakoot → Hamirpur (4. Shravasti, 5. Lalitpur)
- **Density (829) — highest: Ghaziabad** (3,971) → Varanasi (2,395) → Lucknow (1,816)
- **Density — lowest: Lalitpur** (242) → Sonbhadra (270) → Hamirpur (275)
- **Decadal growth (20.22%) — highest: Gautam Buddh Nagar** (49.1%) → Ghaziabad (42.27%) → Shravasti (30.5%)
- **Decadal growth — lowest: Kanpur Nagar** (9.9%) → Hamirpur (11.1%) → Baghpat (11.9%)
- **Adult sex ratio (912) — highest: Jaunpur** (1,024) → Azamgarh (1,019) → Deoria (1,017) • only these 3 above 1000
- **Adult sex ratio — lowest: Gautam Buddh Nagar** (851) → Hamirpur and Baghpat (861) → Kanpur Nagar, Banda and Mathura (863)
- **Child sex ratio (902) — highest: Balrampur** (950) → Sant Kabir Nagar (942) → Bahraich and Siddharthnagar (935)
- **Child sex ratio — lowest: Baghpat** (841) → Gautam Buddh Nagar (843) → Ghaziabad (850)

Census 2011: top vs lowest districts



Trap: female literacy 1st **Kanpur Nagar (75.05%)**, not G.B. Nagar

- **Total literacy (67.72%) — highest: Gautam Buddh Nagar** (80.12%) → Kanpur Nagar (79.65%) → Auraiya (78.95%) (4. Etawah, 5. Ghaziabad)
- **Total literacy — lowest: Shravasti** (46.74%) → Bahraich (49.36%) → Balrampur (49.51%) (4. Budaun, 5. Rampur)
- **Male literacy (77.3%) — highest: Gautam Buddh Nagar** (88.06%) → Auraiya (86.11%) → Etawah (86.06%)
- **Male literacy — lowest: Shravasti** (57.16%) → Bahraich → Balrampur
- **Female literacy (57.2%) — highest: Kanpur Nagar** (75.05%) → Lucknow (71.54%) → Gautam Buddh Nagar (70.82%)
- **Female literacy — lowest: Shravasti** (34.78%) → Balrampur (38.43%) → Bahraich (39.18%)
- **Urban population (22.27%) — highest:** number — **Ghaziabad** → Lucknow → Kanpur Nagar • percentage — **Ghaziabad** (67.55%) → Lucknow (66.2%) → Kanpur Nagar (65.8%)
- **Urban population — lowest:** number — **Shravasti** → Chitrakoot → Kaushambi • percentage — **Shravasti** (only 3.46%) → Kushinagar (4.72%) → Maharajganj (5.02%)
- **Rural population (77.73%) — highest:** number — **Prayagraj** → Azamgarh → Jaunpur • percentage — **Shravasti** (96.54%) → Kushinagar → Maharajganj
- **Rural population — lowest:** number — **Gautam Buddh Nagar** → Mahoba → Hamirpur • percentage — **Ghaziabad** (32.45%) → Lucknow → Kanpur Nagar
- **Scheduled Castes (20.69%) — highest:** number — **Sitapur** → Prayagraj → Hardoi • percentage — **Kaushambi** (34.72%) → Sitapur (32.26%) → Hardoi (31.14%)
- **Scheduled Castes — lowest (both number and percentage): Baghpat** (11.44%) → Bareilly (12.53%) → Balrampur (12.90%)
- **Scheduled Tribes (0.57%) — highest:** number — **Sonbhadra** → Ballia → Deoria • percentage — **Sonbhadra** (20.67%) → Lalitpur (5.86%) → Deoria (3.54%)
- **Scheduled Tribes — lowest (both number and percentage): Baghpat** (only 14 persons — 0.001%) → Kannauj (15 persons) → Budaun (58 persons)

Urbanisation and Million-Plus Cities PYQ 10

- **Total urban population:** UP **second** after Maharashtra • **4.44 crore** • **22.27%** of state population
- **Million-plus statutory towns (7, descending):** **Kanpur → Lucknow → Ghaziabad → Agra → Varanasi → Meerut → Prayagraj**
- **Million-plus Urban Agglomerations:** also 7 • Kanpur bigger than Lucknow here
- **Municipal Corporations:** total **17** • **17th Shahjahanpur** (April 2018) • 16th Mathura-Vrindavan • 15th Ayodhya

Major Tribes and Their Geographical Distribution PYQ 18

- **Scheduled Tribes Order 1967:** The Constitution (Scheduled Tribes) (Uttar Pradesh) Order by the President • 5 tribes — Tharu, Buksa, Bhotia, Jaunsari, Raji
- **After Uttarakhand's creation:** only **Tharu and Buksa** remained
- **Year 2003:** Centre gave ST status to **10 new tribes** • currently **12 listed tribal groups**
- **Gond group (Dhuria, Nayak, Ojha, Pathari, Rajgond):** Maharajganj • Siddharthnagar • Basti • Gorakhpur • Deoria • Mau • Azamgarh • Jaunpur • Mirzapur • Sonbhadra
- **Gond: UP's largest tribal group** • population order: Gond → Kharwar → Tharu
- **Tharu — area: 5 Terai districts — Maharajganj • Siddharthnagar • Shravasti • Bahraich • Lakhimpur Kheri**
- **Tharu — features:** Kirata lineage • short stature, yellowish complexion • **joint family system** • **Deepawali as a 'festival of mourning'** (practice now fading) • **'Bajhar'** festival
- **Tharu — others:** feast at widow remarriage called **'Lathbharwa Bhoj'** • liquor **'Jaad'** from rice • a degree college for them in Lakhimpur Kheri
- **Buksa (Bhoksa): Bijnor** • Agra • claim link with Patwar Rajput clan • head of panchayat called **'Takht'** • main goddess **Chamunda Devi** • varna division like Hindus
- **Kharwar (Khairwar):** Sonbhadra • Mirzapur • Deoria • Ballia • Ghazipur • original home Palamu (Jharkhand) • Katha (catechu) trade • **'Karma dance'** • physically very strong and fierce
- **Sahariya: Lalitpur** (only district) • very backward tribe of Bundelkhand
- **Pahariya and Baiga: Sonbhadra** (only district) • also **Pankha (Panika), Agariya, Patari, Chero, Bhuiya/Bhuinya** mainly in Sonbhadra (Chero also in Chandauli and Varanasi)
- **Mahigir: Bijnor (Najibabad area)** • follow **Islam** • main occupation **fishing** • mentioned in Mahabharata • not in ST list — part of fishing community

Mnemonic (5 least populous districts, ascending): "Mahoba ke chitra mein Hamir ne Shravasti ke Lalit ko dekha" — 1. Mahoba, 2. Chitra = Chitrakoot, 3. Hamir = Hamirpur, 4. Shravasti, 5. Lalit = Lalitpur

Mnemonic (tribes only in Sonbhadra): "Pahari Baiga ne Agariya ke pankhe se Bhuiya par Patari rakhi" — Pahari = Pahariya, Baiga, Agariya, Pankhe = Pankha/Panika, Bhuiya, Patari

Points to Note

1. **First in female literacy:** not Gautam Buddh Nagar — **Kanpur Nagar (75.05%)**, Lucknow second, Gautam Buddh Nagar third (Gautam Buddh Nagar first in total and male literacy)
2. **Largest tribe:** not Tharu — **Gond group** (after 2003; earlier Tharu) • Kharwar second, Tharu third • choose Tharu only if Gond is not an option
3. **Districts with zero tribes: Ayodhya (Faizabad) and Jalaun** — no recognised Scheduled Tribe

One-liner Facts

- **Male-female literacy gap:** highest **Maharajganj** • lowest **Kanpur Nagar**
- **0-6 age group:** **15.41%** of total population
- **Karma dance:** main folk dance of **Kharwar and Gond** tribes of Sonbhadra and Mirzapur
- **Reserved seats:** ST Lok Sabha seats **zero (0)** • SC Lok Sabha **17** • SC Vidhan Sabha **84** • ST Vidhan Sabha **2 (Obra and Duddhi)**
- **Population Policy (2021-2030):** Total Fertility Rate (TFR) from 2.7 to **2.1** by 2026 • **1.9** by 2030

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